Archives of Computational Methods in Engineering (2026) 33:6177–6274
https://doi.org/10.1007/s11831-025-10370-0

REVIEW

Enhancing Stock Market Prediction with LSTM: A Review of Recent Developments and Comparative Analysis

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Received: 26 February 2025 / Revised: 30 March 2025 / Accepted: 16 August 2025 / Published online: 3 September 2025
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Abstract

Artificial Intelligence can be used to predict stock prices because dissatisfaction is a prerequisite to progress. Those individuals at that time dreamed of forecasting stock prices flawlessly, but this remained just a dream. Today, we use machine learning techniques to predict stock prices with high accuracy as a result of these people's visions. It was found that the Recurrent Neural Network (RNN) is a useful tool for approximating dynamic systems. The Long Short-Term Memory (LSTM) is a RNN with multilayer cells and state memory. In this article, we highlight the importance of LSTM networks in predicting stock market movements. Time series data such as financial time series are modeled by analyzing temporal dependencies. The objective of this paper is to synthesize existing knowledge on LSTM applications, focusing on comparing different approaches to predictive modeling, and providing insight into various areas for further improvement in predictive modeling by utilizing the LSTM model to forecast the stock market. We compare 98 studies based on several important criteria. Its purpose is to inform future research and practical applications in financial decision-making by systematically reviewing these factors. In addition to LSTM networks, another contribution of the article is its investigation of the integration of additional techniques with them, such as sentiment analysis of news and social media data, which may provide an even more comprehensive view of market sentiment than LSTM networks alone. To summarize, this article presents a detailed comparative analysis that can be helpful to researchers and practitioners in the future, in addition to consolidating fragmented knowledge surrounding the use of LSTM in financial forecasting.

1 Introduction

The term stock refers to a form of stock that represents a person's ownership of a company in the form of securities. Investors monitor the stock market when investing in stocks so they can know whether the price of a particular stock is decreasing or increasing. Stock prices are determined by market participants and supply and demand during a particular period. Stock prices are affected by a variety of factors, including interest rates, inflation, and others [1]. In order to forecast stock prices, one must take into account the dynamic and noisy characteristics of stock data influenced by external factors [2]. Financial markets are dynamic and complex, making it difficult to predict stock prices effectively. Innovative approaches to improving stock market decision-making are continually explored by investors, traders, and financial analysts. In market forecasting, traditional technical analysis plays an important role. New techniques are emerging, such as deep learning algorithms, to enhance prediction accuracy and capture market patterns [3]. Machine Learning (ML) is an increasingly important methodology that has been successfully applied across a variety of fields. In the financial field, ML is used for predicting stock prices, designing trading strategies, and evaluating assets, among other applications [4]. In machine learning models, it is challenging to predict stock market movements because the relationship is nonlinear, requiring additional processing power. In order to overcome these drawbacks, researchers began using deep learning models to improve prediction precision [5]. Due to the fact that financial time series are noisy and non-stationary, it is difficult to predict stock price behavior in the future despite the long-lasting debate.

Accurate stock price prediction is a highly relevant research topic within the financial industry and has drawn

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the interest of investors and researchers. Stock prices are influenced by numerous variables that are difficult to predict, including economic conditions, political activity, and market outlooks, making the stock market a highly complex market. Often, traditional models fail to capture the data's nonlinearity and dynamic nature. Due to advances in ML and Deep Learning (DL), neural network models have become widely used for financial forecasting. Because they can handle both short-term and long-term dependencies, LSTM networks are useful for financial forecasting.

The aim of this paper is to provide an overview of 98 research studies that employed LSTM networks to predict stock prices. This review identifies the challenges and presents a coherent review that will benefit new and experienced researchers alike, contributing to the consolidation of fragmented knowledge. According to the Efficient Market Hypothesis (EMH), investors cannot achieve greater returns than the market average through stock prices. Inefficiencies and anomalies can be identified and exploited by LSTMs according to various research works. Sequences are predicted by learning features from large amounts of data, such as stock prices and volumes.

This survey covers various aspects of LSTM networks, including preprocessing, feature extraction, and adjusting the network architecture. We discuss how these traits affect the model's accuracy and performance across different markets and sectors. Also discussed are sentiment analysis techniques, where investors' sentiments are derived from news and social media platforms, and hybrid models, which incorporate LSTMs with other machine learning techniques. Due to the complexity of financial markets, this review discusses not only the advantages of using LSTM networks but also their drawbacks and improvements. The purpose of this paper is to summarize the current state of the art and suggest future research directions. Finally, I hope this paper can contribute to the discussion on the use of deep learning in financial markets, especially in stock price prediction. Figure 1 shows the structure of this article.

2 Background

The following sections aim to present a comprehensive comparative study based on our reviewed papers by providing basic information about them. LSTM is a type of recurrent neural network (RNN). We will first introduce neural networks, and then recurrent neural networks (RNNs). Table 1 shows the common abbreviations of this article.

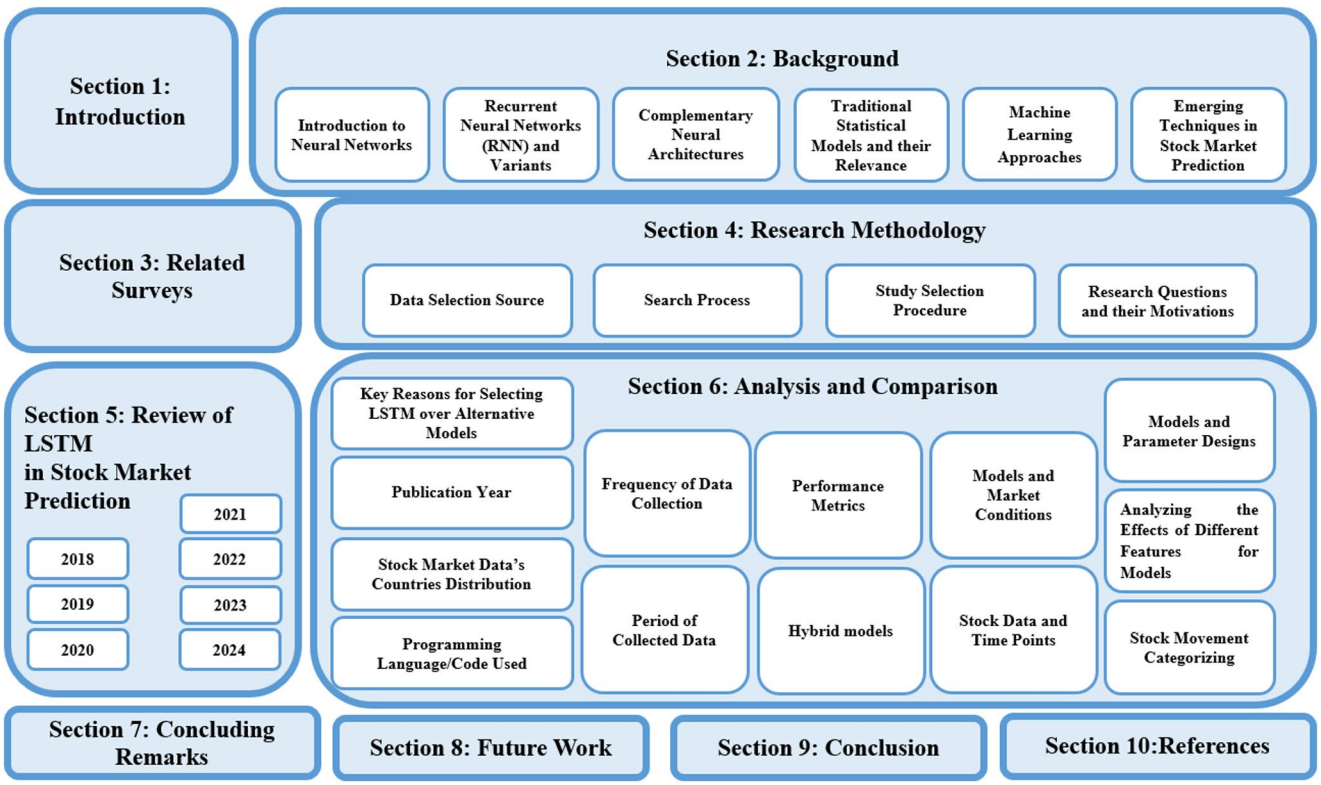


Fig. 1 The article's structure

Table 1 Abbreviations

Abbreviation	Definition	Abbreviation	Definition
LSTM	Long Short-Term Memory	BERT	Bidirectional Encoder Representations from Transformers
Bi-LSTM	Bidirectional Long Short-Term Memory	ANN	Artificial Neural Network
DABiLSTM	Deep Attention Bi-directional Long Short-Term Memory	RNN	Recurrent Neural Network
ConvLSTM	Convolutional Long Short-Term Memory	CNN	Convolutional Neural Network
EA-LSTM	Evolutionary Attention-based Long Short-Term Memory	DNN	Deep Neural Network
SBU-LSTM	Stacked Bidirectional Unidirectional Long Short-Term Memory	GRU	Gated Recurrent Unit
AM	Attention Mechanism	GA	Genetic Algorithm
SoLTD	Sequence-Oriented Long-Term Dependent	DBN	Deep Belief Network
AMPM	Autoregressive Moving Pointer Model	KNN	K-Nearest NeighborsANN
GCN	Graph Convolutional Network	SVM	Support Vector Machine
EMA	Exponential Moving Average	SVR	Support Vector Regression
RSI	Relative Strength Index	RF	Random Forest
ML	Machine Learning	MLP	Multilayer Perceptron
DL	Deep Learning	MDP	Markov Decision Process
WOA	Whale Optimization Algorithm	XGBoost	Extreme Gradient Boosting
PSO	Particle Swarm Optimization	VMD	Variational Mode Decomposition
GWO	Grey Wolf Optimization	VAR	Vector Autoregression
ARO	Artificial Rabbit Optimization	SGD	Stochastic Gradient Descent
PPO	Proximal Policy Optimization	MEEMD	Modified Ensemble Empirical Mode Decomposition
IBBO	Improved Binary Butterfly Optimization	ARIMA	Autoregressive Integrated Moving Average
TLBO	Teaching-Learning Based Optimization	GARCH	Generalized Autoregressive Conditional Heteroskedasticity
RMSE	Root Mean Square Deviation	LightGBM	Light Gradient Boosting Machine

2.1 Introduction To Neural Networks

2.1.1 Artificial Neural Networks (ANN)

Human brains are the inspiration for the ANN, a learning-based information processing unit. This ANN has one hidden layer and works on a backpropagation algorithm known as multilayer perceptrons (MLPs). Data enters the ANN from the input layer in which there are *n* neurons that represent *n* inputs $(x_1, \ldots, x_n)$. In the hidden layer $(h_1, \ldots, h_m)$, there are m neurons in which data is passed through nonlinear activation functions including sigmoid, hyperbolic tangent, rectified linear unit (Relu), etc. As a final step, the output is obtained from the output layer where there are Z number of output neurons, $(o_1, \ldots, o_z)$. The connection links between the input layer and the hidden layer are associated with random weights, $w_{ij}$and w_{jk}, respectively. Gradient descent is used to update these weights after each iteration. Epochs are sets of iterations that cover all observations in a dataset. In order for networks to reach pre-set error thresholds, hundreds, thousands, or even millions of epochs may be required. According to Eq. (1), for each hidden layer, a weighted sum is calculated [6].

$$L_m = \sum_{i=1}^{n} x_i w_{mi} \tag{1}$$

According to Eq. (2), L_m is activated via a nonlinear activation function [6].

$$Y_m = \sigma\left(\beta + L_m\right) \tag{2}$$

Activation function σ is a nonlinear function; bias input β is for hidden neurons and output neurons. Feedforward neural networks take data from the input layer, process it in a hidden layer, and produce output from the output layer [6].

The Multilayer Perceptron (MLP) is an Artificial Neural Network (ANN) with three layers, namely an input

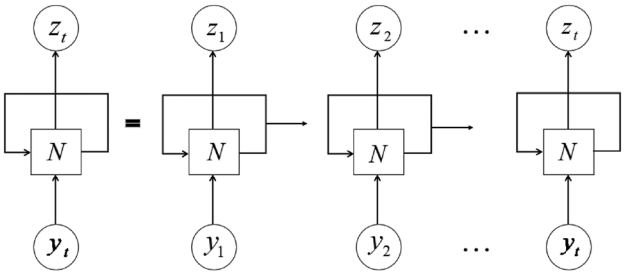


Fig. 2 Recurrent neural network [6]

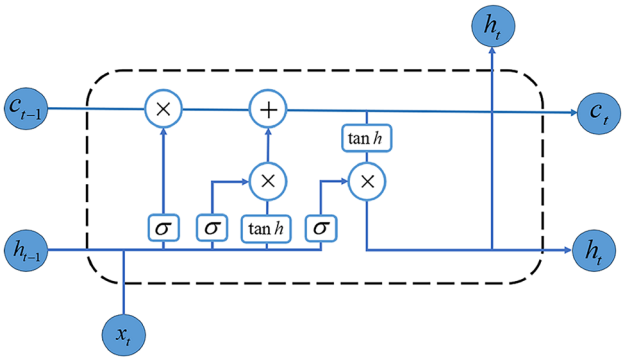


Fig. 3 LSTM architecture [1]

layer, a hidden layer, and an output layer. During the training process, backpropagation is used to implement supervised learning. In MLP training, the result of each hidden layer (referred to as a weight) is backpropagated and these weights are updated iteratively to improve the prediction performance. All parameters of MLP consist of the weights and biases that connect the layers, and finding the optimal values is equivalent to solving the optimization problem, which is usually achieved through gradient descent. First, all parameters are randomly initialized, followed by iterative training. The process involves continuously calculating the gradient and updating the parameters until the stopping criteria are met [7].

2.1.2 Deep Neural Networks (DNN)

DNNs are considered advanced ANNs because they have many hidden layers. Deep neural networks are distinguished by their depth, i.e. the number of layers through which data passes for processing. In order for DNNs to learn effectively, they need to be trained using large datasets. In addition to quantitative data, DNN models also analyze sound, image, video, and any other type of data. During training, DNNs estimate error gradients using a fixed number of samples from the training data. The batch size of DNNs differs from traditional ANNs. The training of a DNN requires a large labeled dataset and a network that can fit the inputs and outputs. GPUs are used to speed up the learning process in most DNNs [6].

2.2 Recurrent Neural Networks (RNN) and Variants

2.2.1 Recurrent Neural Networks (RNN)

A neural network that remembers values from the past can be trained using previous inputs. Figure 2 shows how information is passed between steps. It is difficult for RNNs to handle long-term dependencies. LSTM RNNs are capable of handling such problems. Deep learning's LSTM layer learns long-term dependencies through recurrent connections between memory blocks. There are three multiplicative units that allow reading, writing, and resetting in each of these blocks [6].

2.2.2 Long Short-Term Memory (LSTM)

The Long Short-Term Memory (LSTM) of RNNs is mainly concerned with long-term memory. Due to the possibility of uncertain intervals between significant events, this method can be used to analyze, handle, and forecast time series data. The LSTM also uses a hidden state with three gates to regulate and manage short-term memory. There are three layers in LSTM networks: inputs, hidden layers, and output layers. Figure 3 shows the structure of an LSTM network. Figure 3 shows how information is initially sent from the forget gate. Irrelevant data is discarded in favor of data relevant to the cell state. According to Eq. (3), it can be proved [1].

$$f_t = \sigma\left(W_f\left[ht-1, xt\right] + bf\right) \tag{3}$$

For forgetting gates, the weight matrix and bias vector are determined by three parameters: Wf, Uf, and bf. f_t is the resultant vector containing values between 0 and 1, $\sigma(\cdot)$ is the logistic sigmoid function, and Wf, Uf, and bf are adjustable parameters. The user specifies a length for a vector of zeros that will be used in the first step to initialize hidden states, called ht. The next step is to calculate the potential update vector using the current input x_t and the prior hidden state $ht-1$ [1].

$$\hat{C}t = \tan h\left(Wc.\left[ht-1, xt\right] + bc\right) \tag{4}$$

According to Eq. (4), for each time step, Wc represents the weights, $ht-1$ represents the previous state value, xt represents the current input, bc represents the bias, and $\hat{C}$ represents the cell state function. Multiply and add input gate outputs and cell function results to the cell state [1].

$$Ct = ft \odot Ct-1 + it \odot \hat{C}t \tag{5}$$

As shown in Eq. (5), which is combining the input vector and the candidate memory cell vector, $Ct-1$ is modified and $\odot$ represents an element-wise multiplication of the input vector and the candidate memory cell vector. Input gates, such as forget gates and update gates, hold the results of computation or processing. This layer combines two calculations to get the overall calculation result [1].

$$Ot = \sigma\left(Wo \cdot [ht-1, xt] + bo\right) \tag{6}$$

As shown in Eq. (6), Ot represents the final gate in the current hidden state, which organizes information transfer among all memory cells. This LSTM also produces output in the hidden state. Algorithm 1 shows the LSTM calculation process is outlined within each layer [1].

Pseudocode LSTM Algorithm

Algorithm 1 Pseudocode LSTM Algorithm

1: Input: $x = [x_1, ..., x_n]$, $x_t \in R^m$
2: Given parameters: $W_f, U_f, b_f, W_{\tilde{c}}, U_{\tilde{c}}, b_{\tilde{c}}, W_i, U_i, b_i, V_o, U_0, b_0$
3: Initialize h_0, $c_0 = \vec{0}$ of length p
4: for $t = 1, \ldots, n$ **do**
5: Calculate $f_t, \hat{c}_t, i_t$
6: Update cell state c_t
7: Calculate O_t
8: end for
9: Output $h = [h_1, \ldots, h_n], h \in \mathbb{R}^p$

Pseudocode LSTM algorithm [1]

2.2.3 LSTM Variants

2.2.3.1 Bidirectional LSTM (Bi-LSTM) The Bidirectional LSTM is a variation of LSTM. Dynamic sequence processing is achieved by using LSTMs, where one LSTM takes input forward, and the other takes input backward. In contrast to a regular LSTM, inputs flow in two directions in a Bi-LSTM. In contrast to a traditional RNN, this method uses dynamic processing to resolve the problems it faces. With Bi-LSTM, all previous information can be accessed simultaneously with predicted future information. Figure 4 shows the model of Bi-LSTM [8].

2.2.3.2 Convolutional LSTM (ConvLSTM) Convolutional LSTM or ConvLTSM is a variation of LSTM. This type of RNN combines input-to-state transitions with state-to-state transitions. ConvLSTM usually uses both convolutional structures for spatial and temporal prediction. Based on inputs and past states, it predicts some cells' future states. As shown in Fig. 5, ConvLSTM stores all spatial information in 3D tensors, unlike LSTM. Its strong representational power makes ConvLSTM suitable for predicting complex

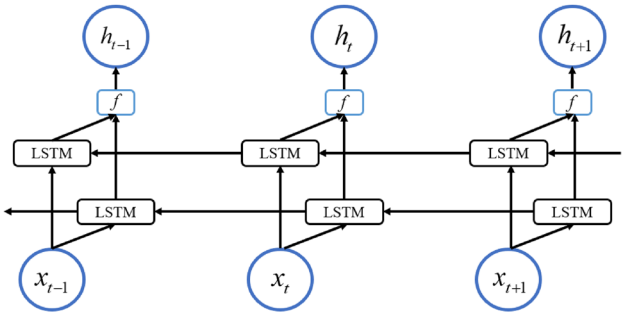


Fig. 4 Bi-LSTM model [8]

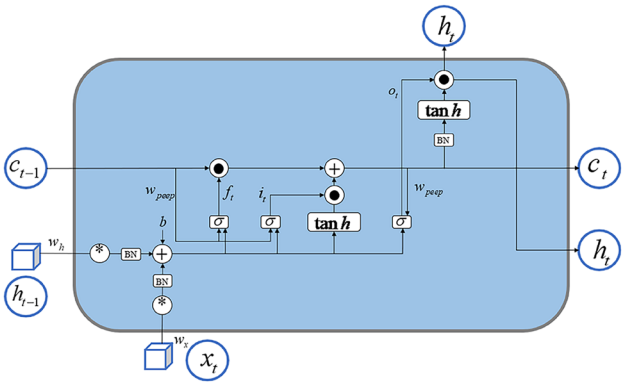


Fig. 5 ConvLSTM model [8]

dynamical systems due to its multiple stacked layers. Convolution instead of matrix multiplication is used in LSTM cells to capture spatial information. In ConvLSTM, LSTM and CNN are combined [8].

2.2.3.3 Stacked Bidirectional Unidirectional LSTM (SBU-LSTM) Deep neural networks are combined with the performance of LSTM to form SBU-LSTM. Modeling complex sequence data with deep neural networks is more effective. By considering both past and future data points, the BDLSTM is able to better capture temporal dependencies. In particular, stock prices benefit from this, since their current value is closely correlated with historic and future trends. The BDLSTM captures the relationship between news sentiment and the number of articles published each day, as well as the temporal relationships between news items and market reactions. As a final step, an LSTM calculates in the forward direction and uses the learned features from the previous hidden layer to predict the values [9].

2.2.4 Gated Recurrent Unit (GRU)

GRUs are more optimized and simpler than LSTM networks. The remember gate and the input gate in GRU are

combined into one gate, the update gate, which works similarly to LSTM. In reset gates, the amount of information that can be derived from the previous information is determined, and a candidate state is generated. In update gates, previous hidden states are retained to a certain extent. Gate and state candidates are then updated [10].

2.3 Complementary Neural Architectures

2.3.1 Convolutional Neural Network (CNN)

Feed-forward neural networks were introduced in 1998. CNN is capable of image processing and natural language processing. CNNs have also been successfully applied to time series prediction. CNNs minimize parameters through local perception and weight sharing. Three layers make up CNNs: convolution, pooling, and full connections. In convolution layers, multiple convolution kernels are used, and their calculation is outlined in Eq. (7). Feature dimensions of a convolution layer must be reduced to reduce training costs and reduce feature dimensions of a pooling layer [11].

$$l_t = \tan h\left(x_{t^*} k_t + b_t\right) \tag{7}$$

where l_t denotes convolution result, $\tan h$ the transfer function implementation, x_t the input vector, k_t the convolution kernel weight, and b_t the bias convolution kernel [11].

2.3.2 Capsule Networks (CapsNet)

In order to process hierarchical relationships within data, CapsNets are neural networks that overcome some of the limitations of CNNs. These neural networks are capable of recognizing complex patterns due to the spatial relationships between features in an image. When CNNs assign the same values to adjacent data points in a feature map, useful information may be discarded, while CapsNets implement capsules. Pooling is less complex than processing capsules of data. In CapsNets, convolutions are performed first, and then primary capsules are created. As a result, high-level capsules are generated in vector format instead of scalar format [12].

2.3.3 Graph Convolutional Network (GCN)

Layer-by-layer updates of node representations are performed in Graph Convolutional Networks (GCNs). It is also possible to apply GCN to other related fields besides node classification, link prediction, community detection, and graph classification [13] with appropriate adaptations. Graph-structured data can be learned using GCN through semi-supervised learning. The neural network operates directly on graphs using an efficient variant of CNNs. Spectral graph convolutions are approximated locally by a first-order convolutional architecture. In addition to encoding node features, the model scales linearly with graph edges.

2.3.4 Variants of Attention Models

2.3.4.1 Attention Layer In the attention model, limited attention resources are used to select a small amount of key information from a vast amount of information. As part of the attention mechanism, LSTM models take more weight from key parts of input sequences, thereby weakening redundant information and extracting useful features from complex data. This mechanism improves the prediction performance of the model by learning the information in the input data [14].

2.3.4.2 Attention Mechanism (AM) Deep learning models benefit from AM representations and comprehensions. Depending on the task, the model can focus on the most relevant input. This technique is widely used in neural network architectures such as RNNs, Transformer models, and CNNs. Different inputs are assigned varying weights depending on their relative importance as a result of the AM. Selective attention allocation allows the model to prioritize crucial elements while disregarding unnecessary ones. As a result, the model performs and is more efficient [11].

2.3.5 Deep Belief Networks (DBN)

DBNs were designed by Geoff Hinton. In DBNs, deep neural networks are connected to deep beliefs. Restrictive Boltzman machines (RBM) are usually stacked together to create a DBN. Stochastic networks are essentially two-layered RBMs [15]. Machine learning algorithms based on deep neural networks are similar to DBNs, but they are not the same. This is a feedforward neural network that has a deep architecture, i.e., many hidden layers. The hidden layer of a DBN is converted into the visible layer using restricted Boltzmann machines (RBMs) or autoencoders.

2.4 Traditional Statistical Models and their Relevance

2.4.1 Time Series Analysis

Data collected over time is analyzed using statistical methods in time series analysis. It is possible to predict future events or better understand how data is generated by examining patterns and trends in the data. It is possible to forecast

a product's demand, predict the stock market, or analyze trends in economic indicators using time series analysis. Analysis of time series data reveals underlying trends, seasonality, and noise in data by predicting future values. Furthermore, time series analysis can be used to identify and forecast the effects of exogenous variables. Furthermore, a time series can be affected by the following factors:

- Trend: Time series trends are defined by their overall direction.
- Seasonality: In data, seasonality refers to patterns, such as holiday sales spikes.
- Cyclicality: Economic booms and busts are examples of cyclical patterns in the data.
- Irregularity: Unpredictable or random fluctuations are called data irregularities.
- Exogenous factors: Economic policies, natural disasters, or social trends are examples of exogenous factors [16].

2.4.2 Autoregressive Integrated Moving Average (ARIMA)

In addition to predicting the future value of a variable based on past values, autoregressive integrated moving averages were developed from the Box Jenkins method. Generally, ARIMA works best with linear and largely stationary time series data, and requires at least 50 historical data events [17]. Based on a combination of historic values, the ARIMA model predicts the value of a variable [18].

2.4.3 Generalized Autoregressive Conditional Heteroskedasticity (GARCH)

GARCH is an extension of ARCH that is better at describing temporal sequence distributions. In GARCH, lagged values at different periods are brought into ARCH, so that the model has a long memory and its lag structure is flexible. GARCH can be described mathematically as shown in Eq. (8) [19]:

$$v_i = \omega + v_{i-1} \cdot \left(\alpha + \beta\epsilon_i^2\right), \epsilon \sim N\left(0, 1\right) \tag{8}$$

The conditional variance can be represented by v_i, while the normally distributed error term can be represented by ϵ. ω, α and β are parameters to be learned by the mode [19].

2.4.4 Vector Autoregression (VAR)

Multivariate time series variables were examined using the Vector Autoregression (VAR) model in Sims (1980). These variables can be modeled simultaneously with their dynamics [20]. Time series variables can be analyzed using VAR as a statistical tool. In contrast to univariate autoregression models, which forecast only one variable, variable autoregression models examine the interconnectedness of many variables. Each variable is modeled based on its past values as well as those of other variables.

2.5 Machine Learning Approaches

Machine learning (ML) is one of the most commonly used methodologies today, having been applied to a variety of fields with considerable success. There are numerous applications of machine learning in finance, including trading strategy design, stock price prediction, and asset evaluation. The use of machine learning is revolutionizing financial investment strategies, enabling more accurate predictions, improved risk management, and data-driven decisions. As ML approaches become more popular, they enable us to predict asset prices, identify hidden patterns, and classify market trends. In addition, alternative data sources such as satellite imagery, social media metrics, and global economic trends can enhance financial investments [4].

Machine learning is a common application of AI for predicting stock prices. Forecasting future patterns is done using algorithms that are trained from historical data. These methods include random forests, support vector machines (SVMs), and linear regression. Linear regression models the relationship between one or more independent variables (past stock data) and a dependent variable (stock price) to make predictions. Also, stock performance decisions are often dynamic, similar to the maximum profit decision in a game, which requires an understanding of market conditions and operational decisions at different times. Predicting stock price data thus offers valuable insights to consider [1, 4].

Machine learning algorithms, especially deep learning algorithms, have enhanced prediction accuracy and captured market patterns thanks to the advent of machine learning techniques. It is possible to analyze sequential data, such as stock prices, using machine learning models such as deep neural networks, RNNs, LSTMs, and GRUs. In reinforcement learning, trading strategies are dynamically optimized based on market interactions to capture the nonlinear and time-dependent dynamics of financial markets. Furthermore, it is widely used to forecast volatility and price dispersion using tools such as GARCH models and hybrid models, which combine traditional statistical methods with machine learning techniques. In addition to feature engineering, alternative data sources, such as social media metrics and sentiment scores, are also used to improve model accuracy. It is possible to minimize risk and maximize returns using optimization algorithms like gradient descent [2, 21].

For the automatic analysis and decision-making of equity investments, algorithms using machine learning are

an important tool. In general, algorithmic trading involves using computers throughout the trading process to perform intricate calculations. The analysis of data before trades (data examination), signal generation for trades (buying or selling recommendations), and the execution of trades (order management) are all part of this process. Data and simulations can be assimilated by deep learning today, going beyond mere data predictions. In this manner, ML can be viewed as a stockbroker who both forecasts and refines strategies for executing transactions. The financial sector can benefit from algorithmic trading because it is effective and efficient in analyzing big data [4].

There are several key benefits of machine learning for financial investments, including increased precision, automated processes, cost savings, the reduction of emotional bias, real-time analytics, and streamlined risk management. The challenges posed by dynamic market conditions, noisy datasets, and interpretability problems are still present. In many cases, deep learning requires extensive computing power and training time, which makes it impractical. In addition, investment strategies must be transparent and explainable to ensure trust and fairness. For efficient operation in volatile markets, hybrid models combining traditional statistical methods with machine learning techniques are promising. Market operations and investment decision-making will continue to be revolutionized by machine learning [21].

In order to gain a comprehensive understanding of the stock market, it is not just necessary to understand its historical trends, but also to anticipate its future movements. Advances in machine learning techniques, including deep learning, unsupervised learning, reinforcement learning, and hybrid models, are transforming financial investments by refining predictions and automating processes. Enhancing accuracy through the use of alternative data sources, in conjunction with optimization algorithms, reduces risks and maximizes returns. In order to maintain trust and fairness, ethical AI, transparency, and explainable machine learning will be crucial. As a result of ML-powered strategies, market efficiency will be enhanced, investors will be empowered, and global finance will be reshaped [21].

2.5.1 Support Vector Machine (SVM)

The Support Vector Machine (SVM) is a machine learning algorithm used for both classification and regression. However, SVM is best suited to classification tasks rather than regression problems. SVM identifies the optimal geometries in N-dimensional space for separating data points into different classes in the feature space based on their hyperplanes. As a result of the algorithm, support vectors, which measure distances between points of different classes, are maximized. Time series forecasting is not the only successful application of SVM. The training samples are classified into two classes by supervised learning. In a 2-D plane, training samples would be assigned two classes. It's a decision boundary [18].

2.5.2 Support Vector Regression (SVR)

Regression tasks are often handled with Support Vector Regression (SVR), a type of SVM. It determines which function can best predict the continuous output value based on an input value. SVR can also use non-linear kernels in addition to linear kernels. As a result of its complexity, a non-linear kernel captures more intricate patterns in the data. In contrast, a linear kernel is the simple dot product of two input vectors. The kernel used depends on the data's characteristics and the task's complexity. SVR minimizes errors similarly to SVM. Time-series data can be accurately predicted with SVR [22].

2.5.3 Random Forest (RF)

Machine Learning uses Random Forest (RF) algorithm to make predictions and then we vote on all the trees to make predictions. In classification and regression tasks, they are widely used. Random forests are classification methods that utilize multiple decision trees and are classified based on the mode of their outputs. The use of RF is widespread today in banking, stock markets, pharmaceuticals, transportation, and e-commerce [23].

2.5.4 Gradient Boosting Models

2.5.4.1 Extreme Gradient Boosting (XGBoost) XGBoost is among the most popular machine learning techniques. Numerous examples of admirable performance have been demonstrated using various types of data. It is a process by which weak classifiers (WC) are converted into much stronger classifiers (SC). It is repeated based on the model's requirements. As an engineering goal, it aims to maximize boosted tree algorithms' performance with the computational resources [24].

2.5.4.2 Light Gradient Boosting Machine (LightGBM) The LightGBM implements the boosting set model efficiently as a member of the boosting set model. In principle, it works like GBDT and XGBoost. However, the residual approximation is the negative gradient of the loss function irrespec-

tive of the case. Taylor expansions approximate the residual (first and second derivatives) and regularization term [25].

2.6 Emerging Techniques in Stock Market Prediction

2.6.1 Variational Mode Decomposition (VMD)

It is entirely non-recursive and quasi-orthogonal to decompose signals on multiple scales with VMD. Wiener filtering is used for denoising, Hilbert transformation is used to solve marginal spectrum, and alternating direction multipliers are used to solve the unconstrained problem by constructing and solving the variational problem. Decomposition is fundamentally about constructing and solving variational problems. Mode numbers determine how signals are decomposed into limited bandwidths. In a variational problem with constraints, VMD finds the optimal solution by converting the problem into an unconstrained one [26].

2.6.2 Modified Ensemble Empirical Mode Decomposition (MEEMD)

MEEMD does not suffer from modal aliasing or incomplete decomposition problems like EMD and EEMD. As a result, random fluctuations, or white noise, are reduced, resulting in thorough decomposition. In this way, each IMF comes closer to the standard IMF and ensures data rationality. By separating abnormal data from the original series, EMD decomposes the remainder [7].

2.6.3 Genetic Algorithm (GA)

In order to solve optimization and search issues, GA simulates natural selection. The process involves selecting, crossing, and mutating a population of candidates called individuals. Based on a fitness function, the generated solution is evaluated. It is repeated over multiple generations until a solution is found that is acceptable. Even though GA is heuristic, if it finds the best global solution, it provides a good solution for a reasonable price. It can be time-consuming and computationally intensive when training large problems [16].

2.6.4 Bidirectional Encoder Representations from Transformers (BERT)

BERT eliminates repetition and complexity. Self-Attention is pretrained in it, which eliminates both. There are two subnets within the Transformer model: Self-Attention and Feed Forward, followed by residual connections and normalization layers. Self-attention allows BERT to train its bidirectional depth representation as well as its output layer simultaneously; furthermore, the pre-trained BERT model can be fine-tuned without further task-specific training [27].

2.6.5 Wavelet Transform (WT)

The Wavelet Transform (WT) decomposes signals into basic functions. Signal transformation analysis is commonly performed using WT. The wavelet method is capable of denoting local signals and obtaining potential signals without complying with special distributions. In principle, high-pass and low-pass filters are applied to the original data [28].

2.6.6 Autoregressive Moving Pointer Model (AMPM)

It is necessary to include independent variables as well as a dependent variable in an Autoregressive Moving Pointer Model (AMPM). When independent variables are viewed as inputs and their corresponding dependent variables as outputs, time series data are used to calculate such input-output pairs [6].

3 Related Surveys

l-Alawi & Alaali [29] selected various papers from two main sectors, namely the banking and healthcare industries. According to this study, LSTM and gated recurrent unit techniques provide the best results regardless of the sector in which they are applied. In this study, Zhou et al. [30] used LSTM models to enhance stock price forecasting and compared their performance with other models including ANN, RNN, and GRU. Based on the significant performance difference between the LSTM model with and without supply chain factors, the inclusion of supply chain factors is crucial for accurately predicting stock prices. It was found that the GRU model performed best in capturing long-term dependencies among all models tested.

Kumar & Anitha [31] provided an overview of the contemporary research on predicting stock prices and highlight the current best algorithms in AI and ML to do so. Divyavalli & Raghuraman [32] analyzed several neural networks, such as LSTM and RNNs, both of which are useful for predicting stock prices. Praneetha et al. [33] compared different methods that primarily predict stock prices. Their main objective was to determine the accuracy of stock market value using machine learning through the best method out of all available, so they chose LSTM.

Patnaik et al. [34] aimed to pinpoint potential pathways for future research in machine learning-driven stock market forecasting by delving into existing literature.

Employing a systematic literature review approach, relevant peer-reviewed journal articles from the past two decades are identified and grouped based on their shared methodologies and contexts. From this analysis, four key categories emerged. Each category's studies were scrutinized to uncover shared discoveries, distinctive insights, limitations, and areas demanding further exploration. The conclusive segment presented overarching deductions and outlines for prospective research endeavors.

Lumoring et al. [35] aimed to identify the best stock price prediction strategy. In a systematic literature review encompassing quantitative and qualitative analyses, SVM emerged as the most popular machine learning technique for predicting stock prices. The LSTM technique, however, demonstrates the highest level of accuracy, achieving a 99.58% accuracy rate. Soreng et al. [36] delved into the application and efficacy of the BI-LSTM method for sentiment analysis on vaccine-related datasets. Through a comprehensive analysis, they evaluated its performance metrics, benefits, limitations, and its positioning against other prevalent models. The findings suggested that BI-LSTM holds significant potential in providing nuanced insights into public sentiment regarding vaccines, which is instrumental for stakeholders to craft informed strategies and communications.

Using deep learning methods, Raipitam et al. [37] proposed two models for predicting stock prices of companies including AAPL (Apple), Brookfield Asset Management (BAM), and Uber using deep learning techniques. CNN is a one-dimensional CNN (LSTM) used in the paper, while LSTM is a bidirectional CNN. LI & BASTOS [38] reviewed the academic literature on financial time series forecasting using DL and technical analysis. The analysis and discussion were guided by four points of view: predictor techniques, trading strategies, profitability metrics, and risk management. Takkar & Chaudhari [39] compiled a systematic survey based on articles utilizing fusion techniques for stock market applications from 2011 to 2020, categorizing them into three categories: information fusion, feature fusion, and model fusion. Their survey provided potential directions for the future and concluding remarks.

From 2015 to 2023, Bai et al. [40] compiled a comprehensive review of data-driven neural network applications for stock forecasting, discussing both classic neural network structures and innovative ones. These include RNNs, CNNs, Transformers, Graph Neural Networks (GNNs), Generative Adversarial Networks (GANs), and Large Language Models (LLMs). As stated by Lin & Marques [41], the most popular stock market prediction AI methods include SVM, LSTM, and ANN. According to Shah et al. [42], various techniques for stock price prediction, including ARIMA, LSTM, Hybrid LSTM, CNN, and Hybrid CNN, were discussed. Results showed that the CNN-LSTM Hybrid Model is highly accurate at forecasting future stock prices, which may be used to generate portfolios. Das et al. [43] introduced GNNs and sentiment analysis, emphasizing their respective contributions and emphasizing limitations of conventional stock prediction techniques.

Takkar & Chaudhari [44] discussed the importance and need for DNNs as it relates to stock price prediction; they examined DNN variations' applicability to temporal data on the stock market, as well as hybrid and metaheuristic approaches. Kehinde et al. [45] published a pioneering scientometric review. The study examined 220 reputable articles (2001–2021) to identify trends. Analyzing science maps was done with VOSviewer. Sheth & Shah [46] demonstrated that financial markets are linked to economic growth and show how artificial intelligence can be used to provide accurate stock market prices. Smagulova & James [47] discussed the history and reasons for the development of LSTM neural networks. LSTM methods were surveyed and recent developments in memory-based LSTM architectures were highlighted.

Htun et al. [48] examined 32 research works that combined feature analysis with machine learning approaches in various stock market applications. They searched Scopus and Web of Science databases for articles published between 2011 and 2022. The articles describe a number of feature selection and feature extraction approaches that have been successfully applied to stock market analyses. Table 2 stated the information of related surveys. Table 3 shows the analyzation of related surveys.

4 Research Methodology

The methodological section describes how the study was conducted in a systematic manner. In order to ensure rigorous and reproducible research, it provides detailed descriptions of the methods and techniques used to collect and analyze data. This study builds a theoretical framework based on a comprehensive review of existing literature on game theory and distributed systems. The research design specifies the study's overall strategy and plan. In addition to the research questions, hypotheses are outlined. The four key steps are illustrated in Fig. 6.

4.1 Data Selection Source

The sources of data for this study were carefully selected to ensure a comprehensive and representative sample. In addition to peer-reviewed journals and conference proceedings, reputable online databases such as Science Direct, IEEEXplore, and SpringerLink were used as primary and basic sources. In order to ensure that the most recent advancements

Table 2 Information of related surveys

Ref.	Year	Title	Publisher	Taxonomy	Paper reviewed	The main aim
Al-Alawi & Alaali [29]	2023	Stock market prediction using machine learning techniques: Literature review analysis	IEEE	Yes	12	Compare ML techniques for prediction in banking & healthcare
Zhou et al. [30]	2023	Enhancing stock price forecasting: Integrating supply chain factors into LSTM models and comparative performance analysis	IEEE	Yes	Not explicitly counted	Enhance forecasting accuracy with supply chain factors
Kumar & Anitha [31]	2022	A detailed survey to forecast the stock prices by applying machine learning predictive models and artificial intelligence techniques	IEEE	Yes	20	Identify best AI/ML algorithms for stock forecasting
Divyavalli & Raghuraman [32]	2023	A review on LSTM based stock price prediction techniques	IEEE	Yes	20	Review LSTM techniques for stock prediction
Praneetha et al. [33]	2023	Classical review on stock prognosis using LSTM	IEEE	Yes	10	Identify stock prediction accuracy methods
Patnaik et al. [34]	2023	Stock price forecasting using machine learning techniques: A systematic review	IEEE	Yes	36	Evaluate ML for effective stock market prediction
Lumoring et al. [35]	2023	A systematic literature review: Forecasting stock price using machine learning approach	IEEE	Yes	30	Overview of ML in stock price prediction
Soreng et al. [36]	2023	A review of sentimental analysis for vaccine dataset using BI-LSTM method	IEEE	No	31	Evaluate BI-LSTM for vaccine sentiment analysis
Raipitam et al. [37]	2023	Comparative study on stock market prediction using generic CNN-LSTM and ensemble learning	IEEE	No	13	CNN & LSTM techniques for stock prediction
LI & BASTOS [38]	2020	Stock market forecasting using deep learning and technical analysis: A systematic review	IEEE	No	34	Review deep learning for stock market forecasting
Thakkar & Chaudhari [39]	2021	Fusion in stock market prediction: A decade survey on the necessity, recent developments, and potential future directions	Science Direct	Yes	37	Survey on fusion techniques for stock prediction
Bao et al. [40]	2024	Data-driven stock forecasting models based on neural networks: A review	Science Direct	Yes	57	Review neural networks for stock forecasting
Lin & Marques [41]	2024	Stock market prediction using artificial intelligence: A systematic review of systematic reviews	Science Direct	Yes	43	Review AI models for stock market prediction
Shah et al. [42]	2022	A comprehensive review on multiple hybrid deep learning approaches for stock prediction	Science Direct	Yes	36	Integrate social media sentiment into forecasting
Das et al. [43]	2024	Integrating sentiment analysis with graph neural networks for enhanced stock prediction: A comprehensive survey	Science Direct	No	76	Examine deep learning techniques for forecasting
Thakkar & Chaudhari [44]	2021	A comprehensive survey on deep neural networks for stock market: The need, challenges, and future directions	Science Direct	Yes	31	Investigate hybrid models for stock prediction
Kehinde et al. [45]	2022	Scientometric review and analysis of recent approaches to stock market forecasting: Two decades survey	Science Direct	Yes	Not explicitly counted	Machine learning in stock prediction
Sheth & Shah [46]	2022	Predicting stock market using machine learning: best and accurate way to know future stock prices	Springer Link	No	Not explicitly counted	Study accuracy of ML algorithms in stock prediction
Smagulova & James [47]	2019	A survey on LSTM memoristive neural network architectures and applications	Springer Link	Yes	17	Survey on LSTM architectures and applications
Htun et al. [48]	2023	Survey of feature selection and extraction techniques for stock market prediction	Springer Link	Yes	32	Review feature selection techniques for stock prediction

and trends in the field were included, the selection criteria focused on publications from the past decade. The databases used in this study are listed in Table 4. Statistics of final chosen articles in each database is shown in Fig. 7.

4.2 Search Process

In order to capture a wide range of relevant studies, the search process was systematic and iterative. Research began in October 2024. A variety of keywords were used to refine the search results, including "stock prediction," "LSTM,"

Table 3 Analyzation of related surveys

Ref.	Paper selection process	Future works and open issues	Environment analysis	Evaluation tool analysis	Graphical representation	Comparison based on the year	Comparison based on performance metrics	Comparison based on the simulation tool
Al-Alawi & Alaali [29]	Yes	Yes	No	No	Yes	No	Yes	No
Zhou et al. [30]	No	Yes	No	Yes	Yes	No	Yes	No
Kumar & Anitha [31]	No	Yes	No	Yes	Yes	No	Yes	No
Divyavalli & Raghuraman [32]	No	Yes	No	Yes	Yes	No	Yes	No
Praneetha et al. [33]	No	Yes	No	Yes	Yes	No	Yes	No
Patnaik et al. [34]	Yes	Yes	No	Yes	No	No	Yes	No
Lumoring et al. [35]	Yes	Yes	No	Yes	Yes	No	Yes	No
Soreng et al. [36]	Yes	No	No	Yes	Yes	No	Yes	No
Raipitam et al. [37]	No	No	No	Yes	Yes	No	Yes	No
LI & BASTOS [38]	Yes	Yes	No	Yes	Yes	Yes	Yes	No
Thakkar & Chaudhari [39]	Yes	Yes	No	No	Yes	No	Yes	No
Bao et al. [40]	No	No	Yes	Yes	Yes	Yes	Yes	No
Lin & Marques [41]	Yes	Yes	No	Yes	Yes	Yes	Yes	No
Shah et al. [42]	Yes	No	No	Yes	Yes	Yes	Yes	Yes
Das et al. [43]	Yes	No	Yes	Yes	Yes	No	Yes	No
Thakkar & Chaudhari [44]	No	No	Yes	Yes	Yes	No	Yes	Yes
Kehinde et al. [45]	Yes	Yes	No	Yes	Yes	No	Yes	No
Sheth & Shah [46]	No	No	No	Yes	No	Yes	Yes	No
Smagulova & James [47]	No	No	No	No	Yes	Yes	No	No
Htun et al. [48]	Yes	Yes	No	Yes	Yes	No	Yes	No

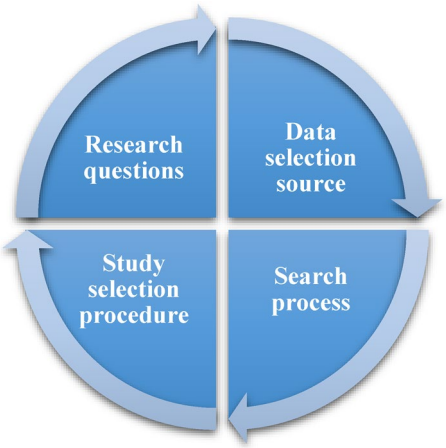


Fig. 6 Steps of research methodology

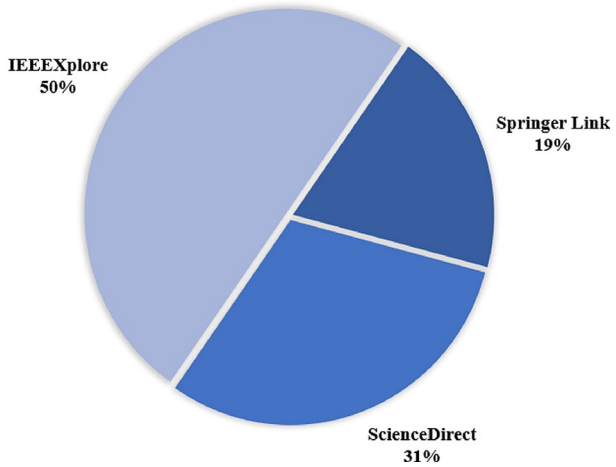


Fig. 7 Articles chosen for each database

Table 4 Data selection sources

No.	Source	URL
1.	ScienceDirect	https://www.sciencedirect.com/
2.	Springer Link	https://link.springer.com/
3.	IEEEXplore	https://ieeexplore.ieee.org/

and"Long Short-Term Memory." Our terms in inclusion/exclusion criteria for selecting a paper are explained here:

- Having a significant proposed LSTM-based model in the stock forecasting field.
- presenting a conspicuous comparison between LSTM-based models in the stock forecasting field.
- presenting a considerable comparison between LSTM-based models and other existing models in the stock forecasting field.
- Exhibiting an excellent analyzation on LSTM-based models on their parameters and performances.

Filters were applied to ensure that irrelevant studies were excluded from the results. To identify additional relevant studies, references list of selected articles was also reviewed.

4.3 Study Selection Procedure

In order to ensure that high-quality and relevant studies were included, the study selection process was meticulously designed. Three major databases were used for our research: IEEE, Springer, and Elsevier. According to Fig. 8, we initially found 71,204 papers using specific keywords. The 11,559 remaining articles after the title and keyword selection were evaluated to deepen our understanding after a detailed review. Inclusion criteria included relevance to LSTM in Stock prediction, peer-reviewed publication status, and full-text availability. Therefore, 9,285 articles were excluded for not meeting the criteria. Inclusion/exclusion criteria were applied to the remaining 2,011 articles. The study focused on 98 papers that addressed the application of LSTM in Stock prediction. A balanced representation of different perspectives and methodologies was achieved in this final compilation.

4.4 Research Questions and their Motivations

In relation to the research question, evaluating the suitability of primary data collection is critical. In Table 5, the research questions and motivations for the study are summarized.

5 Review of LSTM in Stock Market Prediction

5.1 Reviewed Articles of 2018

Liu et al. [49] show that LSTM networks can predict interval time series and long delays. LSTM RNNs were used for filtering, extracting feature value, analyzing, and predicting stock transactions based on their temporal characteristics. However, the experiment showed a better forecasting effect even though their model's accuracy was only 72%. However, they believe the model needs improvement. LSTM models with the right number of stacked layers must be selected within a reasonable timeframe to improve accuracy. To speed up training and testing, LSTM networks should be combined with existing clustering techniques.

Cheng et al. [50] presented a model for predicting stock price movements using attention-based short-term memory. The traders developed trading strategies using historical stock data and technical indicators, then predicted future stock price movements based on those predictions. The deep learning module was used to predict the result. Stocks should be bought when their prediction class is an increasing one, i.e., they predict the stock price will rise. Normally, stocks are sold when a declining class is predicted, which means the stock price is expected to fall. As a result, they developed a trading strategy based on the prediction results.

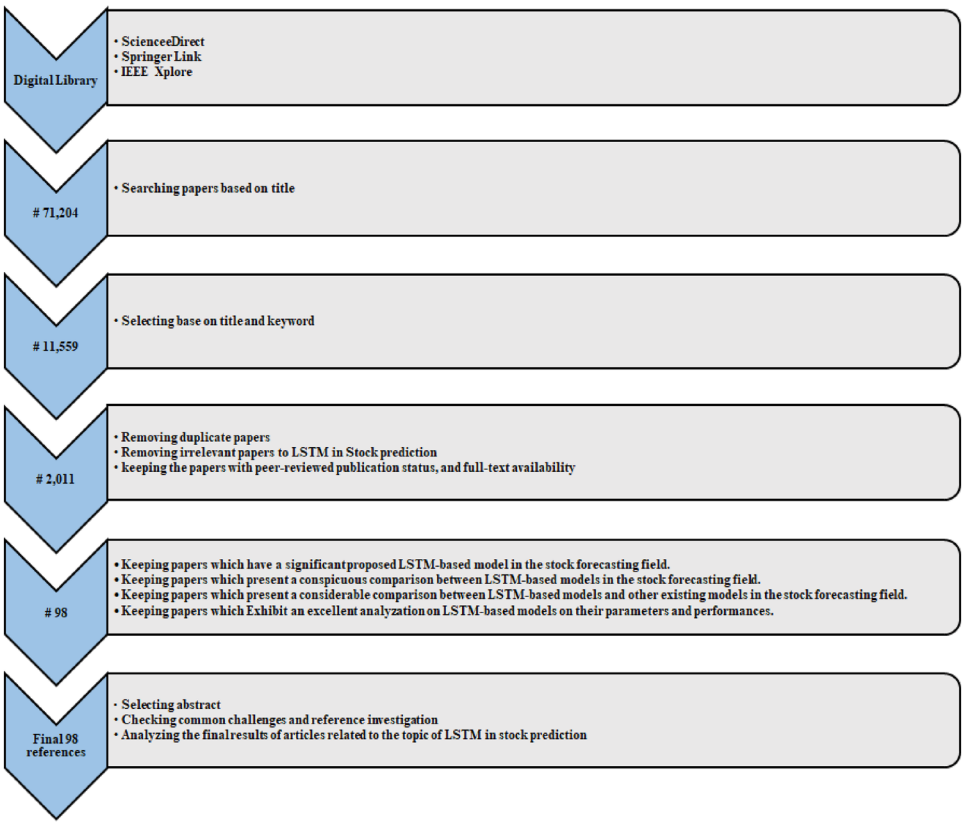


Fig. 8 Process of review methodology

Table 5 Research questions and motivations

No.	Question	Motivation	Section
1.	What are the abilities of LSTM and what is the use of it?	LSTM networks can predict interval time series and long delays. LSTM RNNs were used for filtering, extracting feature value, analyzing, and predicting stock transactions based on their temporal characteristics.	5.1
2.	Which factors influence the stock market's volatility?	The stock market's volatility cannot be mitigated solely based on historical data. Investing and stock market behavior can also be affected by current political and economic events.	5.2
3.	How does LSTM work compared to classical statistical models?	Based on their comparison, both methods capture both the trend and the exact value of the stock data but, LSTM-based models perform better than classical statistical models in balancing portfolios and reducing market risk.	5.2
4.	How do we choose between stateless LSTMs and stateful LSTMs?	Stateless LSTMs are more stable than stateful LSTMs. Because of its higher stability, stateless LSTM models are preferred for time series prediction problems. If successive batches are chunks of text that might be interconnected, stateful LSTMs are recommended.	5.3
5.	How does LSTM work compared to SVR?	By comparing SVR with LSTM-based neural networks and using Several stock indexes in the experiments, including the Dow Jones Industrial Average, the S&P 500, the NYSE, the NSE, and the BSE. Experimental analysis shows LSTM provides better predictions than SVR.	5.3
6.	How does the model's efficiency improve by combining LSTM and CNN?	They are able to conclude that, when sufficient training data is provided, LSTM will perform better over a long-term period of time than CNN. Moreover, CNN models are better suited for predicting trends when there is not a sufficient amount of training data to accurately predict them. Data variation can further increase the robustness of the CNN model. These two basic models were combined to produce LSTM-CNN, which improved performance by 19.3%.	5.4
7.	How does single layer LSTM perform compared to multilayer LSTM?	In comparison to multilayer LSTM models, the single layer LSTM model provides better fit and higher prediction accuracy.	5.5
8.	How does Deep Learning effect LSTM?	Stock price prediction is most effective with LSTM-based deep learning models.	5.6
9.	What's the main difference between LSTM and RNN?	LSTMs can process both long sequences of data and single points, unlike RNNs.	5.6
10.	What input features are required for utilizing time series data by the LSTM?	The characteristics are close price, volume adj-close cost, high price, and open price.	5.7

Jeenanunta et al. [15] predicted daily stock prices for the five largest Thai companies. Forecasting the next daily stock price (High, Low, Open, Close) was done using RNNs. They applied RNN with LSTM for Thai stock price prediction. A comparison was made between the results of the LSTM and the deep belief networks (DBNs). The SET50 index included CPALL, SCB, SCC, KBANK, and PTT. These five stocks were chosen to compare the model's performance with stocks of different volatility levels. There were two experiments with five SET50 stocks. In the first step, MAPE was compared with training data. Training data for one, three, and five years was used in the experiment. The lowest MAPE error values were found for PTT and SCC stocks. CPALL, SCB, and KBANK had the lowest median MAPE errors for one-year training data. The number of looks backs and inputs were varied in the second experiment. The best stock price prediction uses one look back and four inputs. Based on comparisons of the techniques, LSTM predicts CPALL, SCB, and KTB better. LSTM works with low volatility stocks. DBN predicts better with SCC and PTT.

Kumar & Ningombam [51] investigated the effectiveness of LSTM RNNs in predicting Apple stocks from NASDAQ. This paper used a type of RNN called LSTM to predict AAPL stock prices using technical analysis from the NASDAQ exchange. Three output activation layers were tested using the Adam optimization algorithm. A minimal change in the closing price of the selected stock could also explain the improved accuracy. To refine and improve the model,

Table 6 Advantages and disadvantages of 2018's articles

Article	Advantages	Disadvantages
[49]	1. Handles Nonlinear Volatility: LSTM models capture complex stock movements. 2. Long-Term Data Retention: Capably predicts based on historical trends. 3. Flexible Market Contexts: Model adapts to various influencing factors for predictions.	1. Large Data Requirement: Effective training requires vast amounts of high-quality data. 2. Complex Deployment: Implementation can be challenging for non-specialists. 3. Overfitting Risk: Flexibility may lead to model overfitting without proper regulation.
[50]	1. Utilization of Attention Mechanism: This feature allows the model to focus on relevant data segments, enhancing overall prediction accuracy. 2. Integration of Historical Data and Technical Indicators: Merges quantitative and qualitative data, resulting in more robust trading strategies. 3. Multi-Class Prediction Output: Capable of predicting various stock movement classes, supporting informed trading decisions.	1. Complexity of Implementation: The advanced techniques employed complicate both the training and tuning processes. 2. Dependence on Quality Data: The model's performance is heavily impacted by the quality of the input data, particularly in real-time scenarios. 3. Limitations of Historical Data Trends: Sudden market shifts may not be effectively captured, affecting the predictive reliability.
[15]	1. High Forecasting Accuracy: Achieves less than a 2% error margin for numerous stocks. 2. Adaptability to Diverse Data Lengths: Allows for experimentation with various training data lengths. 3. Use of Advanced Neural Network Architectures: Mitigates the vanishing gradient issue of traditional RNNs for efficient learning.	1. Limited to Specific Stocks: Focuses on a narrow set of companies, limiting generalizability. 2. Dependency on Historical Data: Heavily reliant on historical price data, affected by sudden market changes. 3. Computational Complexity: Requires significant computational resources, making real-time applications challenging.
[51]	1. Enhanced Predictive Accuracy: Captures long-term dependencies effectively, improving short-term forecasting. 2. Performance Benchmarking: Provides insights through detailed comparisons across different activation functions. 3. Alignment with Technical Analysis: Complements trader strategies by integrating technical indicators in predictions.	1. Sensitivity to Input Errors: Minor inaccuracies in input data can significantly affect predictive outcomes. 2. Market-Specific Focus: Solely focuses on AAPL stock, limiting broader market applicability. 3. Variable Activation Outcomes: Requires thorough experimentation due to varying results with different activation functions.

they used data from a volatile stock. The accuracy of a price can also be improved by volume, open, and high. Adam backpropagation with LSTM can be used to analyze stock data.

Table 6 shows the comparison between advantages and disadvantages of 2018's articles. Table 7 shows the key attributes and markets of 2018's articles. Table 8 shows the dataset characteristics and feature analysis of 2018's articles. Table 9 shows the comparative analysis and findings overview of 2018's articles.

5.2 Reviewed Articles of 2019

Ojo et al. [18] developed a model based on stacked LSTMs to predict stock market behavior. Based on stacked LSTM networks, this paper presents a model for predicting stock market behavior. Results obtained from training the model showed some accuracy in predicting stock market behavior. The stock market's volatility cannot be mitigated solely based on historical data. Investing and stock market behavior may also be affected by current political and economic events.

Liu et al. [52] exploited cluster stocks to develop a deep learning model for stock prediction. The first model introduced was the LSTM, and the second was the attention model. Cluster stocks were used to predict stock prices using LSTM and attention models. Wind Financial Terminal data was collected to validate the proposed approach. Several experiments were conducted on this data, resulting in a promising development that enhanced stock prediction performance, and the cluster stocks information has a potential for enhancing stock prediction performance.

To improve prediction accuracy, Sanboon et al. [53] proposed a model that captures long-term dependence in stock price data. The Thai Stock Exchange was forecasted using deep learning techniques. Prediction accuracy is enhanced by capturing stock price data. The SET stocks were analyzed using SVMs, Logistic Regressions, Random Forests, Decision Trees, KNNs, and MLPs. In this method, the highest and lowest price are taken from the opening, closing, and highest prices. After 15, 30, 60, and 90 days, all stocks (AOT, BDMS, BEAUTY, CPN, SCC) had the greatest accuracy in SET50. The proposed model failed to outperform all comparison models.

According to Qian & Chen [54], LSTM neural network algorithms were used to predict stock prices under stationary conditions, and multi-experimental data sets were compared statistically. LSTM structures and error analysis methods for stock price prediction are discussed in this article. Despite the fact that experimental results were not affected by stability, the algorithm converged more slowly due to it. Convergence speeds are faster when P values

Table 7 Key attributes and markets of 2018's articles

Reference	Title	Sector	Target	Methods	Market
[49]	Stock transaction prediction modeling and analysis based on LSTM	Specific Stocks	To predict stock transaction prices utilizing the LSTM neural network	LSTM neural networks for filtering and extracting features from stock data to analyze and predict stock transactions	China (CSI 603899 Index)
[50]	Applied attention-based LSTM neural networks in stock prediction	Specific Stocks	To propose and evaluate an attention-based LSTM model for predicting stock price movements, aiming to aid in developing effective trading strategies	Attention-based LSTM model incorporating historical stock data and technical indicators	Taiwan Stock Exchange (TWSE)
[15]	Stock price prediction with Long Short-Term Memory Recurrent Neural Network	Specific Stocks	To forecast daily stock prices (High, Low, Open, Close) of the top five companies in the Thai SET50 index using LSTM and compare with DBN	1. RNN with LSTM. 2. DBN as a comparative method.	Thai stock market, focusing on companies listed in the SET50 index
[51]	Short-Term forecasting of stock prices using Long Short-Term Memory	Specific Stocks	To evaluate the effectiveness of LSTM networks for predicting the end-of-day closing prices of Apple Inc. (AAPL) using technical analysis	Utilized LSTM neural networks with different output activation functions (linear, sigmoid, tanh), and the Adam optimizer as the back-propagation algorithm	United States (NASDAQ Stock Exchange)

Table 8 Dataset characteristics and feature analysis of 2018's articles

Reference	Data	Types of dataset	Features	Period	Frequency	Feature selection/ extraction techniques
[49]	Historical transaction data from the stock data interface of JoinQuant platform	Time series data of stock prices, including open, close, high, low, volume, and others	Features include Moving Average (MA), Exponential Moving Average (EMA), and correlations between open, high, low, and close prices	Data from May 18, 2014, to January 29, 2017 (for training); and from December 26, 2016, to January 29, 2017 (for testing)	Daily data frequency	Utilizes LSTM to extract relevant features based on historical stock data, focusing on patterns and trends in time series
[50]	Historical stock price data sourced from the Taiwan Stock Exchange, including technical indicators like KD, MA, RSV, etc.	Time series dataset	Price data (Open, Close, Low, High, Volume) and technical indicators (KD, MA, RSV, etc.)	The specific years are not mentioned but involve recent and historical stock data	Daily	Integrates an attention mechanism to enhance LSTM's focus on relevant time series data features, improving prediction accuracy and model interpretability
[15]	Stock data sourced from the Stock Exchange of Thailand (SET): CPALL, SCB, SCC, KBANK, and PTT	Time-series dataset comprising daily stock prices including High, Low, Open, and Close prices	Inputs include High Price (HP), Low Price (LP), Open Price (OP), Close Price (CP), and Volume (V). Varied input feature combinations were tested	Tested training data over various periods: - One-year: 30th December 2015 to 30th December 2016 - Three-year: 30th December 2012 to 30th December 2016 - Five-year: 30th December 2011 to 30th December 2016	Daily	Used techniques including moving averages and input combinations to determine effective features for prediction
[51]	Historical stock price data from Quandl's WIKI Prices for AAPL	Time series dataset focusing on stock prices	Daily closing prices of Apple Inc. stocks	From January 1, 2013, to May 1, 2018	Daily	Min-Max scaling of closing prices into (0,1) and (−1,1) ranges to optimize model performance

Table 9 Comparative analysis and findings overview of 2018's articles

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[49]	The performance of single-layer vs. three-layer LSTM models was compared	- Accuracy of single-layer LSTM Model: 66% - Accuracy of three-layer LSTM Model: up to 78%	Short-term stock price prediction	The study found that increasing the number of LSTM layers increases prediction accuracy, although it consumes more computational resources	The model has limitations regarding accuracy (approximately 72%) and requires further improvement; future work includes extracting more features and comparing different neural network models for better accuracy
[50]	Compared with traditional machine learning models and other LSTM variations	Evaluated using accuracy, precision, recall, and F-measure. Exact numerical results are not provided in the summary but indicate the model's effectiveness	Stock price movement prediction	Demonstrates that the attention-based LSTM model outperforms standard models in predicting stock price movements, providing a potential basis for structured trading strategies	Limitations include model complexity and dependence on historical data. Future work suggests improving model interpretability and exploring new feature extraction methods to enhance prediction accuracy
[15]	Compared LSTM performance against Deep Belief Network (DBN) for stock price prediction	Evaluated using: -MAPE -MSE -RMSE Results: LSTM provided less than 2% error for CPALL, SCB, KBANK; DBN yielded best results for PTT and SCC, similarly achieving less than 2%	Focused on price prediction for stocks, specifically forecasting next-day stock prices based on historical data	LSTM performed best for low volatility stocks such as CPALL, SCB, and KBANK with under 2% error; DBN outperformed for high volatility stocks (PTT, SCC). Optimal results were achieved by varying input features and look-back periods	Model limitations include potential overfitting and challenges in capturing market volatility; future work could involve optimizing input features and exploring different LSTM architectures
[51]	Compared LSTM performance with different output activations: linear vs. sigmoid vs. tanh	RMSE: - Linear (0,1): 12.483 - Linear (−1,1): 3.258 (best) - Sigmoid (0,1): 21.769 - Tanh (−1,1): 21.738	Closing price prediction of AAPL	Best accuracy achieved with linear activation function scaled from (−1,1) range, showing potential but still requiring improvements for higher accuracy	Current model tested on a single stock and did not address market volatility. Suggested future work includes integrating data from more volatile stocks, and additional feature inputs like volume data for broader applicability and improved accuracy

are higher. In comparison with the ARIMA algorithm, the LSTM algorithm had a 66.78% lower error rate. In comparison to other neural networks, LSTM networks are more reliable and stable.

According to Faraz et al. [55], autoencoder short-term memory networks are capable of predicting stock market closing prices. Using oscillators and technical indicators, deep learning methods were integrated with technical analysis. Noise was removed from the stock index using wavelet transformation. Anomalies in datasets were also detected using Z-scores. First, the autoencoder was trained to represent the data. The encoder then used the features to build an LSTM network to predict the closing price. Based on previous predictions, the system made new predictions. Simulated results were obtained using the Index of Standard and Poor's 500 (S&P 500) to evaluate the theoretical results. In order to compare results with generative adversarial networks (GANs), several performance criteria were

used. Prediction-by-prediction performed better than GAN in the Python environment.

Peng et al. [56] presented two novel prediction methods embedded in a deep learning framework that made long- and short-term predictions. LSTM and RNN models were embedded with two optimization prediction schemes. In terms of time series analysis, RNNs with LSTM cells were probably the most popular model. These models have been proven accurate in numerous studies. Additionally, LSTM-based models balance portfolios and reduce market risk better than classical statistical models. According to their comparison, both methods capture trend and exact value. They proposed an optimized RNN with LSTM cells. LSTMs can store long-term memories.

Table 10 shows the comparison between advantages and disadvantages of 2019's articles. Table 11 shows the key attributes and markets of 2019's articles. Table 12 shows the dataset characteristics and feature analysis of 2019's articles. Table 13 shows the comparative analysis and findings overview of 2019's articles.

5.3 Reviewed Articles of 2020

Thakkar & Chaudhari [17] developed a CREST-based Long Short-Term Memory (LSTM) prediction method. It has been proposed to use the CREST prediction method for companies listed on multiple stock exchanges in the same country. Based on the movement of Wipro Limited (WIPRO)'s stock price on one exchange, a prediction of the trend in the other exchange was made using the data collected from the daily stock prices. Additionally, CREST applied their approach to Larsen & Toubro Infotech Limited and Infosys Limited to determine its applicability. In order to evaluate the performance of all three companies, root-mean-square error, directional accuracy, precision, recall, and F-measure were used.

Yadav et al. [57] explained that LSTM performance depends heavily on the choice of several hyperparameters, which need to be carefully selected. In addition to time series prediction, LSTM is also used for seasonal and cyclical fluctuations and random noise prediction. It is unknown how LSTM should be configured because it is a relatively new model. This paper aims to fill this research gap. LSTM models were developed using Indian stock market data. A comparison of stateless and stateful models was then performed, along with tuning for the number of hidden layers. According to the results, stateless LSTMs are more stable than stateful LSTMs. Because of its higher stability, stateless LSTM models are preferred for time series prediction problems. If successive batches are chunks of text that might be interconnected, stateful LSTMs are recommended. In most cases, a single hidden layer is recommended because it is more accurate, easier to train, and less likely to overfit. It may be necessary to use more layers for very complex problems that can't be efficiently modeled with one layer, but such cases are unlikely. LSTMs become more stable as the number of hidden layers increases.

Lanbouri & Achchab [58] reported that technical indicators combined with long-short memory can forecast stock prices one-minute, five-minutes, and ten-minutes in advance. Using Kaggle intraday trading data, they analyze technical indicators, and then calculate the regression LSTM model. Analysis of price history, technical indicators, and strategies yields profit metrics. Results of the experiment suggested that the proposed method is effective and reliable.

Moghar & Hamiche [59] used RNNs and especially LSTM models (LSTMs) to predict future stock market values. This paper investigated how much an epoch can improve a machine learning algorithm's prediction precision. As the financial market is abnormally volatile, simple models cannot predict future asset values with better accuracy. Scientists are currently using machine learning in order to make computers perform tasks that normally require human intelligence. They used RNN based on LSTM to forecast future values for GOOGL and NKE assets. As a result of the testing, their model is capable of tracking the evolution of opening prices.

Guo [60] found that sentiments in articles related to company news could predict stock movement on news websites. The LSTM model will be used for predictions. Time-series events, such as stock prices and stock returns, are effectively predicted by LSTM predictors. As a result, this model demonstrated the feasibility of studying public sentiment.

Bathla [22] used SVR to predict stock prices. In order to predict the financial market with high precision, researchers used SVMs. In principle, SVR and SVM are the same, and the aim is to minimize error. Time-series based data can be predicted with SVR, an efficient machine learning technique. This study used deep learning to improve prediction accuracy. The LSTM was optimized using Adam and sigmoid activation functions. This evaluation metric is Mean Absolute Percentage Error. They compared SVR with LSTM-based neural networks. Several stock indexes were used in the experiment, including the Dow Jones Industrial Average, the S&P 500, the NYSE, the NSE, and the BSE. Experimental analysis shows LSTM provides better predictions than SVR.

Gumelar et al. [24] provided historical stock market data for 25 different Indonesian companies. Future stock values were predicted using historical data from both ML algorithms. They successfully built models for Extreme Gradient Boosting (XGBoost) and LSTM. Both predictions have been outstanding. LSTM is slightly more accurate than XGBoost, by 0.005%. While XGBoost iterates 25 times and

Table 10 Advantages and disadvantages of 2019's articles

Article	Advantages	Disadvantages
[18]	1. Extensive Data Utilization: Uses a broad dataset for refined behavioral predictions. 2. Captures Non-linear Patterns: Stacked LSTMs manage complex financial data efficiently. 3. Multivariable Approach: Incorporates diverse indicators for enriched insights.	1. Sensitive Hyperparameters: Requires precise tuning for optimal performance. 2. Requires Significant Data Volume: Needs large datasets for effective training. 3. Dependency on Past Patterns: Relies on historical trends which may not continue.
[52]	1. Cluster Stock Utilization: Leverages group stock behaviors for prediction enhancements. 2. Combination with Attention Mechanism: Focuses data processing for relevant insights. 3. Empirical Validation: Demonstrates robustness through real-world tests.	1. Data Dependence: Quality and specific data are crucial for performance. 2. Overfitting Concerns: Model complexity may lead to overfitting. 3. Limited Market Applicability: Specific to certain stocks, limiting broader application.
[53]	1. High Predictive Accuracy: The model achieves exceptional accuracy rates for specific stock signals, making it highly effective for buy/sell recommendations. 2. Ability to Handle Temporal Dependencies: Efficiently manages long-term dependencies essential for accurate time-series analysis. 3. Robust Evaluation over Varied Time Frames: The performance is assessed across multiple time frames, enhancing the model's credibility and robustness.	1. Complex Model Structure: The intricate architecture complicates training, interpretation, and practical deployment. 2. Data Dependency on Quality: Performance is limited to high-quality data, risking accuracy if the input data is flawed. 3. Limited Generalizability: The findings are specific to the Stock Exchange of Thailand, limiting their applicability to other markets.
[54]	1. Dependence on Quality of Input Data: Accuracy relies on the quality of financial statement data; poor data can hinder performance. 2. Need for Extensive Data: Requires large amounts of historical data for training, which can be scarce or costly to obtain. 3. Model Overfitting: Complexity can lead to overfitting, particularly if training data is limited.	1. Dependence on Quality of Input Data: Accuracy relies on the quality of financial statement data; poor data can hinder performance. 2. Need for Extensive Data: Requires large amounts of historical data for training, which can be scarce or costly to obtain. 3. Model Overfitting: Complexity can lead to overfitting, particularly if training data is limited.
[55]	1. Innovative Hybrid Model: Combines Autoencoder and LSTM, enhancing predictive capabilities. 2. Denoising and Anomaly Detection: Improves data quality through noise removal and anomaly detection, leading to more accurate predictions. 3. Performance Comparison: Shows superior prediction capabilities compared to some generative models.	1. Complexity of Implementation: Requires significant expertise and resources, complicating adoption. 2. Dependence on Feature Selection: The relevance and quality of selected features are crucial for model performance. 3. Generalization Issues: Risk of overfitting due to reliance on historical patterns, leading to poor performance in unpredictable scenarios.
[56]	1. Comprehensive Sentiment Analysis: Uses reliable sentiment from the GDELT dataset, potentially yielding more reliable predictions. 2. Enhanced Predictive Accuracy: Demonstrates significant performance improvements with low MAPE. 3. Temporal Feature Integration: Incorporates important temporal dynamics that enhance robustness.	1. Dependency on External Data Sources: Reliance on the GDELT dataset may create issues with data quality and availability. 2. Sector-Specific Limitations: Results may not generalize beyond the Indonesian market. 3. Complexity of Model Training: The SBU-LSTM architecture adds complexity, leading to potential overfitting.

LSTM iterates 10, LSTM yields the best accuracy after 10 iterations. Hence, XGBoost matches LSTM, which is "naturally" designed to forecast time-series data.

Table 14 shows the comparison between advantages and disadvantages of 2020's articles. Table 15 shows the key attributes and markets of 2020's articles. Table 16 shows the dataset characteristics and feature analysis of 2020's articles. Table 17 shows the comparative analysis and findings overview of 2020's articles.

Table 11 Key attributes and markets of 2019's articles

Reference	Title	Sector	Target	Methods	Market
[18]	Stock market behavior prediction using stacked LSTM networks	Market indices	To predict stock market behavior using stacked LSTM networks based on historical price data	Stacked LSTM networks	United States (NASDAQ Composite Index)
[52]	Exploring cluster stocks based on deep learning for stock prediction	Specific Stocks	To improve stock prediction accuracy by utilizing clustered stock information in a deep learning framework combining LSTM and attention mechanisms	- LSTM model - Attention mechanism - Attention-based LSTM with cluster stock information	China (specifically focusing on stocks from the Shanghai Stock Exchange)
[53]	A deep learning model for predicting buy and sell recommendations in stock exchange of Thailand using Long Short-Term Memory	Specific Stocks (focused on Stock Exchange of Thailand)	To predict buy and sell recommendations for stocks in the Stock Exchange of Thailand using LSTM to improve prediction accuracy	LSTM compared with SVM, MLP, Decision Tree, Random Forest, Logistic Regression, and KNN	Thailand (Stock Exchange of Thailand - SET50)
[54]	Stock prediction based on LSTM under different stability	Specific Stocks	To analyze how different stationary conditions affect the prediction accuracy of the LSTM model in stock price forecasting	Utilized LSTM neural network and ARIMA for comparison; conducted stationary analysis of stock time-series data	China
[55]	Stock market prediction-by-prediction based on autoencoder Long Short-Term Memory networks	Specific Stocks	To propose a prediction-by-prediction method for stock market closing price utilizing Autoencoder LSTM networks (AE-LSTM)	Utilized AE-LSTM for feature extraction and prediction; wavelet transformation for noise reduction and z-score for anomaly detection	United States, focusing on the S&P 500 stock market index
[56]	Stock price prediction based on Recurrent Neural Network with Long Short-Term Memory units	Specific Stocks	To develop novel prediction methods using LSTM units for accurate stock price predictions over short and long-term horizons	1. Recursive Method: Predicts one value at a time using a cascading approach. 2. Hybrid Method: Makes multiple predictions concurrently and averages results	American stock market, focusing on American Airlines data

5.4 Reviewed Articles of 2021

For forecasting stock prices, rather [6] proposed a deep neural network based on Long Short-Term Memory (LSTM). A predictive portfolio model was developed as a result. This model uses an autoregressive moving pointer model (AMPM), which is a new auto-regression scheme. LSTM-DNN generated input-output pairs using AMPM. Based on these predictions, an optimal portfolio was formed. Time series-based stock prediction and portfolio selection are proposed. The stock prices were predicted using LSTM-DNN. These predictions led to the construction of predicted portfolios. In addition to entropy diversification, portfolio risk is represented by odds. The model's access time may be further slowed by non-stationary data.

Srivastava & Mishra [61] explored different forecasting algorithms and compared the most efficient and accurate methods, including Simple Average, Linear Regression, and ARIMA. Weights for each variable are maintained with stochastic gradient descent in LSTM RNN. They used Yahoo Finance to create an efficient and accurate stock prediction model using the TSLA dataset. Their stock forecasts were validated using a data frame analysis, LSTM model

Table 12 Dataset characteristics and feature analysis of 2019's articles

Reference	Data	Types of datasets	Features	Period	Frequency	Feature selection/extraction techniques
[18]	Historical stock market data from Yahoo Finance	Time series data including open, close, high, low prices, and trading volume	Daily stock prices (opening, closing, high, low) and trading volume	January 1, 2009, to July 19, 2019	Daily	Stacked LSTM to learn patterns from historical data
[52]	Data collected from Wind Financial Terminal, focusing on stock prices and trends	Time series data covering various stocks, particularly from the Shanghai Stock Exchange	Features include stock prices such as open, close, high, low, and volume, specifically from a total of 50 selected stocks from the Shanghai Stock Exchange	From January 2010 to September 2019, with 80% data for training and 20% for testing	Daily	Uses cluster stocks as an auxiliary information source and implements preprocessing techniques based on LSTM and attention to extract relevant features
[53]	Stock data sourced from Yahoo Finance API for AOT, BDMS, BEAUTY, CPN, and SCC stocks from January 5, 2015, to December 29, 2017	Time series dataset	High, Low, Opening, Closing prices, and Volume. Output signals are 'buy' (1) or 'sell' (0) based on price movement after n days	January 5, 2015, to December 29, 2017 (730 trading days)	Daily	Data normalization to [−1, 1] using 'tanh' for hidden layers and 'sigmoid' for output, dropout rate of 0.5, and rmsprop optimizer
[54]	Three specific stocks with codes: 399,300, 399,007, 399,009; collected the last 1000 days' dataset for each stock	Time-series dataset containing stock price data	Stock price values extracted over a time period (input features varied, including training and testing sets)	Data from the last 1000 days preceding the experiments	Daily data frequency based on stock market operational days	Stationarity detection using Augmented Dickey-Fuller (ADF) test; Min-Max normalization for standardizing dataset
[55]	Sourced from Yahoo Finance, covering S&P 500 index data over 20 years from March 1, 2000, to November 4, 2019	Time-series dataset comprising stock market data and technical indicators	Features include open, high, low, closing, adjusted closing prices, trading volume, and nine technical indicators	March 1, 2000, to November 4, 2019	Daily frequency of stock prices	Technical indicators calculated included Simple Moving Average (SMA), MACD, RSI, Williams %R, Stochastic Oscillator, Bollinger Bands, and more. Features were extracted using an autoencoder
[56]	Historical stock prices for American Airlines from Kaggle, including columns for date, opening, highest, lowest, and closing prices	Time-series dataset with daily stock prices	Date, Opening Price, Highest Price, Lowest Price, Closing Price	Data spans approximately 20 years from 1970 to present	Daily	Normalization of dataset and calculation of average prices

construction, and RMSE calculations. These results recommend LSTM for stock market forecasts.

Chang et al. [62] used LSTM multilayer price prediction models to determine which correlation parameters affected training efficiency and prediction accuracy. Finally, a specific improvement direction was suggested for LSTM prediction. LSTMs accurately predicted stock prices in experiments. Model training efficiency is improved by input dimension. In the same parameters, a complex network will decrease accuracy as it becomes more complex.

Zhou [63] compared CNN and LSTM models for predicting stock prices as a part of his research. To increase the accuracy of prediction, the researchers combined both models into LSTM-CNN in order to make better predictions. They are able to conclude that, when sufficient training data is provided, LSTM will perform better over a long-term period of time than CNN. Moreover, CNN models are better suited for predicting trends when there is not a sufficient amount of training data to accurately predict them. Data variation can further increase the robustness of the CNN model. These two basic models were combined to produce LSTM-CNN, which improved performance by 19.3%.

Wang & Zuo [64] used a long-short memory model to predict stock prices. The data were normalized from 0 to 1 due to the variability of stock prices. Adjusting the hidden layers, learning rate, and time window improved the model's performance. The LSTM has a control part to take advantage of historical data. For comparing the proposed method with RNNs, they used RRMSE, MAE, and MAPE datasets. In general, more accurate predictions have lower indicators. Its predictions are more accurate than those of RNN with appropriate parameters in three indicators. After

Table 13 Comparative analysis and findings overview of 2019's articles

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[18]	Compared with traditional time series methods	- MSE: 0.0022 - Mean Absolute Deviation (MAD): 0.0495 - Accuracy: 53.6%	Stock market behavior prediction	Stacked LSTM networks can identify patterns in historical data but showed limited accuracy at 53.6%	Limitations include reliance on historical data alone. Future work involves incorporating real-time factors like news and events
[52]	The performance of the proposed attention-based LSTM model is compared against a traditional LSTM model (denoted as LSTM clst) in terms of prediction accuracy and performance	- Accuracy for various indices: - LSTM clst (Shanghai Stock Index): 90.0% - att−LSTM clst (Shanghai Stock Index): 91.0% - Other indices show varied improvements with approximate figures: - Shenzhen Stock Index: 64.8% (LSTM), 73.6% (att−LSTM), - China Stock Index: 57.0% (LSTM), 72.6% (att−LSTM), etc.	Stock price prediction using deep learning approaches that utilize cluster stock information	Implementing cluster stocks information in the predictive models significantly enhances prediction accuracy. The attention mechanism aids in isolating the most relevant data for stock price prediction	Current models rely on historical price data; future research aims to integrate additional information sources that could further enhance stock prediction capabilities
[53]	Compared against SVM, MLP, Decision Tree, Random Forest, Logistic Regression, and KNN	Evaluated using accuracy, true positives (TP), true negatives (TN), false positives (FP), false negatives (FN). LSTM models showed highest accuracy across stocks, e.g., up to 100% accuracy on AOT($n=90$)	Prediction of buy and sell recommendations for stocks	LSTM outperforms traditional models in SET50 predictions, demonstrating the ability to capture dependencies in stock data more effectively	Limitations include reliance on historical data, with future work suggesting the inclusion of external factors like exchange rates and additional indicators for better accuracy
[54]	Compared LSTM model results against ARIMA model results	MAE and RMSE were utilized to assess prediction accuracy. Results: −399,300: MAE: 0.02136 (Test), RMSE: 0.02759 (Test) −399,007: MAE: 0.02413 (Test), RMSE: 0.02004 (Test) −399,009: MAE: 0.03759 (Test), RMSE: 0.02122 (Test)	Price prediction of stocks	LSTM outperformed the ARIMA model, showing an error rate 66.78% lower than ARIMA, indicating better prediction capabilities	Limitations: LSTM requires substantial training time and data samples. Future work: Aim to improve LSTM's training efficiency and accuracy, and apply the method across other fields
[55]	Compared against GAN for daily adjusted closing price predictions	MAE, RMSE, and Average Return: - AE-LSTM: MAE=0.0054, RMSE=0.0073, AR=4.34×10^−5 - GAN: MAE=0.2923, RMSE=0.3553, AR = −9.36×10^−5	Price prediction of stocks, specifically daily adjusted closing prices	AE-LSTM outperformed GAN, with lower MAE and RMSE, demonstrating its efficacy in predicting adjusted closing prices	Limitations include challenges with model robustness and prediction cycles; future work involves improving model adaptability and exploring applications in other financial sectors
[56]	Compared LSTM methods with SMA and EMA. Hybrid vs. Recursive method comparison for accuracy	Evaluated using MSE: - Hybrid Method: MSE=5.631e-4; - Recursive Method: MSE=1.042e-2	Price prediction of stock prices, focusing on both short-term and long-term forecasts	The Hybrid Method outperformed the Recursive Method, demonstrating improved accuracy through averaging despite inherent market prediction challenges	Limitations include overfitting and capturing market volatility; future work aims to adjust prediction weights and reduce cumulative error

considering historical stock prices, LSTMs are more accurate at predicting dynamic non-linear data.

Zolfaghari & Gholami [65] investigated the potential of a hybrid model for predicting stock indices and volatility. In order to forecast stocks, a hybrid model combining adaptive wavelet transform, LSTM, and ARIMAX-GARCH family models is used. Stock volatility is predicted using AWT, LSTM, and heterogeneous autoregressive models of realized volatility (HAR-RV) for two major indexes on the U.S. stock market. When compared to the benchmark models, AWT-LSTM-ARMAX-FIEGARCH models with student's t distributions outperformed them. Both stock indexes were better able to predict different time horizons (1-, 10-, 15-, 20-, 30-, and 60-days in advance) using this model. The

Table 14 Advantages and disadvantages of 2020's articles

Article	Advantages	Disadvantages
[17]	1. Cross-Exchange Insights: Utilizes information from multiple exchanges to make more informed and accurate stock predictions.	1. Geographic Limitation: Primarily focused on the Indian market, which may limit its applicability to other regions.
	2. Validation Across Companies: Tested on various Indian companies, providing versatile and applicable results in different sectors.	2. Correlation Assumption Limitations: Assumes consistent correlations across exchanges, which may not always hold true, especially during anomalies.
	3. Diverse Metric Evaluation: Uses various performance metrics to ensure a thorough understanding of model effectiveness.	3. Data Quality Reliance: The performance of the model heavily depends on the consistency and quality of the data from different exchanges.
[57]	1. Hyperparameter Optimization: Tailored tuning improves prediction capabilities.	1. Dependency on Initial Conditions: Performance significantly impacted by initial settings.
	2. Comparison of Stateful and Stateless Models: Provides insights for more precise forecasting.	2. Limited Guideline Availability: Few precedents for hyperparameter setup.
	3. Use of Local Dataset: Enhances relevance with indigenous data.	3. Computational Complexity: High costs may deter widespread application.
[58]	1. High-Frequency Data Utilization: Captures minute-to-minute market changes.	1. Model and Data Complexity: Large data requirements may limit use.
	2. Incorporates Technical Indicators: Enhances prediction abilities.	2. Sensitivity to Market Noise: Differentiating signals from fluctuations can be challenging.
	3. Profitability Analysis: Evaluates real trading feasibility.	3. Short-Term Focus Limitations: Constrains predictive horizon.
[59]	1. Captures Long-Term Dependencies Efficiently: Improves predictive accuracy.	1. Data Quality Dependence: Historical data quality is crucial.
	2. Technique Combination: Enhances accuracy with varied methodologies.	2. Overfitting Risk: Requires careful regulation for generalization.
	3. Handles Non-linearity: Manages complex financial data relationships.	3. Model Tuning Complexity: Parameter tuning demands expertise.
[60]	1. Integrates Reputable News Sentiment: Uses structured news data for reliable sentiment analysis.	1. Dependence on News Quality: Predictions may be biased by news sentiment sources.
	2. Improved Predictive Accuracy: Combining sentiment with traditional data enhances prediction.	2. Complex Model Implementation: Requires advanced technical skills to integrate models.
	3. Effective Trend Capture: LSTM manages sequential data for better trend analysis.	3. Risk of Overfitting: Complex models may overfit to historical patterns.
[22]	1. Superior LSTM Accuracy: Outperforms SVR in predictive capabilities.	1. Restricted Model Scope: Limited to LSTM and SVR without broader exploration.
	2. Effective Trend Analysis: Captures time-dependent trends in stock prices.	2. Data Quality Reliance: Heavily depends on historical data accuracy.
	3. Diverse Index Validation: Tested across various datasets for robustness.	3. Overfitting Risks: Despite dropout strategies, overfitting remains a concern.
[24]	1. High Prediction Accuracy: Demonstrates outstanding predictive accuracy through the synergy of XGBoost and LSTM.	1. Limited Dataset Representation: Focuses on a narrow range of companies, potentially leading to skewed results.
	2. Advanced Algorithm Combination: Effectively captures complex data patterns with sophisticated algorithms.	2. Heavy Reliance on Historical Data: Predictions depend heavily on the availability of accurate past market data.
	3. Targeted Market Adaptation: Tailors predictions to the Indonesia Stock Exchange, making them highly relevant.	3. Challenges with Dynamic Markets: Real-time market changes can pose challenges for maintaining prediction reliability.

Table 15 Key attributes and markets of 2020's articles

Reference	Title	Sector	Target	Methods	Market
[17]	CREST: Cross-reference to exchange-based stock trend prediction using Long Short-Term Memory	Stock Market, focusing on Market Indices of Indian stock exchanges (NSE and BSE)	To enhance stock trend forecasting by employing a Cross-Reference to Exchange-based Stock Trend (CREST) prediction method using LSTM	LSTM models for trend prediction, utilizing a novel cross-reference approach from data gathered from multiple exchanges	India, specifically National Stock Exchange (NSE) and Bombay Stock Exchange (BSE)
[57]	Optimizing LSTM for time series prediction in Indian stock market	Specific Stocks (Indian Stock Market)	To optimize LSTM models for predicting stock prices in the Indian stock market, focusing on comparing stateful and stateless configurations and tuning hyperparameters	LSTM models with hyperparameter tuning, comparison of stateful vs. stateless models, varying hidden layers	India (NIFTY 50 Index)
[58]	Stock market prediction on High frequency data using Long Short-Term Memory	Market Indices	To forecast stock prices one minute, five minutes, and ten minutes ahead using high-frequency trading data combined with technical indicators through LSTM models	LSTM models with the integration of technical indicators such as EMA, MACD, and Bollinger Bands to predict stock prices based on historical high-frequency trading data	United States (S&P 500 Index)
[59]	Stock market prediction using LSTM Recurrent Neural Network	Specific Stocks (GOOGL and NKE)	To employ LSTM models for forecasting future stock prices and analyzing the impact of epoch variation on prediction accuracy	LSTM as a part of a RNN; optimization via Mean Squared Error	United States (New York Stock Exchange - NYSE)
[60]	Stock price prediction based on LSTM Neural Network: The effectiveness of news sentiment analysis	Stock Market Prediction	To explore the impact of news sentiment analysis, combined with LSTM neural networks, on improving the accuracy of stock price predictions	LSTM neural networks enhanced with sentiment analysis derived from news articles	U.S. Stock Market
[22]	Stock price prediction using LSTM and SVR	Specific Stocks	To apply deep learning techniques, particularly LSTM and SVR, to enhance the accuracy of stock price predictions	- LSTM - SVR	Various global stock markets including S&P 500, NYSE, NASDAQ, NSE, BSE, and Dow Jones Industrial Average
[24]	Boosting the accuracy of stock market prediction using XGBoost and Long Short-Term Memory	Specific Stocks	To enhance stock market prediction accuracy by combining XGBoost and LSTM models on IDX data	Utilized Extreme XGBoost and LSTM for predictions	Indonesia (Indonesia Stock Exchange - IDX)

Table 16 Dataset characteristics and feature analysis of 2020's articles

Reference	Data	Types of datasets	Features	Period	Frequency	Feature selection/extraction techniques
[17]	Collected from Yahoo Finance, focusing on daily historical prices of companies like Wipro Limited	Time-series dataset including daily stock prices: open, high, low, close prices, and trading volume	Input features include daily open prices and potentially derived indicators for forecasting	Data from January 1, 2009, to January 1, 2019	Daily	Selection based on synchronized trading days across NSE and BSE; technical indicators derived as necessary
[57]	Stock price data from selected companies within the NIFTY 50 index, collected from the World Wide Web	Time series dataset	Adjusted closing stock prices for companies, focusing on major companies across different sectors	December 29, 2008, to May 24, 2019	Daily	Adjustments for corporate actions, differencing to achieve stationarity, feature engineering to prepare data for modeling
[58]	High-frequency intraday trading data for the S&P 500, sourced from Kaggle, comprising Open, High, Low, Close prices, and Volume from September 11, 2017, to February 16, 2018	Time series dataset	Open, High, Low, Close prices, Volume, EMA, MACD (Moving Average Convergence Divergence), Bollinger Bands indicators	September 11, 2017, to February 16, 2018	Data recorded at one-minute intervals	Technical indicators calculated to enhance feature richness; data cleaning and standardization techniques applied
[59]	Daily stock prices for GOOGL (August 19, 2004, to December 19, 2019) and NKE (January 4, 2010, to December 19, 2019) from Yahoo Finance	Time series dataset	Daily opening prices for GOOGL and NKE; variations across different epochs during training	GOOGL: 2004–2019; NKE: 2010–2019	Daily	Implemented epoch variations for model training; dropout layers for enhanced accuracy
[60]	Stock price data sourced from Yahoo Finance, along with news articles from The New York Times for sentiment analysis	Time series data for stock prices and textual data for sentiment analysis from news articles	Features include historical stock prices (open, high, low, close, volume) and sentiment scores from news articles (headline composition, article polarity)	Data spans from January 1, 2015, to August 13, 2020	Daily	Incorporates sentiment analysis using VADER for headlines and TextBlob for full article content to derive sentiment scores; data normalized using z-scaling
[22]	Stock index data from sources such as Yahoo Finance, specifically for S&P 500, NYSE, NASDAQ, NSE, BSE, and Dow Jones	Time series data reflecting historical stock prices over a set period	Historical stock price features: Open, Close, High, Low, and Volume	Data from January 2015 to January 2020, with training from January 2015 to June 2019 and testing from July 2019 to January 2020	Daily	Utilizes input, forget, and output gates in LSTM to manage historical data filtering and prediction optimization
[24]	Historical stock data from 25 IDX-listed companies sourced from Kaggle	Time series of daily stock market prices	- Opening prices - High prices - Low prices - Closing prices - Trading volumes	Data from 2000 to 2019, with focus on 2017 data for simulations	Daily	Derived price indicators and trading signals (e.g., golden cross and death cross) for analysis

AWT-LSTM model improves the prediction quality of HAR (3)X-RV for the mentioned time horizons.

Keyan et al. [66] used novel online data sources to predict stock prices. It is often possible to find good investment information on social media platforms (such as China). Thus, social media platforms may provide valuable stock information. In order to create daily social networks, we used data from EastyMoney, China's largest social media platform. With the network variable added to traditional variables, they predicted the close prices of each stock based on the LSTM model. The social network variable significantly improved prediction accuracy. Additionally, their findings can help investors understand the relationship between social networks and stock prices.

Table 17 Comparative analysis and findings overview of 2020's articles

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[17]	CREST method applied to additional case studies on Infosys Ltd and Larsen & Toubro Infotech Ltd to ensure robustness	Evaluated using RMSE, Directional Accuracy (DA), Precision, Recall, and F-Measure. Specific values were not provided from the chunk	Predicting stock price trends and directional changes across exchanges	Demonstrated effectiveness of the CREST method using LSTM networks, emphasizing cross-exchange data utilization for improved accuracy in trend prediction	Future directions include enhancing model performance, exploring other industries, or using additional data sources for broader applicability beyond the IT sector
[57]	LSTM models compared with configurations involving different numbers of hidden layers and statefulness attributes	RMSE as primary metric; results supported by statistical analysis (P-value tests, one-way ANOVA)	Stock price prediction across multiple stocks in the Indian stock market using time-series data	Stateless configurations provided more stability than stateful ones; a single hidden layer was found optimal to reduce errors and overfitting	Further research could explore different LSTM variations, use larger datasets, and enhance epochs for better hyperparameter insights. Focus on computational improvements
[58]	Performance assessment of LSTM models using historical trading data with and without technical indicators	RMSE used for evaluation, indicating: − 1-minute without TI: 0.0018 - With TI: 0.0721 − 5-minutes without TI: 0.0046 - With TI: 0.0201 − 10-minutes without TI: 0.0046 - With TI: 0.0108	Stock price prediction for the S&P 500 index using minute-by-minute trading data	LSTM models show effectiveness in short-term stock price prediction, with better performance observed in the one-minute prediction without technical indicators	Accuracy below 50% poses challenges; proposes exploring deeper learning techniques, sampling methods, and back-testing for enhanced model capabilities
[59]	Analyzed the performance of the LSTM model across different epoch counts for GOOGL and NKE	MSE; different epoch counts resulted in varying predictive capabilities	Price prediction of GOOGL and NKE stocks	LSTM models demonstrated varying accuracies with epoch changes; more epochs introduced prediction volatility	Limitations include reduced predictive ability in volatile markets; future research will explore optimized epochs and model architecture enhancements
[60]	Compared LSTM with sentiment analysis against traditional stock prediction methods and models without sentiment inputs	Performance evaluated using MSE, showing improved prediction accuracy with the inclusion of sentiment analysis	The focus is on stock price prediction, enhancing LSTM predictions by integrating sentiment analysis from news articles to determine overall effect on prediction accuracy	The integration of news sentiment notably enhances LSTM model performance, demonstrating higher accuracy in predicting stock prices, suggesting the value of textual data for financial forecasting	Limited by model's ability to generalize across different market conditions; future work recommended in optimizing model parameters, exploring other sentiment analysis tools, and expanding datasets for broader applicability
[22]	Comparisons between LSTM and SVR models conducted to assess prediction performance across different datasets	MAPE used as the primary evaluation metric, with LSTM showing superior accuracy across most stock indices compared to SVR	Stock price prediction utilizing advanced models like LSTM and SVR for increased forecast accuracy and reliability	LSTM consistently outperformed SVR in prediction accuracy, highlighting its effectiveness in capturing stock price trends over time	The study notes limitations in traditional machine learning techniques, proposing future integration of CNN and hybrid models, and additional metrics like RMSE and MAE
[24]	Compared performance of XGBoost and LSTM models	XGBoost: ~99% accuracy; LSTM: ~99.995% accuracy based on training loss graphs	Stock price prediction	Both models demonstrated high prediction accuracy, with XGBoost slightly outperforming LSTM	Future research should include more technical indicators and other ML models for better predictions

For swing traders, Banik et al. [67] developed a decision support system based on LSTM to fully analyze and predict their stock's future values. In addition to the stock price prediction for the next 30 days, the Decision Support System will also analyze the company's MACD and SIGNAL LINE analyses and five Fibonacci retracement levels. The report calculates a success score to improve the traders' investment decisions. This model provides better results than existing techniques, as the Root Mean Square Error, Mean Absolute Error, and Mean Absolute Percentage Error were 4.13, 3.24, and 1.21%, respectively.

Lin et al. [68] forecasted the prices of S&P500 and CSI300 using a mixture model combining LSTM and Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN). In CEEMDAN, original data is decomposed into several IMFs and a residue. LSTM forecasts are generated using decomposed data. It is necessary to reconstruct the prediction sequences to get the final

prediction. This paper discusses SVMs, Backward Propagation (BP), Wavelet Neural Networks (WAVs), and their mixture models. MCS test was used for evaluation in developed and emerging stock markets, demonstrating CEEMDAN-LSTM's optimal forecasting effects. Predicting price trends and avoiding investment losses can be achieved using a model with high prediction accuracy. Moreover, regulators can use it to identify abnormal market fluctuations.

Table 18 shows the comparison between advantages and disadvantages of 2021's articles. Table 19 shows the key attributes and markets of 2021's articles. Table 20 shows the dataset characteristics and feature analysis of 2021's articles. Table 21 shows the comparative analysis and findings overview of 2021's articles.

5.5 Reviewed Articles of 2022

Liu et al. [19] modelled the volatility of specific stock risks. GARCH was used as a baseline model, and LSTM was explored. In the LSTM model, each stock specific risk factor should be packaged at each timestamp. LSTM is clearly superior to GARCH in empirical experiments on BARRA data. Stock volatility is important for risk management and portfolio allocation since it plays a role in specific risk management. To predict volatility, they used LSTM, a RNN with a long memory, inspired by rapid advances in deep learning. Each timestamp should be packed with all stocks' specific risk factors and fed to the LSTM. Empirical experiments on official BARRA data found the LSTM to be more predictive than the traditional GARCH. However, specific risk appears to be independent of volatility, allowing for better forecasting.

Bhandari et al. [69] proposed an improved neural network architecture for predicting stock index prices. Several sources of data are used to construct the model, including fundamentals, macroeconomics, and technical information. After that, min–max normalization was performed. After that, time steps are used to create input sequences. The neuron parameters include epochs, learning rates, batch sizes, and time steps. Regularization techniques have overcome overfitting. The closing prices of stocks can be predicted by LSTMs once hyperparameters have been tuned. In comparison to multilayer LSTM models, the single layer LSTM model provides better fit and higher prediction accuracy.

Yao [70] introduced the LSTMRNN-based stock price prediction model, which can be used by stock speculators to determine when to buy and sell stocks at the right time to maximize profits. For determining their closeness, they correlated the high, low, and opening prices of Kweichow Moutai stocks. The entropy of indicators was used to weight them. Stock price predictions using the LSTMRNN model were reasonably accurate for Kweichow Moutai. Stock price predictions made by LSTMRNN improve investment success rates.

Joshi et al. [71] examined the pricing prediction aspect of the stock market. It has been a significant area of research for the past decade. The Peephole LSTM was compared with Vanilla LSTMs (a normal LSTM with peephole connections) in this study. Experimental results showed that Peephole LSTM performed significantly better than Vanilla LSTM. In addition to MRF and BAJFINANCE, INFY and BHEL were evaluated based on their daily prices. As a result of the experiments, Peephole LSTM outperformed Vanilla LSTM significantly.

Zhang et al. [72] investigated three influential factors when predicting stock indices. Weibo frequency of increasing and decreasing "positions" is the most accurate factor for predicting Shanghai A-share index value. Additionally, WB was able to predict the rise and fall of stock markets. Stock market variations can be measured or predicted with social media data. Also, it wasn't that the more prediction factors, the better the prediction of Shanghai A-share index value.

WENG et al. [27] collected and analyzed investors' opinions about the stock posted on the Internet on a daily basis. Based on the stock's opening and closing prices, lowest and highest prices, volume, and amount traded, the LSTM predicts the stock's closing price. This paper proposes a BERT-LSTM model that uses investor sentiment as a predictor of stock prices. Additionally, the model is used to predict stock prices for PingAn Bank, ZTE, and MuYuan. It is possible to provide investment advice to stock investors using the method proposed in this paper.

Zhang & Tumibay [73] used LSTMs to predict China's top enterprises' stock prices and solve the long sequence dependence problem. The characteristics of RNNs and stock markets determined whether LSTM layers should be pushed into them. Testing and training of LSTM networks with different hidden neuron numbers. Training and testing data for new energy vehicles are based on BYD's stock price trend. In addition, the prediction accuracy is improved, and the model is evaluated using evaluation indexes and prediction effects. The LSTM model had higher accuracy in predicting stock data when tested against a well-known domestic company. In this paper, a model for predicting stocks is presented, with good results, but it is based only on selected daily stock market data. This model can be tested further using macroeconomic indicators as well as investor sentiment indicators.

According to Cheng et al. [74], attention-based short-term memory can be used to predict stock price movements. In order to predict future stock price movements, the traders used historical stock data and technical indicators. Prediction was based on deep learning. The stock should be

Table 18 Advantages and disadvantages of 2021's articles

Article	Advantages	Disadvantages
[6]	1. Diverse Input Features: Employs a variety of input features to enhance the robustness and accuracy of stock predictions.	1. Overfitting Potential: High complexity and numerous input features increase the risk of overfitting, especially with small datasets.
	2. Focused Performance Indicators: Targets key stock performance metrics, ensuring precise and relevant predictions.	2. Scalability Issues: The model's performance may not scale well when applied to a broader range of stocks or different markets.
	3. Benchmark Comparisons: Provides strong benchmarking against existing models, bolstering confidence in its efficacy.	3. Sensitivity to Input Data Quality: Requires accurate and high-quality input data; any discrepancies can significantly affect predictions.
[61]	1. Long-Term Data Retention: RNN's capability to remember sequential data improves predictive accuracy.	1. Data Quality Dependence: Performance significantly tied to the integrity of historical data.
	2. Robust Performance: Demonstrates significantly lower error rates compared to traditional forecasting methods.	2. Architectural Complexity: Requires detailed understanding of complex parameters for effective model tuning.
	3. Comprehensive Algorithm Exploration: Evaluates multiple variations to identify the optimum approach.	3. Resource Intensive: Demands substantial computational power for training on large datasets.
[62]	1. Long-Term Memory Management: Efficiently captures price trends over lengthy periods, enhancing predictive accuracy.	1. Dependence on Robust Data: Strong reliance on high-quality historical data limits flexibility.
	2. Incorporation of Cyclical Data: Uses cyclical information effectively to refine prediction accuracy.	2. Complex Model Constructs: Multi-layer LSTMs require significant expertise and computational resources.
	3. Focused on Uncommon Variables: Integrates uncommon variables for more nuanced market insights.	3. Sensitivity to Market Changes: May not adapt well to sudden, external market disruptions.
[63]	1. Combined Capabilities: Enhances predictive accuracy by leveraging both LSTM and CNN strengths.	1. Data Quality Sensitivity: Predictive effectiveness heavily relies on the quality of historical data.
	2. Improved Prediction Reliability: Offers more robust and accurate predictions, compensating for individual model weaknesses.	2. Complex Parameter Tuning: Requires careful adjustment of parameters across both architectures, increasing complexity.
	3. Data Utilization Efficiency: Targets reliable stock predictions, minimizing noise and improving accuracy.	3. Specific Stock Focus: Training on a specific stock like Apple limits broader applicability.
[64]	1. Promise of Enhanced Accuracy: Achieves higher predictive accuracy due to advanced LSTM capabilities.	1. Dependence on Historical Data Structures: Relies on high-quality historical prices for reliable predictions.
	2. Handling of Nonlinear Patterns: Successfully navigates nonlinear stock price complexities.	2. Hyperparameter Sensitivity: Model sensitively depends on meticulously chosen hyperparameters.
	3. Long-Term Analysis Enablement: Facilitates better forecasting by integrating extended memory functionalities.	3. Preprocessing Intensity: Requires extensive preprocessing, adding layers of complexity to implementation.
[65]	1. Comprehensive Hybrid Modeling: Merges diverse techniques to boost predictive performance.	1. High Dependence on Data Integrity: Predictive accuracy is closely tied to the quality of input data.
	2. Inclusion of Diverse Financial Variables: Incorporates broader financial information improving forecasting robustness.	2. Technical Integration Complexity: Cross-methodology integration can pose significant challenges.
	3. Adaptability Across Time Horizons: Demonstrates strong capabilities across various prediction timeframes.	3. Market-Specific Effectiveness: General applicability across different markets remains uncertain.
[66]	1. Wide Algorithm Range: Offers a comprehensive overview.	1. Insufficient Practical Focus: Lacks practical implementation insights.
	2. Comparison Framework: Aids in comparing models.	2. Information Overload: Many algorithms can overwhelm readers.
	3. Emerging Trends: Highlights new methodologies.	3. Static Content: Risk of becoming outdated quickly.
[67]	1. Comprehensive Analysis: Extensive literature coverage.	1. Lack of Empirical Findings: Mostly reviews without new evidence.
	2. Model Evaluation: Assesses model strengths and weaknesses.	2. Bias Risks: May reflect author biases.
	3. Future Directions: Suggests areas for further research.	3. Non-Deep Learning Neglect: Limited discussion on alternatives.
[68]	1. Hybrid Architecture: Captures temporal and spatial features.	1. Prolonged Training Time: Longer resource use.
	2. Market Versatility: Robust across various markets.	2. Overfitting: Risk if not managed properly.
	3. Feature Extraction: Improved input quality.	3. Hyperparameter Sensitivity: Requires careful tuning.

Table 19 Key attributes and markets of 2021's articles

Reference	Title	Sector	Target	Methods	Market
[6]	LSTM-based deep learning model for stock prediction and predictive optimization model	Specific Stocks (NIFTY-50)	The article focuses on developing a time-series stock prediction model using LSTM and creating a predictive portfolio based on these predictions	LSTM deep learning model and AMPM	India (NIFTY-50)
[61]	Stock market prediction using RNN LSTM	Stock Market Prediction	To accurately anticipate and analyze stock market prices using advanced machine learning techniques, particularly LSTM networks	Improved LSTM algorithm, stochastic gradient descent for weight maintenance, compared with Simple Average, Linear Regression, and ARIMA algorithms	Tesla Inc. (TSLA) stock data, used within the context of the U.S. stock market
[62]	A perspective of LSTM based stock prediction	Specific Stocks	To apply a multi-layer model of price prediction based on LSTM to determine correlation parameters affecting training efficiency and prediction accuracy	Multi-layer LSTM model, experimental comparisons of different configurations	Potentially Chinese stock market, based on the authors' affiliations
[63]	Stock price prediction using combined LSTM-CNN mode	Specific Stocks	To compare CNN and LSTM models for stock price prediction, and propose a combined LSTM-CNN model that enhances prediction accuracy by utilizing strengths from both approaches	Combined LSTM-CNN model, involving convolutional layers for feature extraction (CNN) and LSTM layers for capturing temporal dependencies in data	U.S. stock market, with a specific focus on Apple Inc. (AAPL) stocks
[64]	Stock price prediction with Long Short-Term Memory model	Specific Stocks	To predict stock prices using an LSTM model, addressing the challenges posed by the non-linear and dynamic characteristics of stock prices	LSTM model, with comparisons made to RNN to evaluate performance enhancements in predictive capabilities	Chinese stock market, specifically using data from the CSI 300 Index
[65]	A Hybrid approach of adaptive Wavelet Transform, Long Short-Term Memory and ARIMA-GARCH family models for the stock index prediction	Market indices	To develop a hybrid model combining adaptive wavelet transform (AWT), LSTM and ARIMA-GARCH family models for improved stock index and volatility prediction	- AWT - LSTM - ARIMA-GARCH family models	U.S. stock market (specifically analyzed indices: Dow Jones Industrial Average (DJIA) and Nasdaq Composite (IXIC))
[66]	Improving stock price prediction using the Long Short-Term Memory model combined with online social networks	Specific Stocks	The study aims to enhance stock price prediction accuracy by integrating data from online social networks with traditional financial data using the LSTM model	LSTM model, GRU model, and RNN model for comparative analysis	China
[67]	LSTM based decision support system for swing trading in stock market	Specific Stocks	To develop a decision support system that enhances swing trading effectiveness by accurately predicting stock prices using an LSTM model integrated with technical indicators	Utilized LSTM model along with technical indicators like MACD, RSI, MFI, Fibonacci retracement levels, and support/resistance levels	India (Indian Stock Market)
[68]	Forecasting stock index price using the CEEMDAN-LSTM model	Market Indices	To improve the accuracy of stock index price forecasting by combining CEEMDAN (Complete Ensemble Empirical Mode Decomposition with Adaptive Noise) with LSTM on the S&P 500 and CSI 300 indices	CEEMDAN-LSTM model, CEEMDAN-enhanced SVM, BP (Back Propagation), Elman network, and WAV (Wavelet Neural Networks) for comparison	United States (S&P 500) and China (CSI 300)

purchased if its prediction class is increasing, that is, if it predicts a rise in the stock price. In general, stocks are sold when a declining class is predicted, which means the stock price will decline. Therefore, a trading strategy was developed based on the prediction results.

AlHakeem et al. [75] developed a hybrid model to increase the accuracy of forecasts for Iraqi stocks. It was developed with a hybridization of CNNs and LSTM. The study community represented six Iraqi financial markets. For banking, telecommunications, tourism, agriculture, industry, and services, weekly data was collected from 2017

Table 20 Dataset characteristics and feature analysis of 2021's articles

Reference	Data	Types of datasets	Features	Period	Frequency	Feature selection/ extraction techniques
[6]	Stock data from the NIFTY-50 index, sourced from the National Stock Exchange of India (NSE)	Time series dataset	Daily closing prices of the 50 constituent stocks of the NIFTY-50 index	January 1, 2018 - April 23, 2021	Daily	Utilized AMPM for creating input-output pairs
[61]	Historical end-of-day stock price data for Tesla (TSLA) from Yahoo Finance	Time series dataset including daily stock prices (OHLCV - Open, High, Low, Close, Volume) and Adj. Close price	Date, Open Price, High Price, Low Price, Close Price, Volume, Adj. Close Price	Specific years not mentioned; refers to historical stock data generally used without precise dates	Daily	Utilization of time series data transformed for supervised learning, specifying no specific scaling mechanism mentioned
[62]	Historical stock data, unspecified dataset names	Time series data including stock prices, turnover rates, etc.	Opening price, closing price, highest and lowest prices, turnover rate, etc.	Not explicitly mentioned, focused on recent and historical data	Daily data, focusing on time spans within 7 days	Hyperparameter tuning—input size, time step, forget bias
[63]	Data sourced from Kaggle, encompassing daily stock features for Apple from December 15, 1980, to July 23, 2021	Time series dataset including daily stock indicators like Date, Volume, High, Low, Closing, and Adjusted Closing prices	Included Date, Volume, High, Low, Closing prices, Adjusted Closing prices, and derivative features such as price variation	From December 15, 1980, to July 23, 2021, with a focus on utilizing recent data to optimize model performance	Daily data frequency was used for model training and assessment	Employed min-max normalization to scale features, combined with exploratory CNN architectures for feature extraction
[64]	Data sourced from the CSI 300 Index, covering metrics like opening price, closing price, high price, low price, trading volume, and turnover from January 2005 to April 2020	Time series dataset, including various trading metrics over the specified period	Features include opening price, closing price, high price, low price, trading volume, and turnover, all normalized for better model convergence	January 2005 to April 2020	Daily data frequency	Data normalization using MinMaxScaler to scale features within a consistent range for analysis
[65]	Data sourced from Bloomberg for stock indexes and dollar index; West Texas Intermediate (WTI) oil price data from the U.S. Energy Information Administration (EIA)	Time series data for stock market indices, dollar index, and WTI oil price	- Stock trading volume - MA5, MA10 (moving averages) - MACD (Moving Average Convergence Divergence) - Daily lags and trading indicators from XGBoost algorithm	From January 4, 2010, to December 31, 2020	Daily	Utilized XGBoost for measuring the importance of stock features to select the best trading indicators
[66]	Data was obtained from Eastmoney, a social media platform, involving user interactions and stocks they follow, generating daily social networks	Two-mode networks between investors and stocks transitioned into one-mode networks based on stocks as nodes	Degree centrality of stocks, traditional variables including opening prices, closing prices, trading volume, price-to-earnings ratio (P/E), turnover rates, and illiquidity ratio	May 1, 2017, to July 28, 2019	Daily data collection (530 trading days)	Degree centrality as a proxy for social interaction, combined with traditional financial indicators
[67]	Data sourced from the National Stock Exchange of India (NSE) and Kaggle, covering specific stocks like ICICI Bank and the NIFTY Banking sector	A time-series dataset featuring daily stock prices and technical indicators	Features include opening price, closing price, highest and lowest prices, traded volume, MFI, RSI, MACD, support and resistance levels, and Fibonacci retracement levels	January 1, 2020, to July 31, 2020	Daily	Calculated technical indicators from stock price data, focusing on price point identification through levels like Fibonacci retracements and support/resistance

Table 20 (continued)

Reference	Data	Types of datasets	Features	Period	Frequency	Feature selection/ extraction techniques
[68]	Data from the Wind database, including daily closing prices for CSI 300 (November 10, 2008, to February 20, 2019) and S&P 500 (March 17, 2009, to February 20, 2019), each comprising 2500 observations	Time-series datasets of daily closing prices of the indices	Daily closing prices decomposed into Intrinsic Mode Functions (IMFs) using CEEMDAN	CSI 300: November 10, 2008, to February 20, 2019; S&P 500: March 17, 2009, to February 20, 2019	Daily	Decomposition of original time series into several IMFs through CEEMDAN provides different temporal features for the LSTM model

to 2021. Experiments conducted on the proposed model have shown that it has a minimum error rate of 0.000404.

Singh et al. [76] proposed a machine learning algorithm based on SGD that uses RNNs, commonly called LSTM, with weights and biases adjusted based on singleton data points. In essence, they were trying to improve the accuracy of stock predictions, which isn't the simplest task since it requires several layers of data processing. An enhanced neural network was used to develop a plan and a model for the development of a plan. Moreover, they compared the enhanced version to other models in terms of accuracy, profitability, and user experience, and found that the enhanced version provided better results and higher profits than other models.

Using a forecasting model, Tengku et al. [77] identified the next trends in the stock market. The FBM KLCI closing prices were used to build LSTM models. Model performance was measured by root mean squared error (RMSE) and mean absolute error (MAE). Using Yahoo Finance data, researchers predicted the stock market. Google Colab was used to analyze data. The LSTM model was improved by 1.87% by using RNNs. FBM KLCI uses this method to predict future closing stock prices. Profit predictions were improved based on the results. Economic growth can be sustained by stock market forecasts, so they can support long-term economic growth.

Wijaya et al. [78] recommend using the price, golden cross, and death cross on technical analysis indicators to improve stock price predictions. This method uses LSTM to predict sequence data, which is a reliable method. Prices for 10 shares of the LQ45 group were forecast using 1200 days of Indonesian Stock Exchange data. The predicted close price next day was calculated using Root Mean Squared Error and Mean Absolute Percentage Error values. Prediction accuracy is improved with golden crosses and death crosses in technical analysis. Based on the proposed method, the best prediction results were obtained with the lowest RMSE. When compared with price-only or price-and-volume predictions, MAPE was 2.30%. Because of its 0.33% standard deviation, this method is more consistent than others with 1.00 and 0.38% standard deviations.

Thakkar and Chaudhari [79] proposed a fusion-based GA approach for predicting stock prices and trends with inter-intra crossovers. Using genetic diversity, their proposed approach aimed to optimize parameters and select a set of features for predicting long-term short-term memories; by integrating inter-chromosome crossovers and conditional intrachromosome crossovers, as well as adaptive mutation, a variety of chromosome solutions were explored. This study demonstrated and evaluated how GA with ICAN is used to predict stock prices and trends one day in advance. The prediction model can be replaced with a surrogate model. They can potentially be extended by using models such as multivariate adaptive regression splines (MARS) and M5Tree.

MD et al. [80] proposed a new method for predicting stock prices. Normalized time series data is also used by MLS LSTM to predict future values based on the relationship between past and future values. Therefore, simple RNNs no longer suffer from the vanishing gradient problem. Forecasts were based on past performance and patterns. In training, the MLS LSTM algorithm predicted 95.9% accurately, and in testing, it predicted 98.1% accurately, outperforming other deep learning algorithms. There was a mean absolute percentage error of 1.79% in the training set and 2.18% in the testing set. Additionally, the proposed model provides an accurate forecast with 0.019 normalized root mean squared error.

For stock market price trends in four major Asian countries, Wang et al. [14] developed a hybrid model based on secondary decomposition, multi-factor analysis, and attention-based Long Short-Term Memory (ALSTM). In addition to reducing noise, two decomposition algorithms captured non-linear features. The data was also analyzed using multifactor analysis. Prediction stage increased weights of effective information. The proposed model was validated using four Asian stock market datasets. Experimentally, the proposed model outperforms the general Long Short-Term Memory. Based on the average percentage errors presented in this paper, the hybrid model showed the lowest mean average (0.612%), 0.903%, 0.606%, and 0.402%.

Chen et al. [81] proposed a hybrid deep learning approach for improving prediction performance. With K-means

Table 21 Comparative analysis and findings overview of 2021's articles

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[6]	Compared LSTM with Multi-Layer Perceptron (MLP) and statistical models: Exponential Smoothing, Holt-Winters, and ARIMA	RMSE values reported for LSTM-DNN with lower errors compared to MLP and traditional models for selected stocks	Stock price prediction and portfolio optimization using predicted data	LSTM-DNN model outperformed conventional models, enabling effective stock prediction and portfolio diversification	Non-stationary data increased computational time. Future work to explore hybrid models and genetic algorithms for enhanced portfolio optimization
[61]	Techniques including Moving Average, Linear Regression, ARIMA, and LSTM were evaluated for their prediction performance based on RMSE values	RMSE was the primary performance metric; RMSE values demonstrated: - Moving Average: 6077.708 - Linear Regression: 51.996 - ARIMA: 96.5 - LSTM: 18.899, identifying LSTM as most effective	Stock price forecasting	The LSTM model outperformed traditional algorithms in accuracy for predicting stock prices, with significant improvements in RMSE scores across models compared	Limitations include the need for larger datasets for improved model accuracy, potential overfitting, and exploring enhanced models. Future work aims to implement more robust models and increased dataset size for better prediction capabilities
[62]	Analyzed different LSTM configurations including input size, time steps, and layer complexity	Training loss, deviation percentage between predicted and actual values	Stock price prediction focusing on improving accuracy and training efficiency	LSTM models are effective for stock price prediction; simpler models can sometimes perform better than complex ones	The model may not perform well with overly complex structures; further dataset testing and better evaluation criteria needed for practical applications
[63]	CNN models (RAW-CNN and DIFF-CNN) compared with standalone LSTM models, as well as with the combined LSTM-CNN model	MSE as the performance evaluation metric, highlighting improvements in prediction accuracy, notably an increase of approximately 19.3% with the hybrid model	The primary focus is on price prediction for Apple's stock, emphasizing the enhancement of accuracy using a combined model approach	LSTM-CNN model exhibited significant improvements over standalone CNN and LSTM, especially under conditions with limited training data, demonstrating robust prediction capabilities	Identified backpropagation complexities within LSTM architectures; future work suggests refining model architecture and expanding data diversity to augment predictive accuracy further
[64]	Compared LSTM model performance against RNN using performance metrics such as Mean Absolute Error (MAE), Mean Square Error (MSE), and Mean Absolute Percentage Error (MAPE)	Performance evaluated using metrics like RRMSE, MAE, and MAPE, with LSTM showing better accuracy compared to RNN	Focus on predicting stock prices, assessing the impacts of historical data and feature configurations on prediction performance	LSTM significantly outperformed RNN in predicting stock prices, indicating improved accuracy for dynamic and non-linear stock price data	Challenges in capturing complex stock dynamics were noted; future work suggests hybrid models or alternative approaches to enhance accuracy
[65]	Compared the proposed models against benchmark models including traditional ARIMA-GARCH models, ANN, WT-ANN, and various combinations of GARCH models	Evaluated using MSE, Diebold-Mariano (DM) test, and Asymmetric Diebold-Mariano (ADM) test to compare predictive performance across models	Price prediction and realized volatility prediction of stock indices	The hybrid AWT-LSTM-ARIMA-GARCH model showed improved forecasting accuracy over benchmark models, successfully predicting stock index volatility across various time horizons	The study acknowledges the potential for improving model performance by incorporating more diverse economic indicators or exploring other deep learning architectures in future work

Table 21 (continued)

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[66]	Models compared include LSTM, GRU, and RNN to evaluate prediction effectiveness	RMSE and MAPE for prediction accuracy. Results indicated improvements when using the network degree centrality variable	Stock price prediction	Incorporating social network variables significantly enhances the accuracy of stock price predictions. The LSTM model outperforms both RNN and GRU models, but GRU showed comparable results	The model's reliance on limited social interaction data could lead to bias in predictions. Future work could explore broader data sources and additional proxies for calculating social network influences on stock prices
[67]	Compared against baseline models like Linear Regression, Moving Average, XGBR, SVR, ARIMA, and Seasonal Naive, among others	RMSE: 4.13, MAE: 3.24, MAPE: 1.21% for stock price predictions, demonstrating superior performance compared to alternative models	Price prediction for swing trading applications	The LSTM-based system provides accurate stock price forecasts, aiding swing traders by utilizing a comprehensive analysis of the stock's technical performance	Limitations include high computational demands for processing vast historical data. Future work suggests using high-performance computing and improving prediction models for broader market indices
[68]	Compared to CEEMDAN-enhanced baselines like SVM, BP, Elman network, and WAV to evaluate effectiveness	Evaluated using MAPE, MAE, MSE, RMSE, indicating CEEMDAN-LSTM's superior predictive accuracy across indices	Price prediction of stock indices focusing on U.S. and Chinese markets	CEEMDAN-LSTM model achieves greater forecasting precision by combining sophisticated data decomposition and neural network strategies, applicable to both developed and emerging market indices	Model limitations include handling high-frequency noise. Future work suggests improving robustness and examining diverse data inputs or hybrid models for enhanced forecasting accuracy

clustering, DTW is modified to cluster banks with similar price trends. After that, a long- and short-term memory (LSTM) neural network model was trained to predict static and dynamic stock prices. The LSTM network output was transformed into a multi-step output to predict multiple time intervals at once, improving long-term forecast performance. As a result of R-SQUARE, MAE, MSE, the hybrid model outperforms the single model. Additionally, static predictions perform better than dynamic rolling predictions. It helps investors and companies make more money by predicting stock prices.

Teng et al. [82] modeled stock price fluctuations by using four distinct local descriptors. Piecewise aggregates approximate average values; short-term series derivatives reflect price change ratios; slope indicates short-term price trends; hog-1D aggregates different gradients into histograms. Stock price series provided a variety of clues. They captured the underlying price trend patterns using multi-scale local cues and hierarchical attention-based LSTMs (MLCA-LSTM). There were two advantages: (1) The sliding windows can induce a variety of local descriptors, (2) Interactions at multiple scales and temporal dependencies are jointly attended to by hierarchical attention mechanisms and multi-branch LSTM structures. It was confirmed that the model worked on real-time stock prices.

Swathi et al. [83] proposed a Teaching and Learning Based Optimization (TLBO) model with LSTM sentiment analysis to predict stock prices using Twitter data. It is essential to pre-process the data in order to get rid of unnecessary data and transform it into a meaningful format since tweets are generally short and have unusual grammar structures. Additionally, the LSTM model was applied to categorize tweets regarding the stock price into three categories - positive, negative, and neutral. They investigated how tweets and stock prices correlated as a result. LSTM model learning rate was determined using the Adams optimizer for optimal prediction results. To ensure optimal performance, the output unit of the LSTM model was also tuned using the TLBO model. Experiments using Twitter data have shown that the TLBO-LSTM model makes better stock price predictions. Experimental results showed that TLBO-LSTM model had promising results when compared to state-of-the-art methods in diverse aspects. The TLBO-LSTM model had a maximum precision of 95.33%, a recall of 85.28%, and an F-score of 90%. TLBO-LSTM model achieves a greater accuracy of 94.73% than other techniques.

Noubakhsh & Habibi [84] improved the performance and reduced model error in stock price trend prediction by combining a CNN with a LSTM method, along with fundamental analysis elements such as the ratio P/E, profitability, and company transactions. For four financial, petroleum, basic metal, and nonmetallic minerals groups, the model was evaluated using parameters for evaluating mean absolute error and mean absolute percentage error, resulting in a reduction in error and an increase in performance. According to the study, the mean absolute percentage error was 0.49 and the mean absolute error was 4.30. This affects performance and error prediction.

Gao et al. [85] used multivariable LSTM network to predict stock market values. There are two serial modules in this network: the first is an MV-LSTM recurrent layer, which encodes information from each variable exclusively. By introducing a variable attention mechanism, the second module introduces a latent variable that measures the importance of each variable. Multivariable financial sequences can be handled by these two modules. Investment strategy based on prediction model. MV-LSTM networks have been shown to offer superior statistical arbitrage investment strategies using Chinese stock data.

For predicting stock price fluctuations, Tian et al. [25] developed LSTMs and Bayesian-optimized LightGBMs. This prediction model was compared with LSTM-BOXGBoost hybrid model, LSTM-LightGBM hybrid model, LSTM-XGBoost hybrid model, RNN network model, and single LSTM network model. The RMSE of LSTM-BO-LightGBM's prediction for the "ES=F" stock was 596.04, the MAE was 15.24, the accuracy was 0.639, and the f1_score was 0.799. In comparison with previous models, LSTM-BO-LightGBM provides better approximation and generalization of stock fluctuation forecasts.

Thu et al. [86] found two approaches to analyze historical stock data that incorporated LSTM. LSTM model inputs were enriched with the historical stock prices of their closest neighbors; LSTM model prediction was enhanced with a dynamic sliding window based on significant price changes. Their models consistently outperformed their competitors across three stock datasets.

Table 22 shows the comparison between advantages and disadvantages of 2022's articles. Table 23 shows the key attributes and markets of 2022's articles. Table 24 shows the dataset characteristics and feature analysis of 2022's articles. Table 25 shows the comparative analysis and findings overview of 2022's articles.

5.6 Reviewed Articles of 2023

Rahmadeyan & Mustakim [10] compared LSTM and GRU algorithms to predict Bank Rakyat Indonesia stock price movements. This study uses new datasets, algorithms, and experiments. This research contributed significantly to this field through its innovative approach. This model can be used to recommend investments to investors. Invested assets can provide additional capital for operational business activities. This study is expected to benefit the company and the community as it provides strategies to increase company value as well as analyze and support stock trading investment decisions. Time series data has been predicted using LSTM and GRU algorithms in previous research. For instance, LSTM outperformed SVR to predict stocks such as the Dow Jones Industrial Average, S&P 500, NASDAQ, NYSE, NSE, BSE, and NYSE. The GRU prediction was more accurate during peak periods, whereas the LSTM prediction was more accurate during stable periods. The GRU network outperformed LSTM by 33% in another study of nickel price forecasting using LSTM and GRU.

Simone et al. [87] proposed a deep learning-based methodology to answer the research questions. As a result, maintenance interventions can be planned ahead of time, optimizing overall costs and duration. In this model, LSTM networks are used to learn long-term dependencies. Data is collected and analyzed by numerous subsystems on railway vehicles. An experiment investigating the cooling subsystem of traction converters. Data was analyzed over several months. It is possible to identify coarse-grained relationships between dataset features and failures using classic statistical techniques, such as trend estimation and correlation. The LSTM network helps maintenance teams understand rolling stock failures. The ML-based predictive maintenance literature outperforms other techniques. They also achieved a false alarm rate of 0.4% and a mean absolute error of 104, both of which are promising findings.

Gülmez [16] demonstrated an ARO-optimized deep LSTM network. It is called LSTM-ARO. The model predicted stock market prices. Stock prices are predicted using the DJIA index. Over five years, 30 different stock prices were collected. Five years of data collected. 20 previous days of data were used to predict the next day's price. As a result of the conversion, the data was split 80% into train data and 20% into test data. New parameters were created for LSTM networks. ARO algorithm variables were connected to the parameters. The LSTM architecture changed when the variables changed. The best architecture was found by ARO. The new proposed model was compared with LSTM1D, LSTM2D, LSTM3D, ANN, and LSTM-GA networks to evaluate its quality. Based on the results, LSTM-ARO was the best model. Traders and investors can benefit from the predictions.

Yuan et al. [88] suggested using PSO algorithms to optimize LSTM stock prediction models, solving the LSTM hyperparametric optimization problem, optimizing LSTM network structures, reducing human interference in LSTM prediction, and improving their ability to capture stock data characteristics. The results showed that PSO-LSTM had the lowest evaluation value, was accurate, and was better fitted. Modeled stock price curves fit better with real stock price curves in the prediction model. PSO solves the LSTM hyperparametric optimization problem, optimizes the network structure of the LSTM, reduces human interference on LSTM prediction, and improves the model's ability to capture stock data characteristics. It can predict different types of stock data more accurately than ANNs and LSTMs, and is generalizable.

Table 22 Advantages and disadvantages of 2022's articles

Article	Advantages	Disadvantages
[19]	1. Multivariate Risk Analysis: Considers multiple risk factors simultaneously, leading to more comprehensive risk assessments.	1. High Sensitivity to Data Quality: Performance can degrade significantly with poor quality or inaccurate input data.
	2. Enhanced Volatility Forecasting: Targets volatility specifically, which is crucial for effective risk management in investments.	2. Complex Configuration: Requires careful tuning of model parameters, complicating its setup for users without deep expertise.
	3. Real-World Application: Validated in dynamic environments, making the model more credible for practical use.	3. Limited Scope on Volatility: While it effectively forecasts volatility, it may miss other influential market variables.
[69]	1. Advanced Neural Network Architecture: Enhances predictive performance.	1. Challenges in Interpretability: Complexity may hinder stakeholder trust.
	2. Comprehensive Feature Selection: Incorporates multiple predictors for accuracy.	2. Overfitting Risk: Multiple layers can lead to poor generalization.
	3. Comparative Model Analysis: Provides insights through performance evaluation.	3. Dependence on Predictor Quality: Effectiveness hinges on quality inputs.
[70]	1. Comprehensive Data Handling: Effectively processes missing data, increasing overall model accuracy.	1. Dependence on Historical Data: Performance reliant on the availability and quality of past data.
	2. Targeted Dataset: Application to known stocks enhances reliability and granularity of the analysis.	2. Limited Generalizability: The model may not perform equally well across different stocks or market conditions.
	3. High Predictive Accuracy: Demonstrated accuracy improves investor decision-making capabilities.	3. Implementation Complexity: Requires substantial expertise in machine learning for effective deployment.
[71]	1. Enhanced Performance with Peephole LSTM: Demonstrates superior prediction accuracy compared to Vanilla LSTM.	1. Increased Implementation Complexity: Peephole LSTM is more challenging to implement compared to Vanilla LSTM.
	2. Effective Long-Term Dependency Management: Improved ability to handle historical data patterns and predictions.	2. Higher Computational Demands: Requires more computational resources and time for training.
	3. Robustness Against Market Noise: Better suited to manage volatile and noisy market data.	3. Dependence on Data Quality: Success reliant on the quality and quantity of available historical data.
[72]	1. Integration of Multiple Influences: Combines various factors to form a comprehensive predictive model.	1. Limited Data Time Frame: The short analysis period affects the robustness of the findings.
	2. Empirical Validation of Sentiment Analysis: Incorporates social sentiment analysis to enhance prediction reliability.	2. Historical Data Dependence: Performance heavily influenced by historical data quality.
	3. Clear Model Comparisons: Highlights the strengths and weaknesses of different predictive models.	3. Potential Overlook of Critical Variables: May miss out on other significant macroeconomic influences.
[27]	1. Sentiment and Price Data Integration: Enhances accuracy by combining different data types.	1. Reliance on Data Quality: Both sentiment and stock data must be reliable for accuracy.
	2. Efficient Sequential Data Handling: LSTM captures stock price patterns effectively.	2. Complex Model Requirements: Implementation needs substantial expertise in both areas.
	3. Practical Market Validation: Tested with real data from Chinese markets for relevance.	3. Risk of Overfitting: Complexity can cause over-adaptation to historical data.
[73]	1. Enhanced Prediction Accuracy: Demonstrates superior accuracy of the LSTM model compared to traditional RNN methods in forecasting stock prices.	1. Data Bias from Specific Company: Results may not generalize to other firms or markets, limiting the broader applicability of findings.
	2. Handling Long-Term Dependencies: Effectively manages long-term dependencies in stock data, a critical feature for financial forecasting.	2. Complexity in Model Implementation: The intricate nature of LSTM models can present challenges, especially for users with limited technical background.
	3. Comprehensive Evaluation Metrics: Employs a variety of metrics (MSE, RMSE, MAE) for a thorough performance assessment, providing a clear picture of model effectiveness.	3. Risk of Overfitting: Longer sequences in training data can lead to overfitting, resulting in diminished performance when applied to new data.
[74]	1. Comprehensive Sentiment Integration: The model effectively combines emotional analyses from various sources, improving prediction accuracy.	1. Data Quality and Reliability Concerns: Performance heavily relies on the reliability of sentiment data, which can be inconsistent or noisy.
	2. Utilization of Advanced Machine Learning Techniques: Employs a hybrid of SVM, LSTM, and GRU, enhancing overall prediction capabilities.	2. Complexity of Model Architecture: The hybrid approach complicates both training and interpretation, requiring advanced expertise.
	3. Focus on Real-Time Data: Utilizes current sentiment data, increasing the model's relevance and responsiveness to market changes.	3. Risk of Overfitting: The complex structure poses a significant risk of overfitting to training data, which may hinder real-world application.

Table 22 (continued)

Article	Advantages	Disadvantages
[75]	1. Hybrid Model Effectiveness: Achieves minimal error rates by leveraging both CNN and LSTM, significantly enhancing prediction accuracy.	1. Complexity of Implementation: Significant resources and expertise are required for effective implementation.
	2. Comprehensive Data Utilization: Utilizes extensive datasets across various sectors, contributing to the robustness of the findings.	2. Dependence on Historical Data: Heavy reliance on past data can be problematic during market shifts, reducing predictive effectiveness.
	3. Advanced Neural Network Techniques: Combines the strengths of CNN for feature extraction and LSTM for sequential analysis.	3. Potential Overfitting: The complexity of the model raises the risk of overfitting if not managed properly.
[76]	1. Integration of SGD with LSTM: Enhances short-term stock price forecasting accuracy through weight fine-tuning.	1. Data Quality Dependency: Performance is contingent on the availability of high-quality input data.
	2. User-Friendly Implementation: Development of a predictive web application for practical use.	2. Complex Data Processing: Intensive preprocessing requirements might hinder usability and accessibility.
	3. Dynamic Data Use: Combines real-time and historical data for robust predictions.	3. Limited Applicability: Findings are primarily geared towards specific stock markets and may not generalize well.
[77]	1. Effective Nonlinear Pattern Recognition: Captures both linear and nonlinear relationships, refining modeling techniques.	1. High Data Quality Requirement: Predictive accuracy is dependent on accessing high-quality historical stock data.
	2. Solid Predictive Accuracy: Demonstrates high accuracy in stock price forecasts, utilizing extensive historical data.	2. Market-Specific Limitations: The model's applicability may not extend beyond the Malaysian market.
	3. Region-Specific Relevance: Tailored to the Malaysian market, enhancing local applicability.	3. Resource-Intensive Training: LSTM models require substantial computational resources for training.
[78]	1. Objective Trading Signals: Automates stock forecasting by applying systematic technical indicators, reducing human bias.	1. Narrow Technical Indicator Use: Focuses primarily on a specific set of indicators, potentially missing other relevant signals.
	2. Enhanced Forecasting Accuracy: Leverages crossing points for meaningful data transformations, improving accuracy.	2. Market Focus Limitation: The model's data reliance limits its generalizability to other markets.
	3. Rigorous Evaluation: Employs robust testing metrics for reliable results, supporting trading decisions.	3. Risk of Overfitting: Highly complex models may overfit the training data, reducing predictive robustness.
[79]	1. Versatile Data Integration: Combines information from diverse sources to enhance analysis robustness.	1. Quality Dependency: The accuracy of predictions depends heavily on the data quality from varied sources.
	2. Genetic Algorithm for Optimization: Improves parameter tuning through evolutionary techniques.	2. Sensitivity to Initial Parameters: Initial setup can significantly affect prediction outcomes, requiring careful configuration.
	3. Adaptive Market Strategies: Adjusts prediction strategies dynamically to follow market trends.	3. Elevated Computational Demands: Higher computational needs might hinder broad applicability.
[80]	1. High Predictive Accuracy: Excels in predictive precision compared to other algorithms.	1. Intensive Data Requirements: Strong reliance on extensive, high-quality data.
	2. Vanishing Gradient Management: Successfully addresses deeper network learning challenges.	2. Training Complexity: Requires substantial computational resources and time for training multi-layer models.
	3. Normalized Data Utility: Utilizes normalized series to enhance correlation insights and forecast reliability.	3. Limited Generalization: Results excel with specific datasets but may not generalize well across different markets.
[14]	1. Multi-Method Synergy: Combines various methodologies to significantly enhance forecasting precision.	1. Dependence on Data Quality: Heavy reliance on data quality for performance optimization.
	2. Effective Noise Filtering: Utilizes decomposition techniques to improve data clarity and prediction quality.	2. Implementation Complexity: The complexity of integrating multiple methodologies requires significant expertise.
	3. Substantial Accuracy Improvement: Outperforms traditional methods with a notable increase in accuracy.	3. Generalization Challenges: Specificity to certain markets might limit broader applicability.
[81]	1. Innovative Hybrid Model: Seamlessly integrates K-means clustering with LSTM for effective pattern identification.	1. Influence of Clustering Quality: Prediction reliability is intimately tied to clustering accuracy.
	2. Multi-Output Prediction: Capable of generating predictions for several future points, enhancing strategic planning.	2. Implementation Complexity with DTW: The use of Dynamic Time Warping adds another layer of complexity.
	3. Enhanced Generalization: Shows better performance in generalization compared to single-model approaches.	3. Volatility Sensitivity: Performance can vary widely in highly volatile market environments.
[82]	1. Multi-Scale Extraction: Captures features across time scales.	1. Complex Interpretation: Hinders understanding feature impacts.
	2. Hierarchical Attention: Prioritizes relevant signals.	2. Computational Demands: Limits accessibility.
	3. Innovative Applications: Versatile in scenarios.	3. Data Requirement Issues: Needs high-quality data.

Table 22 (continued)

Article	Advantages	Disadvantages
[83]	1. Sentiment Integration: Combines Twitter data with metrics.	1. Sentiment Noise: Twitter's volatility introduces inaccuracies.
	2. Social Trends Adaptability: Adjusts to public sentiment.	2. Data Quality Reliance: Information variability affects outcomes.
	3. Enhanced Insights: Understands market dynamics.	3. Trend Dependency: Misleading for long-term predictions.
[84]	1. Integration of Diverse Data Types: The model effectively combines technical indicators from LSTM and CNN with fundamental analysis, enhancing prediction accuracy by leveraging a broader dataset.	1. Complex Architecture: The combination of LSTM, CNN, and fundamental data can lead to a complicated model structure, complicating the training and tuning process.
	2. Improved Predictive Performance: Demonstrates a significant reduction in prediction errors compared to traditional methods, showcasing the effectiveness of hybrid approaches.	2. Risk of Overfitting: The intricate model may fit noise in the training data, especially with smaller datasets, reducing generalizability.
	3. Robustness in Different Market Conditions: The model's ability to adapt to various market scenarios improves its reliability in volatile environments.	3. Dependence on Quality of Fundamental Data: The success of the model heavily relies on the availability and quality of the fundamental analysis data, which can vary significantly.
[85]	1. Effective Handling of Complex Data: The model's ability to process multiple variables allows it to capture the intricate dynamics of stock market behavior.	1. Overly Complex Model Configuration: The incorporation of numerous variables can lead to an overly complex model, requiring significant expertise for effective tuning.
	2. Long-Term Dependencies: LSTM networks excel in retaining long-term information, making them particularly effective for time-series predictions in finance.	2. Heavy Dependence on Historical Data: The reliance on historical data may limit the model's responsiveness to sudden market changes or anomalies.
	3. Validation Across Diverse Datasets: The robustness of the model is confirmed through validation on multiple datasets, enhancing its credibility.	3. Risk of Overfitting: With many variables and parameters, there is a heightened risk of overfitting, especially with limited training data.
[25]	1. Efficiency of Hybrid Models: The combination of LSTM and LightGBM optimizes both accuracy and computational efficiency, catering to large datasets effectively.	1. Complex Training Process: The integration of two different model types complicates the training process, requiring careful calibration.
	2. Robust Performance Across Datasets: The hybrid model demonstrates adaptability, maintaining performance across various datasets and conditions.	2. Interpretation Challenges: The complexity of the model can obscure the contributions of specific features, making it difficult to derive insights from the predictions.
	3. Scalability: LightGBM's design allows for handling larger datasets without significant loss of performance.	3. Assumptions of Historical Patterns: The model's reliance on historical data may hinder its performance in rapidly changing markets.
[86]	1. Dynamic Sliding Window Technique: This feature allows the model to adapt based on significant price fluctuations, enhancing prediction accuracy in volatile markets.	1. Threshold Dependence: The model's performance can be affected by user-defined thresholds for significant price changes, introducing variability in results.
	2. Incorporation of Neighboring Stock Data: By analyzing correlations with similar stocks, the model may gain additional insights, which could be particularly beneficial in identifying trends.	2. Correlational Limitations: The reliance on historical correlations between stocks may diminish effectiveness during changing market conditions or unexpected events.
	3. Multi-Criteria Prediction Capability: Capable of predicting various price types (open, close, high, low), providing a comprehensive overview of market conditions.	3. Complex Distance Calculations: The need for complex computations to assess correlations can slow processing times, impacting real-time trading capabilities.

Sarika et al. [89] used a LSTM algorithm to forecast stock prices. RNNs have been used to predict stock values using LSTM. The popularity of LSTM networks in this field can be attributed to their ability to deal with dependencies. In order to predict stock prices, LSTM models are trained on historical data and then used to forecast future prices. LSTM network learned long-term dependencies in a given sequence based on an input series of previous stock prices. Next, the network's output was used to predict stock prices. These elements could be correlated non-linearly using LSTMs to produce more precise predictions. Its ability to handle noisy and non-stationary data was another advantage of using LSTM to predict stock prices. It is difficult to estimate future prices based on conventional statistical techniques because of unpredictable fluctuations and abrupt changes in market sentiment. The effectiveness of LSTM comes at the price of meticulous feature engineering and rigorous preprocessing.

He & Xu [90] developed CL-Net (Confrontation-LSTM Network), which exploits the "gate" structure in LSTM to solve long-term dependency problems in RNNs. In order to overcome the problems caused by static features, they developed an adversarial training model. Trend forecasting schemes are ill-suited to handling dynamic and volatile markets in real-time.

Table 23 Key attributes and markets of 2022's articles

Reference	Title	Sector	Target	Methods	Market
[19]	Forecasting the volatility of specific risk for stocks with LSTM	Stock Market Volatility, Financial Risk Management	To model and predict the volatility of specific stock risks using LSTM and compare it with the GARCH model	Uses LSTM for predicting stock volatility, with the GARCH model as a baseline for comparison	Global markets; specific countries not-detailed, data is sourced from BARRA
[69]	Predicting stock market index using LSTM	Market Indices	To develop a predictive model utilizing LSTM networks for forecasting the next-day closing price of the S&P 500 index by incorporating various market data features	LSTM neural network architecture with evaluations based on single layer and multilayer configurations; performance assessed using RMSE, MAPE, and Correlation Coefficient	United States (S&P 500 Index)
[70]	Data analysis on the computer intelligent stock prediction model based on LSTM RNN and algorithm optimization	Specific Stocks (focus on Kweichow Moutai)	To investigate and establish a stock price prediction model based on LSTM RNN for Kweichow Moutai with improved accuracy	LSTM RNN for modeling stock prices, utilizes correlation analysis (Pearson, Spearman, and Kendall methods) and an entropy-weight method for indicator selection	China (Kweichow Moutai stock)
[71]	Comparative analysis of Vanilla LSTM and Peephole LSTM for stock market price prediction	Stock Market Prediction	The target of the article is to compare the performance of Vanilla LSTM and Peephole LSTM in predicting stock market prices and identify which model provides more accurate predictions	Vanilla LSTM and Peephole LSTM models	Indian Stock Market, focusing on stocks such as MRF, BAJFINANCE, INFY, BHEL, and SAIL
[72]	Research on influential factors in stock market prediction with LSTM	Stock Market Prediction	To analyze the influential factors impacting stock market prediction using LSTM models, determining the effectiveness of single versus multiple factors in predicting stock prices	LSTM networks, exploring single and multiple influential factors	China's Shanghai A-share index and the United States' NASDAQ index
[27]	Stock price prediction based on LSTM and Bert	Specific Stocks	To propose a stock price prediction model that incorporates investor sentiment with LSTM to predict closing prices	LSTM combined with BERT to analyze investor sentiment and predict stock prices	China (focus on A-share market including PingAn Bank, ZTE, and MuYuan)
[73]	Stock market prediction model based on LSTM Deep Learning: The case of top corporate company in China	Specific Stocks	To predict the stock market value of BYD, a prominent Chinese new energy vehicle company, using LSTM neural network methods to enhance prediction accuracy	LSTM neural network	China (focusing on the stock market of BYD)
[74]	Emotion analysis of news and social media text for stock price prediction using SVM-LSTM-GRU composite model	Specific Stocks	To develop a composite model (SVM-LSTM-GRU) for predicting stock prices through sentiment analysis extracted from news and social media texts	Uses a combination of SVM for sentiment analysis and LSTM-GRU networks for stock price prediction	India
[75]	Iraqi stock market prediction using Hybrid LSTM and CNN algorithm	Specific Stocks (Iraqi stock market)	To develop a hybrid model combining CNN and LSTM to enhance the accuracy of predicting Iraqi stock market indices	Hybrid model using CNN for feature extraction and LSTM for time-series forecasting	Iraq
[76]	Stock prediction on historical data based on SGD and LSTM	Market Indices	To propose and evaluate an LSTM model utilizing Stochastic Gradient Descent (SGD) for accurate stock price prediction	Utilizes a LSTM neural network in conjunction with SGD for model training and optimization	The document does not specify a particular country's market but generally applies to global stock markets

Table 23 (continued)

Reference	Title	Sector	Target	Methods	Market
[77]	Prediction of FTSE Bursa Malaysia KLCI stock market using LSTM Recurrent Neural	Specific Stocks	To forecast and model the FTSE Bursa Malaysia KLCI (FBM KLCI) stock market prices using LSTM models to predict future stock prices	Used LSTM RNN for stock price prediction	Malaysia (FTSE Bursa Malaysia KLCI)
[78]	Stock price prediction with Golden Cross and Death Cross on technical analysis indicators using Long Short-Term Memory	Specific Stocks	To improve stock price predictions by integrating golden cross and death cross indicators with technical analysis and LSTM models	Used LSTM networks combined with technical analysis indicators like golden cross and death cross signals for prediction	Indonesia (Indonesia Stock Exchange)
[79]	Information fusion-based genetic algorithm with Long Short-Term Memory for stock price and trend prediction	Specific Stocks	To enhance stock price and trend prediction accuracy by integrating a genetic algorithm with LSTM for parameter optimization and feature selection	Information fusion-based genetic algorithm (ICAN) combined with LSTM, utilizing inter-intra chromosome crossover and adaptive mutation for optimization	South Korean and Chinese stock markets, focusing on companies like Samsung, Hyundai, and the China Construction Bank, along with indices like KOSPI and CSI 300
[80]	Novel optimization approach for stock price forecasting using multi-layered sequential LSTM	Specific Stocks	To propose a novel Multi-Layer Sequential Long Short-Term Memory (MLSLSTM) model optimized with the Adam optimizer for accurate stock price forecasting	MLSLSTM model using the Adam optimizer for training and data normalization techniques	South Korean stock market, specifically analyzing Samsung's stock
[14]	Asian stock markets closing index forecast based on secondary decomposition, multi-factor analysis and attention-based LSTM model	Market indices	To improve the prediction accuracy of stock price in Asian markets using a hybrid model combining secondary decomposition, multi-factor analysis, and attention-based LSTM	Secondary Decomposition (SD), Multi-Factor Analysis (MFA), Attention-based LSTM (ALSTM), Variational Mode Decomposition (VMD), Complete Ensemble EMD with Adaptive Noise (ICEEMDAN)	Asian stock markets (specifically studied four major countries: China, Japan, South Korea, Thailand)
[81]	China's commercial bank stock price prediction using a novel K-means-LSTM Hybrid approach	Specific Stocks	To present a novel hybrid approach combining K-means clustering and LSTM for improved prediction of commercial bank stock prices in China	K-means Clustering, Dynamic Time Warping (DTW), LSTM	China's stock market, focusing on the commercial banking sector
[82]	Multi-scale local cues and hierarchical attention-based LSTM for stock price trend prediction	Market Indices	To improve the accuracy of stock price trend prediction using a novel model that leverages local descriptors and multi-scale information	1. Multi-scale local cues (using four local descriptors: PAA, derivative, slope, and HOG-1D) 2. Hierarchical Attention-based LSTM (MLCA-LSTM)	China (focus on stock price trends)
[83]	An optimal deep learning-based LSTM for stock price prediction using twitter sentiment analysis	Specific Stocks	To develop a model combining LSTM with Twitter sentiment analysis to improve stock price prediction accuracy	1. LSTM 2. Teaching-Learning Based Optimization (TLBO) 3. Sentiment analysis on Twitter data	General focus (not specific to any particular country's market)
[84]	Combining LSTM and CNN methods and fundamental analysis for stock price trend prediction	Specific Stocks	To improve stock price trend prediction accuracy by integrating CNN with LSTM methods and fundamental analysis	Combination of LSTM, CNN, and fundamental analysis (e.g., P/E ratio, profitability)	Tehran Stock Exchange

Table 23 (continued)

Reference	Title	Sector	Target	Methods	Market
[85]	Long Short-Term Memory networks with multiple variables for stock market prediction	Specific Stocks	To improve stock market prediction accuracy by using a multi-variable Long Short-Term Memory (MV-LSTM) network that effectively distinguishes and utilizes multiple input variables	MV-LSTM network, enhanced by a variable attention mechanism	Chinese Stock Market (CSI 300 Index)
[25]	Stock price prediction based on LSTM and LightGBM Hybrid model	Specific Stocks	To develop an effective model for predicting stock price fluctuations using a hybrid approach that combines LSTM and LightGBM optimized by Bayesian methods	A hybrid model integrating LSTM and LightGBM, along with Bayesian optimization for parameter tuning	U.S. Stock Market (references to indices like "ES=F," "YM=F," and "AAPL")
[86]	Dynamic sliding window and neighborhood LSTMBased model for stock price prediction	Stock Price Prediction/ Financial Analytics	To develop a model that integrates historical stock prices and their similarities with neighboring stocks using LSTM to improve stock price prediction accuracy	LSTM algorithm combined with a neighborhood approach (LSTM-kNN) and a dynamic sliding window technique to enhance predictive performance based on historical data trends	United States, Germany, and Vietnam

Loday et al. [8] used four deep learning models mostly based on LSTM. LSTM was applied to Royal Securities Exchange of Bhutan (RSEB) tourism data. Bi-directional LSTMs (Bi-LSTMs), Convolutional LSTMs (ConvLSTMs), and Gated Recurrent Units (GRUs) were also compared on the same dataset. LSTMs can process both long sequences of data and single points, unlike RNNs. Stock price prediction is most effective with LSTM-based deep learning models. The models' performance was compared with RMSE, MAE, and R-squared. Stock price prediction was significantly affected by Deep Learning algorithms in this study. Bi-LSTM has the highest R-squared and fewer errors than LSTM, ConvLSTM, and GRU in the project.

Nikhil et al. [91] investigated four models for stock price prediction: LSTM, CNN, LSTM-CNN, and Genetic Algorithm-Assisted LSTM-CNN. This study used daily stock prices for a specific company. The root mean square error (RMSE) was used as a performance metric. The models were trained and evaluated using historical stock market data. Training and testing datasets were used to measure RMSE. GA-LSTM-CNN predicts stock prices more accurately than other models based on RMSE, according to their findings. Findings from this study can be used to inform stock market judgements. Comparisons between GA-LSTM-CNN and other models show that it has the lowest RMSE. In LSTM-CNN, low RMSE values were also observed. According to the testing data, the LSTM model had the highest RMSE. The plots showed that all models could represent stock prices, but the GA-LSTM-CNN model fit best. According to the testing data, the CNN model also captured the trend well, but overfitted the training data.

According to Shinde et al. [92], predicting stock prices can be done using machine learning algorithms. The proposed work evaluated the effectiveness of LSTM models for forecasting stock prices. In order to create the LSTM model, historical stock price data was preprocessed and trained. In addition to evaluating its accuracy, test sets are also used. LSTM models capture stock price trends and patterns more accurately than traditional forecasting techniques like SVR, RNN, etc. Due to the volatility of the stock market, investors may have difficulty making informed decisions based solely on SVR's predictions. As a result of this study, financial analysts and investors may find the model useful. In contrast to emotions, computer-based algorithms are more accurate because they are based on facts and figures.

Chen [93] proposed Evolutionary Attention-based LSTM (EA-LSTM), which had never been used for stock prediction. With this model, multiple time steps are used to pay different degrees of attention to the sampling window's features to offset the limitations of LSTM. Forecast values increase as errors decrease. A comparison was conducted between EA-LSTM prediction efficiency and RNN, GRU, and LSTM as baselines, which are popular time series prediction models. RMSE and MAE results showed that EA-LSTM performed better than other methods.

Nashir et al. [9] used a deep SBU-LSTM neural network to improve the accuracy of predicting the Indonesia Stock Exchange (JKSE) index price. JKSE stock index predictions were improved through deep learning, sentiment analysis, and feature enhancement. Stock prices could be predicted accurately with SBU-LSTM. Stock index movement and sentiment are closely related, as revealed by hyperparameter optimization. Forecasting models can be improved by incorporating sentiment variables, and price movements can be influenced by sentiment variables.

Table 24 Dataset characteristics and feature analysis of 2022's articles

Reference	Data	Types of dataset	Features	Period	Frequency	Feature selection/extraction techniques
[19]	Volatility and risk factor data from BARRA for multiple stocks	Time series dataset comprising historical volatility data of stocks	Specific risk factors, notably measures of volatility	Multi-year dataset, precise years not explicitly mentioned in the summary	Daily	Min-max normalization for scaling specific risk factors to ensure effective learning by the models
[69]	Historical stock market data from Yahoo Finance and economic data from the Federal Reserve Economic Data (FRED)	Time series dataset	Includes features such as closing price, open price, VIX, EFFR (Effective Federal Funds Rate), UNRATE (Unemployment Rate), UMCSENT (University of Michigan Consumer Sentiment Index), USDX (U.S. Dollar Index), MACD (Moving Average Convergence Divergence), ATR (Average True Range), and RSI (Relative Strength Index)	2006–2021, encapsulating major economic events such as the 2008 financial crisis and the 2020 COVID-19 pandemic	Daily for most features; monthly economic data converted to daily using forward filling techniques	Features were selected based on correlation analysis and economic impact; denoising performed via Haar wavelets, and normalization applied using min-max scaling
[70]	Historical stock prices from Sina Finance, including maximum, minimum, and opening prices	Time series dataset	Maximum, minimum, opening prices collected daily from January 2021 to June 2021	January 2021 to June 2021	Daily	Correlation analysis and entropy-weight method used to understand relationships between variables
[71]	Historical stock prices from Yahoo Finance	Daily time series dataset including stock prices (Open, Close, High, Low, Adjusted Close, and Volume)	Close Price was used for prediction; data was normalized using MinMaxScaler and transformed using a rolling window approach for sequences	Specific periods aren't detailed, but they use long-term historical data sets	Daily frequency	Features were extracted using a rolling window approach, with a lookback period set between 10 to 60 timesteps, focusing on Close Prices
[72]	Historical stock prices from the Shanghai A-share index and NASDAQ index, and social media data from Weibo on investor sentiment (terms like "increase positions" and "decrease positions")	Time series dataset with daily values for stock indices and frequency of specific social media terms	Daily values of Shanghai A-share index (Close, Open, High, Low), NASDAQ index values, Weibo term frequencies ("increase positions", "decrease positions")	Data covers January 1, 2018, to December 31, 2020	Daily frequency for stock indices and social media data	Combination of stock indices and social media sentiment as influencing factors; preprocessing includes data normalization methods
[27]	Historical stock transaction data and investor sentiment data from EastMoney Post Bars	Time series data for stock prices and textual data for sentiment analysis	Stock features include opening, closing, highest, lowest prices, trading volume, and amount; sentiment features classified using BERT	January 2, 2019, to September 24, 2021 (665 trading days)	Daily	Sentiment analysis using BERT, integrated with LSTM to enhance prediction accuracy

Table 24 (continued)

Reference	Data	Types of dataset	Features	Period	Frequency	Feature selection/extraction techniques
[73]	Historical stock price data for BYD, sourced from a Chinese stock market trading software website	Time series data with daily stock prices	- Open (Opening Price) - High (Highest Price) - Low (Lowest Price) - Close (Closing Price) - Previous (Previous Closing Price) - Volume	June 30, 2011, to May 28, 2022	Daily	Pearson correlation coefficient for analyzing correlation between stock data features; data preprocessing included interpolation and wavelet denoising
[74]	Collected news data from six financial news portals and one social media platform (Twitter) spanning from 2017 to 2019	Time series dataset	Sentiment-scores derived from textual data, including features like historical prices, volume, and sentiment polarity	2017 to 2019	Daily	Use of sentiment scores extracted through SVM from textual data to serve as additional input features for the LSTM-GRU model
[75]	Collected from the Iraqi stock market, covering six sectors over five years, sourced directly from the market	Time series dataset	Stock index data from sectors: banking, telecommunications, tourism, agriculture, industry, and services	2017 to 2021	Weekly	Feature extraction using CNN layers; LSTM layers used for capturing temporal data patterns
[76]	Historical stock price data, presumably from sources like Yahoo Finance, but specific datasets are not explicitly mentioned	Time-series datasets comprising historical stock prices	Historical stock prices, possibly including key financial indicators, though these are not detailed in the document	The exact period of the data is not specified, but it is implied to cover a substantial historical range to enable effective training	Daily frequency is likely used for stock predictions, although this is not explicitly stated	Techniques like Recursive Feature Elimination (RFE) and Principal Component Analysis (PCA) are mentioned for feature selection and dimensionality reduction
[77]	Collected historical closing prices from Yahoo Finance for the FBM KLCI	Time series dataset focusing on stock market daily closing prices	Daily closing prices of the FBM KLCI index	Data covers from October 20, 2016, to October 20, 2021	Daily	Data was preprocessed by scaling prices to optimize the model's performance
[78]	Historical data from 10 LQ45 stocks on the Indonesia Stock Exchange over 1,200 trading days	Time series dataset comprising stock prices and technical analysis indicators	- Daily closing prices - Stochastic values (with golden/death cross signals) - RSI (with crossing signals) - MACD values (with crossing signals)	The dataset covers from November 9, 2021, for 1,200 trading days	Daily	Transformation of technical indicators into binary signals indicating golden cross or death cross, enhancing the feature set for better prediction
[79]	Daily stock price data from Yahoo Finance, covering Samsung, Hyundai, and KOSPI from 2000–2016 and China Construction Bank and CSI 300 from 2010–2020	Time series dataset of historical stock prices with daily metrics	Include opening price, closing price, trading volume, technical indicators such as SMA, EMA, RSI, and others	2000–2016 for South Korean stocks; 2010–2020 for Chinese stocks	Daily	Genetic algorithm used for optimizing feature selection and model parameters, performed in parallel

Table 24 (continued)

Reference	Data	Types of dataset	Features	Period	Frequency	Feature selection/extraction techniques
[80]	Historical stock data from Yahoo Finance for Samsung, including columns like low, high, open, close, adjusted close, and volume	Time series dataset of historical stock prices	Includes closing price, low, high, open, adjusted close, and volume, all normalized for training	November 23, 2016, to November 23, 2021	Daily	MinMaxScaler for data normalization to scale feature values between 0 and 1, ensuring stable model convergence during training
[14]	Stock price data from the Wind Database covering the period from July 2012 to June 2020 for the Shanghai Stock Exchange Composite Index, Tokyo NIKKEI Index, Seoul KOSPI Index, and SET Index	Time Series data for stock market indices	Closing prices, various technical indicators (e.g., MACD, RSI, CCI), open, high, low prices, and other relevant variables from historical data	2012 to 2020	Daily	F-test for linear relationships, Mutual Information (MI) for nonlinear relationships, Autoencoder (AE) for dimensionality reduction
[81]	Stock price data of 16 commercial banks obtained from the Wind database	Time series data for stock prices	Historical stock prices, returns on equity (ROE), market capitalization statistics	August 19, 2010, to September 16, 2019	Daily trading data	DTW for clustering similar time series shapes, followed by LSTM for learning complex patterns
[82]	Real stock price data including features such as close price, open price, high price, low price, and volume from 50 stocks across various sectors from 2007 to 2016	Time Series Data (financial time series for stock price), categorized into training and validation/testing datasets	Local descriptors derived from original stock price series including: 1. Piecewise Aggregate Approximation (PAA) 2. Derivative 3. Slope 4. Histogram of Oriented Gradients (HOG-1D)	2007 to 2016 (data from 2007 to 2014 used for training, with 2015 and 2016 used for validation/testing)	Daily (stock price data extracted daily)	Four local descriptors for feature creation: 1. PAA for stability 2. Derivatives for change ratio 3. Slope for trend detection 4. HOG-1D for statistical gradient information in time series
[83]	Twitter data alongside historical stock price data sourced from unspecified financial sources	Time Series Data and Textual Data (from Twitter)	Sentiment scores extracted from tweets, historical stock prices	Not explicitly stated	Daily (based on stock price and sentiment aggregation)	Sentiment analysis includes preprocessing tweets, POS tagging, and sentiment polarity scoring
[84]	Financial datasets from groups: financial, petroleum, basic metals, non-metallic minerals. Specific dataset sources not mentioned	Time Series Data	Fundamental analysis features and technical indicators (e.g., P/E ratio, profitability, transactions)	between 2009 and November 2019 (10 years)	Daily	Utilizes fundamental analysis to improve the accuracy of trend predictions
[85]	Data sourced from the CSI 300 index and its constituent stocks	Time Series Data	Closing prices and various technical indicators related to stock price movements	April 2005 to August 2019	Daily	Enhanced LSTM that separates and encodes contributions from multiple input variables, with an attention mechanism highlighting variable importance

Table 24 (continued)

Reference	Data	Types of dataset	Features	Period	Frequency	Feature selection/extraction techniques
[25]	Stock price datasets, specifically mentioning "ES=F," "YM=F," "AAPL," indicating a focus on futures and major stock tickers	Time series data representing stock price fluctuations	Various stock attributes assessed through Pearson's correlation coefficient	Not explicitly stated but suggests recent analyses leading up to 2022	Daily	Uses the Pearson correlation coefficient for selecting relevant stock features
[86]	Datasets include stock price data from the US market (10 years), the German market (10 years), and the Vietnamese market (5 years)	Time series datasets containing historical stock price data, including daily open, high, low, close, and volume data	Input features include historical stock prices of target stocks and their nearest neighbors (in terms of temporal similarity), as well as features determined by dynamic sliding window analysis	10 years for US and German markets, 5 years for the Vietnamese market	Daily frequency for stock price data	Uses a neighborhood approach by calculating temporal similarity among stock time series, enriching the input data for LSTM using a dynamic sliding window technique for significant price changes

Mantravadi et al. [94] developed a model that predicts share prices using real-time NSE data from Yahoo Finance. LSTM algorithm was used to forecast future trends in an efficient and accurate manner, along with an analysis of share prices. In comparison with linear regression, LSTM is the most accurate algorithm for retrieving historical data in short-term epochs. LSTM model tuning was performed on SBI data collected from NSE, which showed an accuracy loss of 0.0015. With machine learning, fiscal solutions can be improved for decision makers and stock markets.

Mroua & Lamine [95] showed that LSTM neural networks could predict several time series from financial classes during the COVID-19 crisis. Furthermore, S&P GSCI commodity indices and stock market indices from different regions were considered. LSTM networks can be used to generate robust prediction models based on daily price data. In COVID-19, their LSTM network achieved RMSEs and MAEs that were too small for the selected financial assets. They also outperformed ARIMA-type models with their LSTM network.

Beneish et al. [96] incorporated LR and LSTM models into this article to estimate and forecast stock prices, thereby reducing investment risk. With the help of the LR algorithm, the prediction model became more stable. Three different equities (AAPL, MSFT, and AMZN) were studied, and the LR Network provided accurate estimate of price changes, as well as low false alarms. The findings provided evidence to support this point.

Wang [97] proposed CNN-BiLSTM-Attention models for stock prediction. As a result of its ability to capture key features and patterns in time-series data comprehensively, CNN-BiLSTM-Attention performed exceptionally well in predicting stock market movements. In this study, BiLSTMs and CNNs were used to estimate local dependencies. The accuracy of predictions and understanding of market dynamics have improved. Stock price trends can be predicted more accurately as the model learns and adapts to different stocks and markets. Insights into market behavior support investment decisions. Additionally, two sets of data were tested, one with and one without wavelet analysis. On the basis of improved data, this paper proposes a CNN-BiLSTM-Attention model showing an average prediction accuracy of 62%.

Kumaresan et al. [98] proposed a comparative study of different LSTM-based models. The models included historical stock prices as well as univariate and multivariate data. Multivariate models include additional variables, like technical indicators, in addition to historical stock prices. They compared the accuracy of these models against a real-world stock price dataset and assessed their performance using MSE, RMSE, MAE, and accuracy metrics. Additionally, different hyperparameters were investigated, such as the number of hidden layers, batch size, and learning rate. The authors also provided insight into LSTM interpretability. Making financial decisions requires stakeholders to understand what factors influence predictions. LSTM-based stock price prediction models are useful for trading strategies, portfolio management, and risk management. Furthermore, they emphasized the limitations of these models, including complexity and data availability. Models based on LSTMs performed better than traditional machine learning models when detecting temporal patterns and correlations.

Sharma & Kaur [99] used machine learning to predict stock values. Stock values predicted machine learning. It was to make more accurate financial decisions by forecasting stock values. They developed a stock forecasting

Table 25 Comparative analysis and findings overview of 2022's articles

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[19]	Comparison made between LSTM and traditional GARCH models for volatility prediction	Evaluated using MAE and MSE: GARCH achieves MAE of 0.1904 and MSE of 0.0380, while LSTM achieves MAE of 0.0730 and MSE of 0.0063, showing LSTM's superior accuracy	Volatility prediction for stocks, focusing on improving risk management strategies through modeling price movements	LSTM demonstrates significantly better performance than GARCH in predicting stock volatility, leveraging its ability to capture relationships among multiple stock risk factors	Although LSTM performs well, further refinement is necessary; future work suggests exploring implicit risk connections among stocks to improve forecasting accuracy further
[69]	Compared the performance of single-layer LSTM models to multilayer configurations to determine optimal network architecture for predictive accuracy	RMSE, MAPE, and Correlation Coefficient were used as primary metrics, where single-layer LSTM models demonstrated higher prediction accuracy	Price prediction of the S&P 500 index	The study concluded that a single-layer LSTM model yielded the best results for predicting the stock index closing prices	The study highlighted the complexity of feature selection; it suggests exploring alternate neural network configurations and incorporating additional financial indicators for future work
[70]	Correlations among maximum, minimum, and opening prices using Pearson, Spearman, Kendall methods	Not explicitly stated, but correlation coefficients are referenced for assessing fit	Stock price prediction	LSTM model shows reliable prediction capabilities, with strong interdependencies among price variables identified through correlation analysis	Limitations from data anomalies and interpolation challenges; future work might explore broader indices and advanced forecasting models
[71]	Comparison between Vanilla LSTM and Peephole LSTM	RMSE and MAE. Peephole LSTM showed an average improvement of 6.04% in RMSE and 9.85% in MAE over Vanilla LSTM	Stock price prediction for the Indian Stock Market	Peephole LSTM significantly outperforms Vanilla LSTM in predicting stock prices due to its enhanced ability to capture nonlinear and periodic patterns	Limitations include the need for larger datasets and the exploration of more frequent data intervals. Future work may involve combining both LSTM versions for improved prediction capabilities and implementing advanced normalization techniques
[72]	Single factors (e.g., social media alone) versus combinations (e.g., A-share index with Weibo) compared for predictive accuracy	MAE, RMSE, and MAPE. Social media (WB) alone showed best metrics: RMSE = 167.04, MAE = 113.45, MAPE = 0.0326	Price and trend prediction of the Shanghai A-share index using historical and sentiment data	Weibo sentiment is a strong predictive factor, often outperforming combinations with other indicators in prediction accuracy	Limitations include the small sample size for sentiment data and potential overfitting. Future research aims to incorporate more macroeconomic indicators (e.g., GDP, PPI) to enhance model predictions
[27]	Compared LSTM + BERT with standalone LSTM and Informer models	- BERT-LSTM accuracy metrics: - PingAn Bank: MAE: 0.46, MAPE: 2.24%, MSE: 0.36, RMSE: 0.60, Accuracy: 46.94% - ZTE: MAE: 0.59, MAPE: 1.73%, MSE: 0.67, RMSE: 0.82, Accuracy: 61.22% - MuYuan: MAE: 1.33, MAPE: 2.49%, MSE: 3.29, RMSE: 1.81, Accuracy: 57.14%	Predicting stock closing prices based on investor sentiment and historical stock data	Incorporating investor sentiment via BERT significantly enhances prediction accuracy compared to solely using LSTM	Current model's limitations include incomplete sentiment classification and omission of certain data sources. Future work aims to integrate more methods and influencing factors

Table 25 (continued)

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[73]	LSTM compared with RNN models to evaluate prediction accuracy	- MSE: RNN = 53.24, LSTM = 60.76 - RMSE: RNN = 2.39, LSTM = 2.46 - MAE: RNN = 5.72, LSTM = 6.07	Stock price prediction for BYD to assess the impact on stock trends with deep learning models	LSTM models excel in capturing long-term dependencies but are slightly overshadowed by RNNs in this setup; LSTM improved handling of temporal dependencies	Limited by the dataset focusing solely on stock market data; suggests incorporating broader economic data, macroeconomic indicators, and sentiment analysis in future research
[74]	Compared against other models, specifically testing the effectiveness of the new composite model in context of existing machine learning models	The model was evaluated with a mean prediction error of 13%, indicating efficient correlation between sentiment and stock prices	Stock price prediction using sentiment analysis of news and social media	The model shows that integrating sentiment analysis into stock prediction models significantly enhances prediction accuracy by effectively correlating public sentiment with market movements	The model's limitation includes reliance on a limited dataset scope causing narrow price window predictions which could lead to overfitting. Future work involves expanding data diversity and incorporating additional macroeconomic variables
[75]	Compared results across sectors and model iterations (10, 50, 100 iterations)	Evaluated using MSE, MAE, Mean Squared Logarithmic Error (MSLE), and RMSE. Achieved an accuracy of 99.74%	Stock index prediction	The hybrid CNN-LSTM model achieved high accuracy in predicting stock indices, outperforming traditional models	Limitations include potential overfitting due to model complexity; future work suggests incorporating economic indicators and expanding dataset features
[76]	The effectiveness of the LSTM model is compared against traditional Artificial Neural Networks (ANNs)	Performance evaluated using metrics such as RNS, MAPE, and MAE; specific numerical results are not disclosed in the excerpt	Primarily focused on stock price prediction using machine learning and deep learning techniques	An improvement in prediction accuracy was noted with the use of LSTM-SGD over standard ANN approaches, although specific quantitative findings are not provided	Points out challenges in dealing with non-stationarity and high volatility of stock prices; suggests future exploration of enhanced models and integration of more sophisticated features for improved accuracy
[77]	LSTM model's performance was evaluated and compared with traditional statistical methods like ARIMA, as referenced in the literature review	MAE: 0.08 and MAPE: 2.9%, indicating the model's predictions are generally slightly pessimistic but still highly accurate at 97.1%	Prediction of stock prices (specifically closing prices) for the FBM KLCI index, focusing on price trends over time	The LSTM model achieved a high level of predictive accuracy, with a predicted price increase indicating market growth potential on specified test dates	Limitations mainly concern analyzing a single stock without accounting for broader market volatility; future work could involve integrating more diverse data inputs and additional financial indicators for improved model accuracy and broader applicability in volatile stock markets
[78]	Compared LSTM predictions using different feature sets: price only, price with volume, and the proposed method with technical indicators	- RMSE: Lower RMSE in proposed method, showing superior prediction accuracy. - MAPE: Proposed method achieved 2.30%, better than 3.03% from price-only features	Price prediction based on technically derived indicators, focusing on capturing enhanced patterns through LSTM for better forecasting	The addition of golden cross and death cross indicators significantly improved predictive accuracy over traditional methods reliant solely on price data	Focused on Indonesian stocks; future work could explore additional datasets or combine with other neural networks to further enhance predictive capabilities and accuracy
[79]	Compared with other GA-based optimization methods and algorithms like Ant-Lion Optimization	MSE, MAE, and MAPE used as metrics. The proposed model showed improvements of 43%, 27%, and 26% respectively over traditional methods	Prediction of one-day-ahead closing prices and market trends	The combined ICAN-LSTM model provided significant improvements in prediction accuracy, outperforming traditional and other GA-based methods	The model faces challenges with volatility prediction. Future work plans to extend the methodology to broader financial markets and enhance model robustness

Table 25 (continued)

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[80]	Compared MLSLSTM against Linear Regression and SVM models over various epochs to evaluate performance	R^2: 0.959 (training), 0.981 (testing); Adjusted R^2: 0.958 (training), 0.979 (testing); Normalized RMSE: 0.019 (training), 0.028 (testing); MAPE: 1.79% (training), 2.18% (testing)	Stock price forecasting	The proposed MLSLSTM model achieved a high prediction accuracy of 95.9% on the training set and 98.1% on the testing set, surpassing traditional methods significantly	Model requires substantial computational resources; suggested future work includes enhancing model efficiency and applying it to broader datasets
[14]	Compared with nine other models, including LSTM, GRU, various econometric models, and previous studies of VMD-based forecasting models	MAE, RMSE, MAPE, Area Under the Curve (AUC). Performance results for the model were MAE values of approximately 18.653 (SSE), 196.414 (NIKKEI), 13.941 (KOSPI), 5.869 (SET), which show improvement over benchmarks	Price prediction of Asian stock market indices	The proposed hybrid model significantly outperformed comparative models, achieving higher accuracy and lower error metrics, indicating its effectiveness in stock index prediction	The model can be further enhanced with more advanced algorithms in the future and integrate unstructured textual data such as sentiment analysis for richer information
[81]	Compared against traditional LSTM models trained on individual bank data	R-SQUARE, MAE, MSE. R-SQUARE values ranged from 0.8864 to 0.9606 for the hybrid model, showing better performance	Price prediction of commercial bank stocks for mid and long-term trend forecasting	The hybrid K-means-LSTM model outperformed traditional models in prediction accuracy and generalization ability, especially in long-term predictions	Identified potential improvements in long-term dynamic prediction due to cumulative errors, suggesting further exploration of clustering methods and data sources
[82]	Compared against decision tree (DT), CNN, LSTM, Attention-LSTM, XceptionTime, InceptionTime, OmniScaleCNN, Adv-LSTM, and MLSTM-FCN	Accuracy (ACC) and Macro F1 Score (mF1) used as evaluation metrics. The results showed MLCA-LSTM had: - At T = 24: ACC = 0.430, mF1 = 0.410 - At T = 32: ACC = 0.426, mF1 = 0.400	Price trend prediction in stock markets	The MLCA-LSTM model outperformed other models in predicting stock price trends, especially in noisy environments, by effectively leveraging local descriptors and hierarchical attention mechanisms	The model faces challenges with noise filtering in high-dimensional financial data. Future work may explore further optimization of feature selection and methods to adapt predictions for different market conditions
[83]	Compared with models like Deep Learning models (DNN) and sentiment analysis methodologies such as DeepClue	Accuracy, Precision, Recall, F-Score, AUC Score, with TLBO-LSTM showing notable improvements in these metrics over other models	Stock price prediction influenced by social media sentiment	The integration of sentiment analysis with stock prediction models significantly enhances accuracy and reliability	Suggests further exploration into multi-source sentiment features and market adaptability across diverse economic conditions
[84]	Compared the integrated CNN-LSTM method with traditional models without fundamental analysis	MAPE, MAE for different groups	Stock price trend prediction	CNN-LSTM with fundamental analysis improves trend prediction accuracy compared to standalone methods	Potential overfitting; future work may explore more extensive fundamental datasets and application to a wider range of stocks
[85]	Traditional LSTM models and other forecasting methods	Likely uses MSE and other standard accuracy metrics (exact metrics not detailed in the excerpt)	Stock price prediction and statistical arbitrage	MV-LSTM network shows improved prediction accuracy and provides a robust statistical arbitrage investment strategy	Model complexity and potential overfitting; suggests future applications with broader datasets or in different markets
[25]	Compared LSTM-BO-LightGBM with LSTM-BO-XGBoost, LSTM-LightGBM, LSTM-XGBoost, single LSTM, and RNN network models	RMSE: 596.04, MAE: 15.24, Accuracy: 0.639, F1 Score: 0.799	Stock price fluctuation forecasting	LSTM-BO-LightGBM outperforms other models in stock price prediction	Future work may explore further applications of the hybrid model and refining it for different stock behaviors

Table 25 (continued)

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[86]	Compared against traditional, simpler models and other machine learning models; the LSTM-kNN model shows superior performance	Includes accuracy rates and RMSE. The neighborhood LSTM (LSTM-kNN) model outperforms competitors in prediction accuracy	Stock price prediction, specifically forecasting future stock prices based on historical data and relationships among stocks	The LSTM-kNN model significantly improves prediction accuracy by considering relationships between stocks and using dynamic data selection for price fluctuations	Limitations include challenges in capturing market volatility and external factors. Future work may integrate more complex datasets or external indicators for further accuracy improvements

innovation that anticipates stock prices and makes successful transactions using fine feature mechanisms. Generally speaking, there are two types of transactions: intraday dealing, or day trading. The stock market is a complex place with many variables that influence the decision-making process. Investors need timely and accurate information to make informed decisions. It takes time and energy to forecast share values. In order to obtain the closest estimate with the least amount of error, it is necessary to modify the most efficient forecasting model. To determine one's stock market position, investors, analysts, and bureaucrats are provided with information.

Anuradha et al. [100] used LSTM, a type of RNN that selectively stores or forgets information based on the current input and previous state. Stock predictions are made using LSTM models based on Genetic Algorithms. Only fundamental data was considered, such as profit and loss, balance sheet, and cash flow. The dataset includes data from CSI 300 stocks and China Construction Bank from January 1, 2010 to April 1, 2020. Genetic algorithms were used to rank the elements. LSTM, PCA SVM, and random forest are inferior to GA-LSTM.

Kunathananon & Kantavat [101] developed a tool for stock market investors and institutions to predict stock prices using LSTM. It was developed with the intention of increasing profits for investors and reducing losses. LSTM is unbiased, fast, and low-cost compared to traditional stock price prediction methods. In the period 2006 to 2021, Yahoo Finance was used to retrieve SET50 and MAI stock prices based on financial statements. The LSTM models are also better suited to handling noisy data, reducing human bias, and providing low-cost solutions for predicting stock prices. According to the study results, LSTMs are effective at capturing complex relationships in data and predicting stock prices. Investors and institutions may benefit from LSTMs according to this study.

Cui et al. [28] used wavelet and LSTM network groups to process stock price data. To obtain time frequency features with varying scales, the original data was decomposed using Wavelet Transform (WT). In the next step, each component of the model was fed into an LSTM model for further processing. Each LSTM model has different topological structures and parameters. The prediction results were then reconstructed using inverse WT. To evaluate the predictability of the WT and W-LSTM networks and the W-LSTMG, mean square errors (MSEs) were calculated. Based on the results, W-LSTMG shows the lowest prediction error in stock price prediction.

According to Almusawi et al. [102], LSTM-based stock market price prediction is improved by using the Improved Artificial Rabbits Optimization (IARO) algorithm. LSTM parameters were optimized using IARO algorithm to maximize predictive accuracy. Using collected data, stock market prices were predicted. To evaluate the proposed method, Mean Square Error (MSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) and Coefficient of Determination (R2) were used. Compared with other existing methods, such as HMM, PCA-EMD-LSTM, and PCC-BLS, the proposed method was superior in terms of MSE, MAE, MAPE, and R2, compared to other existing methods.

Using the tushare tool, Yu et al. [103] extracted seven highly correlated indicators using principal component analysis. The final prediction model was built using seven extracted indicators, then adjusted and iterated numerous times. With the training results, the LSTM model can be used for stock prediction with high accuracy. To validate the model's effectiveness, short-term and long-term predictions were verified. The LSTM model performed better than the neural network for stock predictions, while the neural network did better for long-term predictions. When there is a large difference between actual data and forecasted data, the neural network will have a large fluctuation, which will make it closer to reality. Neural networks fluctuate quite a bit to reflect reality. For decision-makers, the model is a great reference for long-term prediction.

Based on the cascaded Long Short-Term Memory (CLSTM-PPO Model), Zhou et al. [104] proposed a DRL-based stock trading system. LSTM networks are used in two stages of the model: in the first stage, LSTMs extract

time-series features from daily stock data. LSTM networks are also used by actors and critics of the agent to train the agent using reinforcement learning. In terms of cumulative returns and maximum earning rate, their model performed better than baseline models. Stock indexes improved by 5–52%. Their method can be used to build an automated stock trading system.

Gao & Zhang [26] started by introducing investor sentiment before establishing a hybrid VMD-LSTM model for assessing short-, medium-, and long-term trends in Shanghai Stock Exchange Composite Index (SSEC). The model was trained using the sequence of each trend for predicting SSEC and SENT. The predicted value of the original sequence was obtained by comparing several predicting methods. Econometric models were used to establish a correlation between SENT and SSEC through the establishment of VAR models. Stock market returns are strongly correlated with investor sentiment. There is a positive correlation between stock market returns and investor sentiment during the lag period. The stock market has a lag effect on investor sentiment, which is influenced by stock returns.

Baek [105] proposed a deep learning technique based on an artificial intelligence model that predicts future stock prices. The CNN model extracted features related to stock price prediction, while the LSTM model reflected long-term historical trends. To predict the next day's closing price, we used basic stock price and technical indicator data from the past 20 days and a CNN-LSTM hybrid. This model was optimized using GA in combination. The Korean Stock Index (KOSPI) was used for model evaluation. There is a higher accuracy with CNN-LSTM models with GAs than with CNNs, LSTMs, and CNN-LSTM models without GAs. Deep learning models are used to provide investors and policymakers with more accurate predictions.

Yang et al. [23] examined several factors that played a role in the Wenchuan earthquake. LSTM is adopted for dealing with the influencing factors of landslide disasters. Based on the results, RF-based methods are the most accurate when predicting landslide disaster points. MDBN is the most accurate algorithm when comparing MDBN, DNN, LR, and RF. Landslide warning and prevention strategies can now be selected scientifically.

Shi et al. [13] proposed that major fluctuations in stock prices are mainly caused by high-volume transactions among stocks sharing similar characteristics. They developed a GCN-LSTM method for more accurate capital flow predictions. The stock (node) embeddings in multiple time periods were extracted from graphs with various relational knowledge (edges). Long Short-Term Memory RNNs (LSTMs) are used to discriminate price movements. They have conducted several experiments on major Chinese stock indices, demonstrating the model's accuracy of 57.81%. The results of the GCN-LSTM have also been compared under various graphs and node embedding dimensions, indicating key parameters to select. It presents an insight into how graph-based neural networks and graph data can be used for predicting stock price movements and how they can be applied to quantitative finance, risk management, and business analysis, along with insights into how these techniques might be applied.

Balamohan et al. [106] developed an upgraded LSTM-UP model based on Wavelet transforms to predict stock prices based on human sentiment. To extract the features of financial time series, wavelet transforms, short-term memory, and improved mechanisms were used. Human emotion was investigated by using sentiment polarity scores. By analyzing ADBL, NIB, NABIL, and SCB stock datasets, this model was evaluated against LSTM and GRU. It is possible to compare models' efficiency by using metrics such as root-mean-square error, mean absolute error, R2, and MDA. The Root-Mean-Square Error (RMSE) and Mean Absolute Error (MAE) for all stocks in the experiments were both less than 3.5, while R2 and MDA were both greater than 0.95. When human judgment was added to the model, it performed better than comparable models.

Table 26 shows the comparison between advantages and disadvantages of 2023's articles. Table 27 shows the key attributes and markets of 2023's articles. Table 28 shows the dataset characteristics and feature analysis of 2023's articles. Table 29 shows the comparative analysis and findings overview of 2023's articles.

5.7 Reviewed Articles of 2024

Zhang et al. [12] integrated a 1D CapsNet with an LSTM network for multi-step stock index forecasting. Therefore, a hybrid 1D-CapsNet-LSTM model has been developed, which utilizes a 1D CapsNet to represent high-level capsules and an LSTM network to capture temporal relationships. Throughout different forecasting horizons, MIMO maintained stochastic dependency. Furthermore, the model's performance was compared to baseline models such as LSTM, RNN, and CNN-LSTM using metrics such as RMSE, MAE, MAPE, and TIC. Two key dimensions are outperformed by CapsNet-LSTM. Based on the model, forecasting errors were significantly reduced. Furthermore, it exhibits reduced error escalation with increasing forecast horizons.

Ricchiuti & Sperlí [107] proposed LSTM-based Informative Stock Analysis as an Advisor Neural Network concept. Forecasting stocks was done using a LSTM-based model combining technical indicators, contextual information, and financial data. Therefore, next-day advice was provided by the advice unit using heuristic stock selection based on predicted information. This study examined 417 stocks and

Table 26 Advantages and disadvantages of 2023's articles

Article	Advantages	Disadvantages
[10]	1. Banking Sector Focus: Tailored for banking stocks, providing specialized and relevant insights for this specific sector.	1. Sector Limitation: The model is primarily applicable to the banking sector and may not generalize well to other sectors.
	2. Comparative Method Analysis: Offers a clear comparison between LSTM and GRU, helping in selecting the most suitable approach for stock prediction.	2. Adaptation to Market Movements: Might struggle with sudden market trends or anomalies, reducing its flexibility.
	3. Strong Predictive Power: Shows significant improvements in forecasting accuracy compared to traditional approaches.	3. High Computational Demand: Resource-intensive, making it less accessible to entities with limited computational capabilities.
[87]	1. High Accuracy: Achieves over 99% in predicting equipment failures.	1. Dependence on Historical Data Quality: Poor data negatively impacts prediction accuracy.
	2. Low False Alarm Rate: Reduces unnecessary maintenance activities.	2. Implementation Complexity: Requires expertise in advanced algorithms.
	3. Effective Long-Term Dependency Learning: Captures gradual changes in condition effectively.	3. High Initial Investment Costs: Can be prohibitive for initial adoption.
[16]	1. High Prediction Accuracy: Surpasses traditional methods.	1. Historical Data Dependence: Biases from past information can impact accuracy.
	2. Innovative Optimization Method: Enhances performance with new techniques.	2. Limited Generalizability: Findings may not apply universally.
	3. Comprehensive Dataset Utilization: Supports varied market behaviors.	3. Complex Algorithm Setup: Advanced techniques may be overwhelming.
[88]	1. Improved Forecasting Accuracy: Combines PSO with LSTM for better results.	1. High Computational Demand: Substantial resources required during optimization.
	2. Versatility: Applicable across different stocks and conditions.	2. Historical Data Quality Dependency: Effectiveness heavily reliant on data quality.
	3. Automated Parameter Tuning: Streamlines model development.	3. Overfitting Risk: May overfit on training data.
[89]	1. High Prediction Accuracy: Excels in time-series handling.	1. Model Complexity: Requires expertise for effective deployment.
	2. Multivariable Integration: Incorporates diverse data sources.	2. Historical Data Dependence: Quality of input data is crucial.
	3. Robust Performance: Lower error rates than traditional models.	3. Overfitting Potential: High complexity can lead to overfitting.
[90]	1. Effective Long-Term Pattern Learning: Manages complex dependencies.	1. Sensitivity to Market Volatility: May hinder reliability during spikes.
	2. Dynamic Feature Integration: Adjusts to market changes for improvement.	2. Training Complexity: Requires significant computational resources.
	3. Performance Enhancements: Exceeds standard methods.	3. Feature Selection Dependency: Critical reliance on selected features.
[8]	1. High Predictive Accuracy: Modified Bi-LSTM achieves R^2 values above 95%, showcasing robust performance.	1. Data Quality Constraints: Limited historical data quality impacts predictions.
	2. Comprehensive Model Comparison: Offers valuable insights through the comparative analysis of different models.	2. Complex Application: Practical implementation of the advanced model can be challenging.
	3. Emerging Markets Focus: Application to Bhutan's stock market provides unique regional insights.	3. Overfitting Risk: Detailed parameter tuning may lead to overfitting on training data.
[91]	1. Enhanced Forecast Accuracy: Genetic algorithm improves predictive capabilities over traditional methods.	1. Complex Implementation: Hybrid models involving multiple methodologies can be resource-intensive.
	2. Long-Term Historical Data Utilization: Strengthens the robustness of the study by leveraging extensive data records.	2. Data Quality Dependency: Diverse data quality crucial for reliable predictions.
	3. Performance Comparison: Evaluates GA-LSTM-CNN versus other models, providing comprehensive insights.	3. Limited Generalizability: Findings may not apply across different stocks or market environments.
[92]	1. Superior Forecast Accuracy: Consistently outperforms traditional models like SVR, showcasing robust predictive abilities.	1. Market Volatility Sensitivity: Struggles with sudden market changes, leading to potential inaccuracies.
	2. Effective Complexity Management: Captures intricate patterns in stock price movements.	2. Dependence on Historical Data Quality: Relies on good quality historical data for accurate performance.
	3. Visual Outputs for Analysis: Provides graphical representations enhancing results interpretation.	3. Complex Implementation: Requires detailed technical understanding and expertise.

Table 26 (continued)

Article	Advantages	Disadvantages
[93]	1. Enhanced Predictive Performance: Incorporates attention mechanisms yielding better accuracy.	1. High Computational Requirements: Extensive resources needed for effective model training.
	2. Attention Mechanism Usage: Focuses on critical historical data to improve predictions.	2. Dependence on Data Quality: Predictions heavily reliant on historical data integrity.
	3. Adaptability to Various Investment Strategies: Flexibility in look-back periods supports different trading approaches.	3. Limited Real-World Benchmarking: Primarily evaluated on specific datasets, limiting generalizability.
[9]	1. High Performance: Achieves low error rates with innovative model architecture.	1. Sentiment Data Dependency: Performance is sensitive to accuracy of sentiment data.
	2. Utilizes GDELT Data: Leverages comprehensive sentiment data for predictions.	2. Complexity in Implementation: Combining advanced models requires expertise.
	3. Captures Long-Term Dependencies: Stacked LSTM effectively models intricate patterns.	3. Limited Applicability: Results might not be generalizable across different markets.
[94]	1. Hybrid Model Approach: Combines LSTM with linear regression for enhanced predictions.	1. Data Quality Sensitivity: Heavily reliant on the accuracy of historical data.
	2. Local Market Focus: Specific to Indian stock exchanges for increased relevance.	2. Limited Model Exploration: Focuses primarily on LSTM and linear regression.
	3. Potential Sentiment Analysis Integration: Opens pathways for future improvements.	3. Complex Model Management: Hybrid approach adds practical complexity.
[95]	1. Robust Performance Amid Volatility: Handles pandemic-induced market fluctuations well.	1. Data Dependency: Must have accurate and continuous historical data.
	2. Outperforms ARIMA: Demonstrates superior capabilities over traditional methods.	2. Risk of Overfitting: Model complexity might lead to excessive adaptation.
	3. Comprehensive Historical Data Usage: Uses extensive data for stronger robustness.	3. Setup Complexity: Advanced setup is daunting for new practitioners.
[96]	1. High Logistics Regression Accuracy: Proven superior to LSTM in some settings.	1. LSTM Overfitting Risk: Prone to over-adaptation with increased complexity.
	2. Diverse Methodology Analysis: Offers insights by comparing different model approaches.	2. Data Quality Requirement: Performance tied to input data quality.
	3. Market Dynamics Consideration: Accounts for various factors affecting prices.	3. Variable Complexity: More factors can complicate interpretation and modeling.
[97]	1. Improved Prediction Accuracy: Achieves a notable average prediction accuracy of 62% using a CNN-BiLSTM-Attention model.	1. Complexity of Model Structure: The combination of various architectures increases complexity, making it difficult to implement and interpret.
	2. Integration of Multiple Techniques: The combination of CNN, BiLSTM, and attention mechanisms enhances the model's ability to capture both local features and long-term dependencies.	2. Dependence on Quality of Data: Performance is highly reliant on the accuracy and quality of both historical stock data and sentiment data.
	3. Incorporation of News Sentiment Analysis: This integration improves prediction capabilities by utilizing external sentiment data, providing a more holistic view of market dynamics.	3. Potential Overfitting: The intricate structure raises the risk of overfitting, particularly during prolonged training phases.
[98]	1. Comparative Analysis of Models: Systematic evaluation between univariate and multivariate LSTM models provides valuable insights into their forecasting efficacy.	1. Data Limitation Acknowledgment: The paper notes that availability and quality of data significantly impact model training and applicability.
	2. Exploration of Hyperparameters: In-depth analysis on how variations in hyperparameters affect model performance, guiding future studies.	2. Model Complexity: The computational demands and technical challenges associated with advanced LSTM architectures are substantial.
	3. Real-World Applicability: Highlights practical applications in finance, emphasizing the model's interpretability and relevance to portfolio management strategies.	3. Risk of Overfitting: Although mentioned, the discussion on overfitting lacks depth, limiting insights on how to mitigate this issue.
[99]	1. Utilization of LSTM for Sequential Data: Effectively captures sequential dependencies essential for accurate stock predictions.	1. Limited Applicability of Findings: Results are focused on specific tickers, which may not generalize to the broader market context.
	2. Integration of External Variables: Incorporates external economic indicators alongside stock prices, enhancing model robustness.	2. Dependence on Quality of Input Data: Performance is at risk if the quality of data is compromised, particularly in real-time applications.
	3. Comprehensive Methodological Framework: Provides a clear roadmap from data collection to model training, beneficial for future research.	3. Complexity in Hyperparameter Tuning: Extensive tuning is required for optimal performance, complicating practical implementation.

Table 26 (continued)

Article	Advantages	Disadvantages
[100]	1. Long-term Forecasting Capabilities: Utilizes LSTM networks to capture long-term dependencies, making it suitable for predicting stock returns over extended periods.	1. Data Dependency: Performance is heavily reliant on the quality and quantity of historical data; inadequate data can lead to poor predictions.
	2. Market Insight for Investors: Provides valuable insights into potential returns, aiding in informed financial planning.	2. Complexity of Implementation: Requires careful tuning of parameters and knowledge in machine learning, complicating the implementation process.
	3. Risk Management: Assists in identifying investment risks, refining asset allocation strategies.	3. Performance Variability: Results can vary based on market conditions, with predictions being less reliable in volatile markets.
[101]	1. Bias Reduction: LSTM helps maintain objectivity in predictions, reducing biases common in traditional analysis methods.	1. Dependence on Quality of Input Data: Accuracy relies on the quality of financial statement data; poor data can hinder performance.
	2. Complex Relationship Capture: Effectively captures complex relationships within financial data, enhancing accuracy in volatile markets.	2. Need for Extensive Data: Requires large amounts of historical data for training, which can be scarce or costly to obtain.
	3. Cost-Effective and Time-Efficient: Offers a low-cost and fast alternative to traditional prediction methods.	3. Model Overfitting: Complexity can lead to overfitting, particularly if training data is limited.
[28]	1. Improved Prediction Accuracy: Wavelet Transform enhances prediction accuracy compared to traditional models.	1. Complexity of Model Design: Increases overall complexity and may hinder training effectiveness.
	2. Robustness to Noise: Effectively separates high-frequency noise, allowing LSTMs to focus on underlying trends.	2. Dependency on Wavelet Parameter Selection: Performance is sensitive to the selection of wavelet basis functions.
	3. Adaptability: Employs multiple LSTMs tailored to specific data components, enhancing adaptability.	3. Overfitting Risk: Increased complexity may lead to overfitting if not adequately managed.
[102]	1. Information Fusion: Combines multiple data sources for richer analysis and enhanced prediction accuracy.	1. Dependency on Data Quality: Heavily relies on the accuracy and quality of the integrated financial data.
	2. Genetic Algorithm Optimization: Uses genetic algorithms for better adaptation to volatile market conditions.	2. Initial Configuration Sensitivity: Performance can be affected by initial parameter configurations.
	3. Dynamic Adaptability: Incorporates strategies to adjust predictions dynamically based on market changes.	3. High Computational Demand: Requires significant computational resources for both genetic operations and LSTM training.
[103]	1. Advanced Pattern Recognition Capabilities: Uses sophisticated LSTM models to recognize complex market patterns.	1. Dependency on Historical Data Quality: Model effectiveness relies on high-quality historical stock data.
	2. Market Shift Adaptation: Quickly adapts predictions based on changing market conditions, enhancing flexibility.	2. Parameter Selection Complexities: Requires precise tuning for optimal model performance, complicating the training process.
	3. Trader Insights Provision: Provides valuable predictive insights to inform financial decision-making.	3. Limited to Specific Datasets: Results are often specific to the datasets used, potentially restricting general use.
[104]	1. Enhanced Feature Extraction: Utilizes cascaded LSTM structures for superior temporal data analysis.	1. Market Dynamics Sensitivity: Vulnerable to unexpected external market factors which can affect performance.
	2. Robust PPO Application: Utilizes proximal policy optimization for improved reliability in dynamic trading environments.	2. Historical Data Quality Dependency: Heavily reliant on high-quality historical data for training success.
	3. Superior Return Rates: Demonstrates better financial returns compared to traditional trading models.	3. Implementation Complexity in Real Environments: Adapting the system for real-world trading needs may present challenges.
[26]	1. Attention Integration: Enhanced feature focus and accuracy.	1. Complex Implementation: Increases difficulty for practitioners.
	2. Scalability: Adaptable to dataset sizes.	2. Quality Data Dependence: Needs high-quality data.
	3. Predictive Performance: Improves over traditional models.	3. Generalization Limits: Variable performance across stocks.
[105]	1. Enhanced Model Optimization: The use of genetic algorithms for hyperparameter tuning allows for more efficient and effective model optimization, potentially leading to better performance.	1. High Computational Requirements: The combination of deep learning and genetic algorithms demands significant computational resources, which may not be feasible for all users.
	2. Synergistic Model Benefits: The integration of CNN and LSTM leverages both spatial and temporal features of the data, resulting in a comprehensive predictive capability.	2. Interpretability Issues: The complexity of the model makes it difficult to interpret predictions, posing challenges for practical implementation and user trust.
	3. Superior Empirical Results: The model outperforms traditional stock prediction methods in various empirical tests, indicating its practical applicability.	3. Sensitivity to Data Quality: The model's performance is heavily dependent on the quality of input data, making it vulnerable to noise and outliers.

Table 26 (continued)

Article	Advantages	Disadvantages
[23]	1. Holistic Risk Assessment Approach: The integration of LSTM with Random Forest and MDBN provides a comprehensive framework for landslide susceptibility prediction.	1. High Resource Intensity: The combined methodologies demand significant computational resources, which may limit accessibility.
	2. Versatile Data Utilization: The model can incorporate diverse environmental data, enhancing its predictive capabilities.	2. Quality Dependency of Input Data: The model's effectiveness is contingent upon the availability and quality of relevant environmental data, which can vary.
	3. Flexibility for Geographic Application: The methodology can be adapted to various geographical regions, making it widely applicable in different contexts.	3. Extended Training Times: The complexity of the model can lead to long training periods, impacting its applicability in time-sensitive scenarios.
[13]	1. Incorporation of Social Media Sentiment: The use of Twitter sentiment analysis provides real-time insights into market sentiment, enhancing predictive accuracy.	1. Noise in Social Media Data: The inherent noise and bias in social media data can negatively impact model reliability and prediction accuracy.
	2. Timely Decision-Making: Real-time sentiment analysis can be crucial for trading strategies, allowing for quicker reactions to market changes.	2. Complex Data Integration: Merging sentiment analysis with LSTM models increases the complexity of data processing and interpretation.
	3. Improved Model Performance: The combination of sentiment data with traditional financial metrics leads to more accurate predictions.	3. Limited Historical Context: A focus on contemporary sentiment may limit the robustness of the model in more stable market conditions.
[106]	1. Wavelet Transform for Noise Reduction: The application of wavelet transforms improves model efficiency by effectively filtering out noise from the time series data.	1. Quality Dependency: The accuracy of sentiment analysis can vary; if the sentiment scores are inaccurate, they may degrade the model's performance.
	2. Integration of Human Sentiment: The model significantly boosts prediction accuracy by incorporating sentiment scores derived from news articles and social media, providing a holistic view of market conditions.	2. Complex Implementation: The combination of multiple advanced techniques can complicate model implementation and require expertise for effective utilization.
	3. Strong Predictive Performance: Demonstrates low RMSE and MAE values across various datasets, indicating high reliability and effectiveness.	3. Risk of Overfitting with Feature Expansion: The introduction of various features raises the risk of overfitting, particularly if the model is not validated properly with diverse datasets.

67 cryptocurrencies over three years, respectively. Stock market seasonality can also be determined using historical financial information, such as daily price and volume. As a time series, this information was compiled into the LSTM-based model to predict the next day's price. Their algorithm for heuristic stock selection is based on the outputs of the deep learning modules in order to handle investment losses. Although the positive score increased with the forecast error, its magnitude decreased. State-of-the-art approaches are outperformed by the proposed framework in both markets. Additionally, they earned a return on investment of 39.38% for the Cryptocurrency market despite NASDAQ's downward trend.

According to Yadav & Vishwakarma [11], stock prices are predicted by combining convolution and LSTMs with attention mechanisms. Four different stocks were used as inputs to the prediction model: Apple, Facebook, Nike, and Uber. In order to predict future stock values in the short term, many convolutional and LSTM layers were applied to the input data. For final prediction, the proposed model combined the obtained data with LSTM layer output vectors. Past stock values were used to predict the future. RMSE and computational time were used to evaluate performance. As a result, it performed better than benchmark models with average values of 0.091 and 11.57 s, respectively. Security was enhanced by storing the data on blockchain. In contrast to market-specific markets, it is more efficient at predicting other markets. In this paper, CNN, BiLSTM, and AM were combined to predict stock prices. In the CNN and the BiLSTM, stock data are analyzed for features, interdependence is considered in the BiLSTM, and past time series data is accounted for in the AM. For predicting the last hour of trade, a multilevel neural network model was developed based on six-hour stock movement. It may assist intraday traders in making the right decisions on the stock market.

Safari & Badamchizadeh [108] found DeepInvesting performed well across several key AMZN dataset features. There is a gap in stock market forecasting accuracy. On the AMZN dataset, five analyses were performed: Open, Close, High, Low, and Volume. In each modality, DeepInvesting captured intricate patterns. Model accuracy is measured by MAPE, RMSPE, and R2. This paper demonstrated strong price forecasting using SoLTD-BiLSTM (on the AMZN dataset). With multimodal modeling, intricate financial patterns were captured with minimal MAPE and RMSPE and robust R2. Additionally, they benchmarked the model against KNN, LSTM, RNN, CNN, and ANN.

Latif et al. [109] reported that ML methods can be used to analyze stock market data. In this review, dimensionality reduction, time series decomposition, and hyperparameter

Table 27 Key attributes and markets of 2023's articles

Reference	Title	Sector	Target	Methods	Market
[10]	Long Short-Term Memory and Gated Recurrent Unit for stock price prediction	Banking	This research aims to apply LSTM and GRU models to accurately predict the stock price of Bank Rakyat Indonesia	LSTM and GRU models	Indonesia
[87]	LSTM-based failure prediction for railway rolling stock equipment	Method Comparing (Predictive Maintenance for Railways)	To develop a methodology using LSTM algorithms for predictive maintenance to predict failures in railway rolling stock equipment efficiently	LSTM models and statistical data analysis methods like Mann-Kendall test and correlation analysis	Italy (Railway Transport Sector)
[16]	Stock price prediction with optimized deep LSTM network with artificial Rabbits optimization algorithm	Specific Stocks (Dow Jones Industrial Average - DJIA Index)	To develop a deep LSTM network optimized with the Artificial Rabbits Optimization (ARO) algorithm for effective stock price predictions	Deep LSTM with ARO for hyperparameter optimization; comparison against traditional ANN, standard LSTM, and GA-optimized LSTM	United States
[88]	Particle swarm optimization LSTM based stock prediction model	Specific Stocks (Shenzhen Stock Market)	To propose and analyze a PSO-LSTM model for improving the accuracy and applicability of stock price predictions	Combines LSTM neural network with Particle Swarm Optimization (PSO) algorithm for parameter optimization	China (specifically Shenzhen Stock Market)
[89]	An LSTM-Based model for stock price prediction	Specific Stocks	Develop and validate an LSTM model for accurate stock price predictions	LSTM model with data normalization, recursive feature elimination, and PCA for feature selection	United States (Various companies)
[90]	Confrontation-LSTM network for stock trend prediction	Specific Stocks (focus on stock trend predictions)	To develop a new stock trend prediction algorithm (CL-Net) that effectively handles complex nonlinear dependencies in financial time series data	Confrontation-LSTM (CL-Net), an enhancement of traditional LSTM using adversarial training for improved stochastic transformation handling	China
[8]	Stock price prediction using modified Bidirectional Long Short-Term Memory and deep learning models: A case study of Bhutan tourism corporation limited stock data	Stock Market Prediction	The article aims to predict the stock prices of Bhutan Tourism Corporation Limited (BTCL) using various deep learning models and evaluate their performance	Modified Bi-LSTM, LSTM, ConvLSTM, GRU models were used for analysis	Bhutanese Stock Market, focusing on Bhutan Tourism Corporation Limited (BTCL)
[91]	Stock market prediction using genetic algorithm assisted LSTM-CNN Hybrid model	Stock Market Prediction	To predict stock prices of Apple Inc. (AAPL) and compare the performances of various models including a GA-assisted LSTM-CNN model	LSTM, CNN, LSTM-CNN hybrid, Genetic Algorithm-Assisted LSTM-CNN (GA-LSTM-CNN)	U.S. Stock Market
[92]	Stock price prediction using LSTM	Stock Market Prediction	The article aims to analyze the effectiveness of LSTM models in forecasting the stock prices of a specific company	The study employs LSTM neural networks for stock price prediction, comparing it with SVR and RNN	Indian Stock Market
[93]	Stock prediction using evolutionary attention-based LSTM	Stock Market Indices	To propose and assess a novel deep learning method, the Evolutionary Attention-based LSTM (EA-LSTM), for enhancing stock price predictions over traditional models	This article uses Evolutionary Attention-based LSTM (EA-LSTM) and compares it to models like RNN, LSTM, and GRU	U.S. Stock Market, focusing on the Dow Jones Index (DJI)
[9]	Indonesian stock index price prediction using the stacked Bidirectional Unidirectional Long Short-Term Memory (SBU-LSTM) with the GDELT news sentiment	Stock Market Prediction	Enhance accuracy of stock index price predictions on the Indonesian Stock Exchange (JKSE) using SBU-LSTM integrated with sentiment analysis from GDELT	SBU-LSTM combined with sentiment analysis from GDELT, optimized with RMSprop	Indonesia (Indonesian Stock Exchange, JKSE)

Table 27 (continued)

Reference	Title	Sector	Target	Methods	Market
[94]	Prediction of stock prices based on historical data using Linear Regression model through LSTM algorithm	Specific Stocks	To implement an LSTM algorithm combined with a Linear Regression model to predict stock prices based on historical data and enable non-risk investments	LSTM algorithms and Linear Regression	India (National Stock Exchange – NSE)
[95]	Financial time series prediction under Covid-19 pandemic crisis with Long Short-Term Memory (LSTM) network	Specific Stocks, Commodity Indices, Stock Market Indices	To design and apply an LSTM neural network approach to predict various financial asset classes' time series during the COVID-19 pandemic crisis	LSTM neural network model, with comparisons against ARIMA-type statistical models	Global financial markets, specifically referencing the S&P GSCI commodity indices, U.S. bond market indices, and various stock market indices
[96]	A novel technique to minimising Mean Square Error in stock price index prediction utilising Logistics Regression and LSTM model	Specific Stocks	Forecast changes in stock price indices of three equities traded on the Bombay Stock Exchange using LSTM and Logistic Regression	- LSTM model - Logistic Regression (LR) model	India (Bombay Stock Exchange – BSE)
[97]	Long Short-Term Memory (LSTM) network applications in stock price prediction	Specific Stocks	To enhance the accuracy and stability of stock market price predictions using a CNN-Bi-LSTM-Attention model	CNN-BiLSTM-Attention model: combining CNN for feature extraction, Bi-LSTM for dependencies, and Attention mechanism	General applicability, no specific country's market highlighted; focus is on stock price prediction methodologies applicable across various markets
[98]	Stock price prediction model using LSTM: A comparative study	Specific Stocks	To evaluate and compare different LSTM-based models for predicting stock prices based on historical data and optimized hyperparameters	The study implements univariate and multivariate LSTM models, including attention-based and stacked LSTM architectures, and analyzes their performance using a range of evaluation metrics	Indian stock market
[99]	Stock price prediction using Long Short-Term Memory algorithm	Specific Stocks (focused on various stocks in the Indian market)	To predict stock prices using machine learning methods, specifically the LSTM algorithm, to assist investors in making informed financial decisions	Utilizes LSTM algorithm, a type of RNN, for stock price prediction	India (Indian stock market)
[100]	Prediction of Long-Term Stock returns using Long Short-Term Memory Neural Network	Specific Stocks	To forecast long-term stock returns using LSTM neural networks to make well-informed financial decisions	LSTM neural networks for predicting stock returns, with a focus on model training and hyperparameter tuning	India, specifically focusing on NIFTY-50 stocks
[101]	Estimating stock price based on information from financial statements using Long Short-Term Memory network	Specific Stocks	To develop a model using LSTM for predicting stock prices based on financial statement information	LSTM neural networks are used for stock price estimation, leveraging financial statements and macroeconomic indicators	Thailand (focusing on SET50 and MAI stock indices)
[28]	Stock price prediction based on Wavelet Transform and group of Long Short-Term Memory network	Specific Stocks	To predict future stock prices by incorporating Wavelet Transform for data decomposition combined with a group-based LSTM network model	Utilized Wavelet Transform (WT) to decompose time-series data into different frequency components; prediction using grouped LSTM models (W-LSTMG); output reconstruction via Inverse Wavelet Transform (IWT)	Chinese stock market, specifically stock data from Ping An Insurance (ticker: 601318)

Table 27 (continued)

Reference	Title	Sector	Target	Methods	Market
[102]	Long Short-Term Memory optimized with improved artificial Rabbits optimization algorithm for stock market price prediction	Market Indices	To optimize LSTM using Improved Artificial Rabbits Optimization (IARO) for effective stock price prediction	LSTM model optimized with Improved Artificial Rabbits Optimization (IARO)	United States market, focusing on the Dow Jones Industrial Average (DJIA)
[103]	Feature extraction and prediction of stock information based on LSTM	Specific Stocks	To explore the application method and value of LSTM in stock market prediction, assisting investors to avoid risks	LSTM, Principal Component Analysis (PCA) for feature extraction, Adam optimization algorithm for model training	China's stock market, specifically the Shanghai and Shenzhen stock markets
[104]	A novel deep reinforcement learning based automated stock trading system using cascaded LSTM networks	Automated Stock Trading	To develop an automated stock trading system using Deep Reinforcement Learning (DRL) with Cascaded LSTM networks to enhance trading strategies	Cascaded LSTM networks, Proximal Policy Optimization (PPO), Markov Decision Process (MDP)	Stock markets in the United States, China, India, and the United Kingdom
[26]	The fluctuation correlation between investor sentiment and stock index using VMD-LSTM: Evidence from China stock market	Market Indices	To analyze the relationship between investor sentiment and stock index price fluctuations in the Chinese capital market using a VMD-LSTM hybrid model	VMD-LSTM (Variational Mode Decomposition and Long Short-Term Memory) model, GARCH model for volatility, VAR model for causality analysis, and comparative modeling with standard methods	China
[105]	A CNNLSTM stock prediction model based on genetic algorithm optimization	Specific Stocks	To propose a GA optimized deep learning technique using a CNN-LSTM model to predict the next day's closing price of stocks more accurately	CNN, LSTM, and GA for feature selection and model parameter optimization	Korea Stock Index (KOSPI)
[23]	Research on landslide susceptibility prediction model based on LSTMRFMDBN	Environmental Science/ Geological Disasters	To develop a predictive model for landslide susceptibility using an integrated approach combining LSTM, Random Forest (RF), and Multi-Dimensional Bayesian Network (MDBN)	LSTM for temporal prediction, Random Forest (RF) for classification, Multi-Dimensional Bayesian Network (MDBN) for complex relationship mapping	Not applicable (focus is on geographical data, not market-related)
[13]	Integrated GCNLSTM stock prices movement prediction based on knowledgeincorporated graphs construction	Stock Market/Financial Analytics	To develop an integrated GCN-LSTM method for predicting stock price movements using graphs constructed from relational knowledge to enhance predictive accuracy	The article employs a hybrid model combining GCN and LSTM neural networks for stock price prediction	Chinese Stock Market, focusing on major stock indexes in China
[106]	Effective Stock Market Pricing Prediction Using Long Short-Term MemoryUpgraded Model (LSTMUP) on Evolving Datasets	Stock Market Prediction/ Financial Analytics	To develop an LSTM-upgraded model (LSTM-UP) that incorporates human sentiment and performs effectively on non-stationary and non-linear financial time series data	Utilizes Wavelet Transform for data preprocessing combined with LSTM and an upgraded mechanism for stock price prediction	Primarily focuses on the Nepal Stock Exchange (NEPSE) with datasets like ADBL, NIB, NABIL, and SCB

optimization are discussed. Stock market trend prediction was improved by this study. Modeling the future direction of sampled stock markets with PLSTM-TAL. Additional technical indicators (TIs) were added besides Open, High, Low, Close, and Volume (OHLCV) and Close price and Volume. The contractive autoencoder identified important stock market characteristics. In order to improve prediction accuracy, EEMD broke down the closing price series into quasistationary components. The entropy of the highest component of the original series was subtracted. PLSTM-TAL parameters were tuned using Bayesian optimization. Finally, prediction performance metrics were compared with RF and SVM ML techniques.

Kristyanti et al. [1] studied inflation growth in Indonesia and other comparable countries. CRISP-DM was used to forecast Indonesia's transportation stock prices using data mining techniques. During this process, models, evaluations, and deployments are performed. Various hyperparameter activations functions (linear, ReLU, sigmoid, tanh) and optimizations (ADAM, ADAGRAD, NADAM, RMSPROP,

Table 28 Dataset characteristics and feature analysis of 2023's articles

Reference	Data	Types of datasets	Features	Period	Frequency	Feature selection/ extraction techniques
[10]	Bank Rakyat Indonesia stock data obtained from Yahoo Finance	Time series dataset	Historical stock prices, including opening, closing, high, low prices, and trading volume	1 January 2018 to 30 December 2022	Daily	Time steps (windowing strategy) of 10, 20, and 30 were used to create sequences for model input
[87]	Operational data from train traction converter cooling systems collected from sensors	Time series dataset	Cooling liquid level, flow rate, train identifier, GPS coordinates, current speed, and timestamp of measurement	November to December 2019	Sampled at variable rates (1 s and 60 s), standardized to 60 s for analysis	Data cleaning, spike filtering, and sequence extraction used to prepare data for LSTM model training
[16]	DJIA index stock prices; specific companies not provided, likely sourced from platforms like Yahoo Finance	Time series dataset	Stock prices with other relevant financial indicators; exact features not detailed	Not explicitly stated but typically covers several years of historical data	Daily	Optimized hyperparameters searched via ARO; emphasizes the importance of metaheuristics
[88]	Historical stock data for Shenzhen Ping An Bank stock, with processing involved for anomalies and missing values	Time series dataset	Opening and closing stock prices; preprocessing for abnormal data and gaps	Historical data from 2014 to 2016	Daily	PSO algorithm used to optimize LSTM hyperparameters, enhancing model predictions by reducing manual tuning influence
[89]	Historical stock prices pulled from Yahoo Finance; includes open, high, low, close, adjusted close, and volume data	Time series dataset	Date, Open, High, Low, Close, Adjusted Close, Volume	1980 to 2021	Daily	Recursive Feature Elimination and PCA
[90]	Stock price data sourced from real stock market exchanges, focusing on tracking dynamic periodic patterns	Time series dataset	Includes varying features related to stock prices, with specifics not explicitly detailed	Not explicitly stated	Daily	Adversarial training techniques to address static features
[8]	The dataset was collected from the Royal Securities Exchange of Bhutan (RSEB), consisting of 2,502 records of BTCL stock data	Time series dataset with Date and Price as features, stored in CSV format	Date and Price of BTCL shares	April 23, 2020, to July 15, 2022	Daily	Min-Max scaling was used to normalize the data, which is a crucial preprocessing step for neural network models to enhance learning efficiency
[91]	Historical stock prices from Yahoo Finance accessed via 'yfinance' Python package	Time series dataset including daily historical prices with open, high, low, close prices, and trading volume	Daily closing prices alongside financial statistics and stock splits	January 1, 2010, to February 3, 2023	Daily	Min-Max Scaling for normalization, input-output structuring for training with a windowing technique for LSTM
[92]	Historical stock price data sourced from the National Stock Exchange (NSE) of India	Time series dataset consisting of daily stock prices and related financial metrics	The input features include stock price, open price, volume, trading quantity, lowest and highest share values for the day, and technical indicators derived from historical prices	The data period is not explicitly mentioned; however, it typically relates to recent historical prices prior to analysis in 2023	Daily	The model implementation includes techniques like input sequence formation for historical data analysis, LSTM layers to extract patterns, and dropout techniques for regularization

Table 28 (continued)

Reference	Data	Types of datasets	Features	Period	Frequency	Feature selection/ extraction techniques
[93]	Historical data on Dow Jones Index from Stooq, including daily opening, lowest, highest, closing prices, and trading volume	Time series dataset covering financial metrics over several years	Daily financial figures such as opening, lowest, highest, closing prices, and trading volume, processed with statistical indicators and min-max scaling	January 2007 to March 2023	Daily	Techniques include min-max scaling for normalization and preprocessing, with the datasets divided into training, validation, and test sets
[9]	Stock price data from Yahoo Finance for JKSE and news data from GDELT	Time series dataset for stock prices and textual sentiment data	Closing price, tone score of news articles, optimism sentiment score, media attention measure, and tone dispersion	January 1, 2008, to December 31, 2022	Daily	Sentiment analysis metrics calculated from GDELT; tone scores based on word counts
[94]	Historical stock data obtained from Yahoo Finance	Time series data including attributes such as high, low, open, close, volume, and adjusted close	High, Low, Close, Open, Adjusted Close, and Volume	April 1, 2014, to March 31, 2022, for training; April 22, 2022, to August 30, 2022, for testing	Daily	Pre-processing includes cleaning, removing null values, converting data types, and normalizing data between 0 to 1
[95]	Historical daily financial time series data from DataStream, covering a range of commodity, equity indices, and bond markets	Time series data including stock market indices, commodity indices, and bond market indices	Daily closing prices for various asset types including S&P GSCI commodity indices, MSCI equity indices, and U.S. bond market indices	Data covering from January 2, 1998, to September 16, 2020, including multiple financial crises and market events	Daily	Utilizes LSTM memory modules to factor in long-term dependencies and normalizes data between 0 and 1 for effective processing
[96]	Historical stock prices from Yahoo! Finance for Apple, Microsoft, and Amazon	Time series data with daily stock prices	- Date - High price - Low price - Open price - Volume - Adjusted close price	January 4, 2012, to December 10, 2020	Daily	Selected based on relevance for stock price prediction
[97]	Historical stock price data; specific data source not specified	Time series data containing historical stock prices	Price indicators such as opening price, closing price, high price, low price, and volume (specific features might be abstracted in the explanation)	December 21, 2015, to December 19, 2022	Daily	Feature extraction using CNN; employs an Attention mechanism for focusing on relevant data attributes
[98]	Real-world stock price datasets including historical prices, technical indicators, and macroeconomic factors	Time series dataset	Historical stock prices, trading volumes, technical indicators, and macroeconomic variables (for multivariate models)	The exact years are not specified, but the data covers several years of historical stock prices	Daily stock prices (typical for such studies, though not explicitly mentioned)	Techniques include normalization and preprocessing to prepare data for LSTM models, highlighting significant feature interactions

Table 28 (continued)

Reference	Data	Types of datasets	Features	Period	Frequency	Feature selection/ extraction techniques
[99]	Historical stock price data sourced from Google Finance and Yahoo Finance	Time series dataset (closing prices of various stocks)	Date, opening price, daily high, daily low, closing price, volume, and Open Interest (OpenInt)	Historical stock prices prior to the publication date (specific years not explicitly mentioned but includes 10 years of historical data as noted in methods)	Daily (data collected at the end of each trading day)	Data pre-processing includes transforming time-series data into input-output pairs for supervised learning and scaling the data within the range of [−1, +1]
[100]	Historical stock prices and financial indicators sourced from the Screener website	Time series dataset on long-term stock returns based on fundamental data	Sales, price-to-earnings ratio, debt-to-equity ratio, earnings per share, and other relevant financial indicators	Historical data, though specific years are not detailed in the provided information	Daily	Includes data preprocessing with cleaning, handling of missing values, feature selection based on importance, and normalization of selected features
[101]	Financial statement data and stock prices from the Stock Exchange of Thailand (SET) and Yahoo Finance from 2006 to 2021	Time series dataset focusing on quarterly financial statement data and stock prices	Total assets, Total equity, Total liability, Earnings per share, Closing price, Price-to-Earnings ratio, Market Capitalization, SET index, etc.	2006 to 2021	Quarterly	Data normalization to a range of 0 to 1, with additional preprocessing steps including computation of average quarterly prices
[28]	Stock price data (closing prices) from January 4, 2016, to December 28, 2018, focusing on the Ping An Insurance company	Time-series data focused on daily closing stock prices	Features include closing price data and decomposed time-frequency components obtained through Wavelet Transform	Data spans from January 4, 2016, to December 28, 2018	Daily	Wavelet Transform used for feature extraction, breaking down time-domain data into time-frequency space to aid prediction tasks
[102]	Data from Yahoo Finance pertaining to DJIA stock prices	Time-series dataset of stock prices from the DJIA	Stock prices with features extracted using PCA for dimensionality reduction	Historical performance period (specific years not explicitly stated)	Daily	PCA used for feature extraction and dimensionality reduction, enhancing computational efficiency
[103]	Historical trading data from Tushare for the Huaxta Stock Exchange 50ETF from 2010 to 2022	Time series data, including daily and intraday (5-minute, 15-minute, 30-minute, and 60-minute) data	Opening price, closing price, market value, earnings per share, return on equity, circulating capital, high price, low price, trading volume, price change, rise/fall indicators	2010 to 2022	Daily and intraday data (5-minute, 15-minute, 30-minute, and 60-minute intervals)	PCA
[104]	Stock datasets from four major indices: Dow Jones Industrial index (DJI), Shanghai Stock Exchange 50 (SSE50), S&P BSE Sensex Index (SENSEX), and FTSE100	Time series data	- Adjusted close price - Available balance - Number of shares owned - Moving Average Convergence Divergence (MACD) - Relative Strength Index (RSI) - Commodity Channel Index (CCI) - Average Directional Index (ADX)	Historical data from January 1, 2009 to May 8, 2020, divided into in-sample (01/01/2009–12/31/2015) and out-of-sample data (01/01/2016–05/08/2020)	Daily	LSTM for extracting hidden time-series features from daily stock data, feeding these features into the PPO-based DRL agent for training

Table 28 (continued)

Reference	Data	Types of datasets	Features	Period	Frequency	Feature selection/ extraction techniques
[26]	Monthly values of investor sentiment index and Shanghai Stock Exchange Composite Index (SSEC) from March 2003 to May 2022, retrieved from CSMAR and RESSET financial databases	Time-series datasets comprising investor sentiment indices and stock index prices	Investor sentiment derived from quantitative indicators such as Monthly IPO funds raised, first-day IPO return rates, discount rates, new investor accounts, market turnover, and Consumer Confidence Index	March 2003 to May 2022	Monthly data	Potential proxy variables selected based on Pearson correlation analysis; composite index constructed using factor analysis reflecting investor sentiment dynamics
[105]	Daily stock price data collected from Yahoo Finance API for KOSPI	Time Series Data	Basic stock price indicators, technical indicators (SMA, WMA, RSI, Stochastic oscillators, MACD)	From September 14, 2014, to October 31, 2022	Daily	Utilizes GA for optimizing model parameters and selecting relevant technical indicators
[23]	Various geospatial datasets from ten severely affected areas post-Wenchuan earthquake, including terrain, vegetation, and seismic data	Geospatial and temporal data related to landslide susceptibility	Eight primary factors: Terrain information entropy, lithology, distance to rivers, faults, vegetation index (NDVI), proximity to roads, peak ground acceleration (PGA), and annual rainfall	Data relevant to the post-Wenchuan earthquake period, though specific years are not outlined	Continuous data during the analysis period, based on landscape changes and seismic activities	GIS technology used for data selection, with significance given to factors impacting landslide susceptibility
[13]	Historical stock price data from major Chinese stock indexes involved in experiments to assess model performance	Structured time-series datasets reflecting stock prices and trading volumes	Features include trading characteristics, relational knowledge (e.g., industry, region, volatility similarities), and constructed graphs representing stock interconnections	from Jan 3, 2017 to Sep 30, 2021, totally 1155 labelled transaction days	Time frames not explicitly stated; however, likely daily based on typical stock market analysis practices	Graphs were constructed based on prior knowledge and capital flow theories, with GCN used to extract node embeddings before inputting them into LSTM
[106]	Datasets derived from NEPSE stock prices and relevant financial indicators, gathered from public listings and trading records	Time series datasets featuring OHLC (Open, High, Low, Close), Volume, and technical indicators for forecasting	Inputs include stock prices (OHLC), trading volume, turnover, and technical indicators enhanced by sentiment polarity scores from text analysis	2014 to 2022	Daily frequency for stock prices, common in stock market analyses	Uses wavelet transforms for noise filtering and feature extraction from numerical data and sentiment scores from text analysis

ADADELTA, SGD, ADAMAX) were analyzed to refine prediction accuracy. MAE 0.0092918, MAPE 0.06422, and R-Square of 96% demonstrated ReLU activation function and ADAM optimizer's effectiveness. In addition, a web-based application facilitates user inputs like ticker and activation-optimizer selection.

Dhokane & Agarwal [3] combined the Moving Average Convergence Divergence (MACD) and Exponential Moving Average (EMA) indicators with LSTM to predict stock prices. Through the use of technical indicators and deep learning methodologies, this study aimed to improve stock market prediction accuracy and robustness. In this study, LSTM algorithms were used to model long-term time series data. They evaluated various metrics including RMSE, R2, accuracy, precision, recall, and F1. Stock price prediction accuracy is significantly improved when MACD and EMA (26-day EMA, 100-day EMA, and 200-day EMA) features are combined with LSTM. Results showed that the accuracy improved when MACD and EMA indicators were used, along with open, high, low, and close (OHLC).

Kumar et al. [110] explored a way of predicting stock prices as a result of machine learning, using LSTM networks. As a result, these networks, which fall under the learning umbrella, provide a solution by capturing complex patterns that have been accumulated over time from past stock data. In exploring existing research and real-world implementations, their objective was to contribute to the discussion around the use of machine learning, specifically LSTM networks, to improve stock price forecast accuracy and reliability. In their research on LSTM stock price prediction, they came to a notable conclusion. This study demonstrated how advanced machine learning can interpret and

Table 29 Comparative analysis and findings overview of 2023's articles

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[10]	Compared LSTM and GRU models; also compared LSTM with SVR in previous studies	For GRU model: MSE: 4958.9168, RMSE: 70.4195, MAPE: 1.1699%. For LSTM model: MSE: 4997.4083, RMSE: 70.6923, MAPE: 1.1834%	Price prediction of Bank Rakyat Indonesia stock prices	The GRU model outperforms LSTM in predicting stock prices and predicts a decrease in stock prices for the following month	Further research could evaluate additional optimizations and explore more extensive datasets or alternative algorithms for improved prediction accuracy
[87]	Performance compared with classical models like Decision Tree, Random Forest, and Multilayer Perceptron	LSTM achieved over 99% prediction accuracy, with a false alarm rate of ~0.4% and mean absolute error of the order of 10^{-4}	Failure prediction in railway rolling stock cooling systems	LSTM methodology outperforms existing models, leading to high accuracy in failure prediction and optimizing maintenance to minimize downtime	Potential improvements include advanced data-preparation techniques, exploring other deep learning architectures, and applying methods to different train subsystems
[16]	Compared performance between LSTM-ARO, ANN, standard LSTM, and GA-optimized LSTM models	MSE, MAE, MAPE, R-squared (R^2); LSTM-ARO outperformed others	Price prediction	LSTM-ARO model significantly improves predictions over tested models	Predictive limitations under market anomalies; future research could enhance optimization algorithms and dataset scope
[88]	Compared the PSO-LSTM model against traditional ANN and LSTM models	Key metrics included RMSE, MAE, and MAPE; results showed significant improvements with PSO-LSTM over other models. - PSO-LSTM: RMSE: 0.2339, MAE: 0.16228, MAPE: 1.5011% - LSTM: RMSE: 0.23556, MAE: 0.17399, MAPE: 1.5956% - ANN: RMSE: 0.54932, MAE: 0.46551, MAPE: 4.1615%	Price prediction	The PSO-LSTM model significantly outperformed traditional forecasting models in predictive accuracy and demonstrated good general applicability for stock prediction	Limitations include challenges with LSTM tuning; future work could explore further optimization and application expansion to different stocks
[89]	LSTM was compared with traditional methods such as regression and K-nearest neighbors (KNN)	MSE, RMSE, MAE and R^2; LSTM outperformed others	Price prediction	The LSTM model captures long-term dependencies, leading to improved predictive accuracy compared to traditional models	Rigorous preprocessing and feature engineering challenges; future work could explore advanced training strategies
[90]	Compared CL-Net with earlier RNN methods and state-of-the-art forecasting techniques	CL-Net outperforms existing methods; however, specific metric values are not provided	Stock trend prediction	CL-Net effectively captures long-term dependencies in stock price prediction, leading to enhanced forecasting results	Future work could include applying the model across different types of stocks to evaluate its performance further; specific limitations not highlighted in the document
[8]	Performance of models was compared: Bi-LSTM, ConvLSTM, LSTM, GRU	Bi-LSTM: RMSE = 21.468, MAE = 12.039, R^2 = 0.959; ConvLSTM: RMSE = 23.930, MAE = 12.230, R^2 = 0.949; LSTM: RMSE = 27.151, MAE = 13.728, R^2 = 0.935; GRU: RMSE = 30.452, MAE = 17.823, R^2 = 0.918	Predicting prices of BTCL stocks using deep learning models	Bi-LSTM outperformed other models in predicting BTCL stock prices, achieving higher accuracy and lower error rates due to its modified structure	Study limited by the small dataset and potential overfitting issues. Future work involves integrating additional influential factors and utilizing larger datasets to enhance the models' predictive capability

Table 29 (continued)

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[91]	LSTM, CNN, LSTM-CNN, GA-LSTM-CNN	Evaluated using RMSE: - LSTM: Train RMSE = 0.02, Test RMSE = 0.09 - CNN: Train RMSE = 0.012, Test RMSE = 0.053 - LSTM-CNN: Train RMSE = 0.0082, Test RMSE = 0.04 - GA-LSTM-CNN: Train RMSE = 0.0064, Test RMSE = 0.03 (best)	Apple Inc. stock price prediction using various machine learning models	GA-LSTM-CNN model outperformed other models in accuracy	Limitations in model complexity and computation. Future work could involve optimization techniques, exploring different architectures, and real-time predictions
[92]	Models compared include LSTM, SVR, and RNN	The accuracy of models was assessed using actual price comparison. Results show: - AXISBANK: Actual Price = 493.85, LSTM = 494.12, SVR = 442.37 (LSTM outperformed SVR) - INDUSIND: Actual Price = 1231.10, LSTM = 1238.20, SVR = 1159.25 (LSTM outperformed SVR) - MARUTI: Actual Price = 8454.00, LSTM = 8519.00, SVR = 8172.39 (LSTM outperformed SVR)	The primary focus is on stock price prediction, specifically predicting future stock prices leveraging historical data through machine learning methods	The study concludes that the LSTM model is significantly more effective than traditional models (SVR/RNN) in capturing trends and predicting stock prices, suggesting it can aid investors in making informed decisions	Limitations include the model's potential for overfitting due to noise in the data. Future work may involve exploring different model architectures, hyperparameter tuning, and optimizing algorithms to enhance predictive accuracy
[93]	EA-LSTM model compared against RNN, LSTM, and GRU models for performance evaluation	Detailed performance metrics include RMSE and MAE: - EA-LSTM RMSE for 30-day look-back = 0.0196, MAE = 0.0157 - EA-LSTM RMSE for 45-day look-back = 0.0156, MAE = 0.0127 - EA-LSTM RMSE for 60-day look-back = 0.0340, MAE = 0.0305	The article focuses on predicting closing prices of stock indices, specifically the Dow Jones Index, leveraging advanced deep learning techniques	The EA-LSTM model demonstrates superior performance over traditional baseline models by providing more accurate stock price predictions, underscoring its relevance in effective financial forecasting	The paper indicates potential limitations in computational complexity and suggests future exploration of hyperparameter optimization and diverse window sizes to enhance model robustness and predictive capability
[9]	SBU-LSTM vs. traditional LSTM models and study of sentiment impact on predictions	MAPE: average MAPE of 0.81%, best result of 0.80%	Stock price prediction for JKSE utilizing historical prices and news sentiment	SBU-LSTM significantly improved prediction accuracy with sentiment variables compared to traditional models	Model performance affected by market conditions; future work on hyperparameter optimization and dataset expansion suggested
[94]	Compared performance of LSTM and Linear Regression models	Evaluation metrics include MSE and R^2 error for various banks; specifics of errors for training and testing phases were provided for different banks	Stock price prediction using historical data for time series forecasting	LSTM is more effective than Linear Regression for retrieving historical stock prices, yielding reduced prediction errors	Current model focuses on historical data; future work will include sentiment analysis from Twitter to enhance prediction accuracy
[95]	LSTM network compared to ARIMA-type models	-RMSE -MAE Results showed that the LSTM model achieved significantly low RMSE and MAE during COVID-19	Price prediction of financial assets across various financial classes, specifically during the COVID-19 period	The LSTM model showed superior performance in predicting prices of diverse financial asset classes compared to traditional methods, particularly evident during the pandemic crisis	While LSTM provided strong predictions, the study suggests exploring hybrid models and additional factors impacting market conditions for more robust future work

Table 29 (continued)

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[96]	Performance of LSTM compared with Logistic Regression	- MSE - Accuracy: LR (99.8%), LSTM (72.3%) - Statistical tests supporting LR superiority	Price prediction of selected stocks on BSE	Logistic Regression outperformed LSTM significantly in prediction accuracy and MSE	LR effective for smaller datasets; LSTM may overfit larger datasets; future work may integrate more predictive metrics and potentially hybrid modeling approaches
[97]	Compared performance with and without the Attention mechanism; benchmarked against other unspecified models for validation	Achieved average prediction accuracy of 62%, indicating superiority over some unspecified traditional models in predictive performance	Stock price prediction and forecasting trends for improved decision-making in stock markets	The introduced CNN-BiLSTM-Attention model improves prediction accuracy significantly by efficiently integrating feature extraction and sequence processing capabilities	Recognizes market complexities; future work suggests enhancing feature extraction, better risk management integrations, and expanding model adaptability
[98]	Various LSTM architectures: univariate, multivariate, attention-based LSTM, stacked LSTM; compared with traditional machine learning models	MSE, RMSE, MAE and prediction accuracy. Attention-based and stacked LSTMs show superior performance compared to other models	Stock price prediction	LSTM-based models, especially attention-based and stacked LSTM, outperform other models in stock price prediction. Importance of feature selection and model complexity is emphasized	Limitations include data scarcity, model complexity, and interpretability issues. Future work suggests exploring model interpretability, enhancing data preparation methods, and investigating hybrid models combining LSTM with other techniques
[99]	LSTM model compared against previous models and approaches such as RNN and various machine learning methods mentioned in literature surveys	Evaluated for prediction accuracy using RMSE. Specific RMSE results include: - ITC: 5.381 - BPCL: 5.187 - HPCL: 5.680	Stock price prediction using LSTM for accurate forecasting of future stock prices	The LSTM model effectively captures historical stock price patterns and improves the accuracy of stock price predictions compared to traditional methods	The model's limitations primarily revolve around dependence on historical data; future work may involve adding external macroeconomic indicators or exploring enhancements to the LSTM configuration
[100]	Various configurations of LSTM layers compared with traditional models	MSE: 0.2237, RMSE:0.4730, MAE: 0.3892, Estimated Sharpe Ratio: 1.2914, Estimated Sortino Ratio: 1.5029	Long-term stock price prediction based on fundamental data	The LSTM model effectively captures long-term stock trends, aligning predictions closely with actual market data	Challenges include noise in data and need for optimized feature selection; future work aims to enhance model robustness and accuracy
[101]	LSTM compared with Linear Regression and Deep Neural Networks (DNN)	MAE: - LSTM SET50: 9.6848, MAI: 15.9683 - Linear Regression SET50: 10.8037, MAI: 16.2857 - DNN SET50: 10.3686, MAI: 15.9394	Stock price prediction based on fundamental financial data	LSTM outperformed others, capturing complex relationships effectively for more accurate stock price predictions	LSTM's sensitivity to hyperparameters; future work will refine methods and expand datasets for improved accuracy
[28]	Performance of the W-LSTMG model compared with single LSTM networks and basic LSTM models without Wavelet preprocessing	Evaluated using MSE. Key results: - Best performance with db4 wavelet basis: MSE = 0.964332 - Original data's MSE = 2.000617	Price prediction, with a focus on accurately forecasting next-day closing prices for stocks	The W-LSTMG model achieved superior prediction accuracy over single LSTM models, particularly when using the db4 wavelet basis with three decomposition layers	Model complexity and parameter tuning remain challenges; future research will focus on optimizing parameter settings and exploring alternative wavelet and hybrid modeling techniques

Table 29 (continued)

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[102]	Compared LSTM-IARO with Whale Optimization Algorithm (WOA), Particle Swarm Optimization (PSO), Grey Wolf Optimization (GWO), ARO, MLP, ANN, RNN, CNN	Metrics included MSE, MAE, MAPE, R^2. IARO achieved significant performance: MSE of 0.43, MAE of 0.37, MAPE of 0.21	Stock price prediction for DJIA index, with an aim to enhance forecasting accuracy through optimization	LSTM-IARO significantly enhanced predictive performance compared to several existing methods, effectively addressing limitations seen in traditional techniques	Challenges include handling high noise levels and capturing market volatility accurately; future work could explore integrating more recent data and additional feature extraction techniques for deeper insights
[103]	LSTM compared with traditional models such as SVM and ARIMA in terms of prediction accuracy	Mean square error, with improvements showing up to 90% accuracy after training	Price prediction of stock closing prices	LSTM demonstrated superior performance for long-term forecasting with high stability and accuracy compared to traditional models	Model struggles with short-term price fluctuation predictions; future work suggested is to enhance accuracy for short and medium-term forecasts
[104]	Compared with: - Buy-and-hold strategy - PPO model with MLP policy - Recurrent PPO model with LSTM policy - Historical data-based models (MLP, LSTM, LGBM, Ensemble Strategies)	Metrics for evaluation include Cumulative Returns (CR), Maximum Earning Rate (MER), and Average Profitability per Trade, showing improvements from 5–52%	Automating stock trading and improving returns through model optimization	The CLSTM-PPO model demonstrated superior performance compared to various baseline models across multiple stock indices, highlighting its effectiveness for automated trading	The model's ability is limited by market unpredictability and noise in data. Suggested future work includes exploring variations of data weighting and additional data sources for better market dynamics representation
[26]	Compared VMD-LSTM with LSTM, VMD-RBF, and VMD-GRNN; performance compared to GARCH	MAE, RMSE, MAPE indicated significant reduction in errors for VMD-LSTM compared to individual prediction models	Price prediction and correlation analysis between investor sentiment indices and stock market indices	A strong positive correlation exists between investor sentiment and the stock index, particularly in the long term, with VMD-LSTM outperforming other models	Potential modeling scope limitations concerning external economic influences; future work on refining model accuracy with broader data sources
[105]	Compared GA-based CNN-LSTM with single CNN, single LSTM, and basic CNN-LSTM models	MSE; GA-based CNN-LSTM showed higher prediction accuracy	Stock price prediction, specifically predicting the next day's closing price for stocks in the KOSPI index	The hybrid GA-optimized CNN-LSTM model demonstrated superior accuracy in predicting stock prices	Model complexity may challenge parameter tuning; future applications to other markets and broader market behaviors are suggested
[23]	Traditional models like DNN and LR (Logistic Regression) were compared using AUC as a performance metric	The model achieved an AUC of 0.965, outperforming RF and LSTM alone	Prediction of landslide susceptibility in post-earthquake affected regions	The LSTM-RF-MDBN model significantly improves the prediction accuracy for landslide susceptibility compared to traditional approaches	Complexity in modeling multi-dimensional relationships; future work includes expanding datasets and refining model parameters to enhance prediction reliability
[13]	Compared the GCN-LSTM model's performance against baseline models and other methods, focusing on parameter settings for optimal performance	Best accuracy achieved of 57.81% for stock price direction prediction, indicating effective model performance relative to baselines	Stock price movement direction prediction (up or down) based on both internal (trading) and external (relational) factors	The study confirms that the integration of relational knowledge into stock prediction models significantly enhances accuracy and provides a theoretical foundation for capital flow dynamics	The model is complex, implying limitations in interpretability and requiring further validation; future work may explore more diverse datasets and enhance graph construction methodologies

Table 29 (continued)

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[106]	Compared against traditional LSTM and Gated Recurrent Unit (GRU) models for prediction accuracy and performance metrics	Evaluated using RMSE, MAE, R^2, and MDA, with RMSE and MAE below 3.5, and R^2 and MDA above 0.95 across datasets	Predicting stock market prices based on historical data, market trends, and incorporating sentiment for enhanced accuracy	Including human sentiment in the LSTM model significantly improves predictive performance, reducing error metrics compared to traditional models	Limitations include accounting for external market influences and noise. Future work may integrate more data dimensions or explore alternative sentiment analysis

understand financial markets. In the LSTM model, complex patterns were identified and adaptable market conditions were taken into account to predict future stock prices effectively.

Al-Fatlawy et al. [111] proposed a single-layer LSTM model for forecasting closure prices and high prices the following day. Six input features are required for utilizing time series data by the LSTM. The six characteristics are close price, volume adj-close cost, high price, and open price. In the selection model, the model and value closest to 1 are selected based on the evaluation index. For validating the experimental results of the single-layer LSTM against previous state of the art hybrid and multilayer approaches, the Shanghai composite index dataset was used. When compared to other models, LSTM has direct RSME and MAE. It is a suitable method for forecasting stock prices and helps investors to invest at the right time for maximum profit. Two stock parameters are predicted: higher costs and lower prices for the following day. In comparison with existing models, the proposed single layer LSTM model improved MAE by 26.961, RMSE by 12.121, and accuracy by 0.987.

Al-Farouni et al. [112] proposed an Attention-Long Short-Term Memory model (Attention-LSTM) for predicting stock market value. Stock market data was primarily available online via historical data, technical indicators, and financial news. To predict stock market value, they combined historical and technical pointers' data and used the Attention-LSTM network to make superior predictions. When compared to previous models such as Correlation Coefficient-Broad Learning System (PCC-BLS), Deep Learning-GuesSentiment Analysis (DL-GuesS), and LSTM, the proposed model produced a better Mean Absolute Error (0.0479), Mean Squared Error (0.0050), Root Mean Squared Error (4.04), and Mean Absolute Percentage Error (1.0861).

Huang et al. [4] proposed integrating LSTM predictions and Evolutionary Operating-Wights (EOW) algorithms to improve investment returns. The proposed method predicts future stock prices using a multilayer LSTM, then combines those predictions with real-world market data. This method was found to outperform existing methods based on this research.

Liu et al. [113] incorporated multi-source data into a LSTM-GRU-SA-AM model to improve the accuracy of their stock price prediction. In this structure, LSTM, GRU, SA, and AM are linked together as SA and AM represent self-attention and attention mechanism, respectively. Financial news and stock market trading data are among the sources of data. Four stocks from Chinese NEV companies were selected to validate multi-source data with LSTM-GRU-SA-AM. Data from multiple sources improves prediction accuracy significantly. Additionally, their LSTM-GRU-SA-AM model performs better than any other.

Yang et al. [7] presented a decomposition-ensemble model for predicting highly fluctuating stock prices. Stock price indexes are decomposed into subsequences with different frequencies using MEEMD. Subsequences were predicted using MLPs and LSTMs. From various subsequences of the model, the integrated prediction results were reconstructed. The MEEMD-LSTM-MLP model demonstrated excellent prediction performance despite severe market fluctuations caused by COVID-19, not only in emerging markets but in developed markets as well. Its applicability and reliability have been increased by extending the MEEMD-LSTM-MLP model to a variety of financial time series. Governments and enterprises benefit from smart technical support provided by hybrid MEEMD-LSTM-MLP models.

Chandar [2] developed a model to predict future stock prices using LSTM. Three steps were included in the proposed model. Ten technical indicators are computed based on previous data. The feature vectors are normalized in the second phase. Finally, the LSTM model uses normalized characteristics to predict the closing price. The model was evaluated on the NASDAQ and NYSE. Predictions were more accurate with the new model. The proposed model was highly predictive, according to the results.

Liu & Wang [20] found that LSTM models with sliding windows can forecast stock market returns better than LSTM models without sliding windows. The investigation explores three aspects: stationarity, time, and order. In addition, a standard VAR model was developed. In this experiment, the six major stock markets were observed daily between January 2010 and December 2019. Data from stationary inputs enhances LSTM's predictive capability. Shorter time

intervals usually produce better results than long intervals, regardless of input order. LSTMs have a lower predictive capability than standard VARs, which differs from previous research, but compensates for their conditional limitations.

Prakash & Saleena [5] developed a Deep Attention BiLSTM-COOT algorithm that is optimized for the COOT Birds Algorithm. Stock closing prices were accurately predicted using the NSE stock dataset (Nifty 50) and technical and fundamental indicators. Optimizing the features using the Improved Binary Butterfly Optimization (IBBO) algorithm was performed. In both normal and unpredictable conditions, the proposed approach predicts the closing price accurately.

Altarawneh et al. [114] developed a model specifically designed for developing countries to predict COVID-19 vaccination rates. Three stages are used in the proposed model to construct new disease spread features. Feature-dependent prediction improved accuracy. In order to test the model's effectiveness and generalizability, case studies were conducted in Nigeria, Nepal, and Peru. Using CNNs and LSTMs together, the model showed an additional 16% improvement in mean absolute percentage error (MAPE) after the enhanced feature engineering process.

Chen & Sun [115] developed the TG-LSTM (Time Series Ratio Analysis combined with LSTM) model for early warnings about enterprise financial health. Ratio analysis identifies debt-paying capability, operational efficiency, growth prospects, and profitability as important aspects of financial performance. Prediction accuracy is enhanced by TSVD, while data dimensionality is reduced by XGboost. TG-LSTM is demonstrated using real-world data from the CSI 300 and SSE 50 datasets. The Root Mean Square Error (RMSE) for this model is 96.1%.

Agarwal & Dhokane [116] used RSI, EMA, and basic indicators such as open, high, low, close, and volume (OHLCV). Data from the OHLCV database was used as input for the LSTM model. There was a better performance of OHLCV with technical indicators than traditional OHLCV alone. This research was evaluated using the following metrics: RMSE, MAPE, and R2 were used to evaluate the results. It was found that the proposed technique improved the R2 score as well as MAPE and RMSE scores. In addition, the proposed technique might also be able to be used to help improve intraday trading by predicting the closing price of selected stocks the next day, as well as the direction of those trends.

Patel et al. [117] used bidirectional LSTMs with preprocessed data. Pretreatment included data cleansing and sentiment analysis. Data from the past 90 days was selected and fed to a bidirectional LSTM model for prediction. RNNs are used for prediction of time series, among other tasks. In addition to its ability to capture past and future dependencies, bitcoin price predictions can benefit from its ability to capture both. With the bidirectional LSTM model, they have been able to improve prediction accuracy when forecasting Bitcoin values hourly through sentiment analysis. Actual values aligned significantly with sentiment input characteristics.

Parreño [118] combined chaos theory and machine learning to predict the Philippine Stock Exchange Index (PSEi) by using LSTM networks. The embedding theorem was employed here in order to fully understand PSEi's nonlinear, complex behavior. This study provided a robust platform for performing LSTM analysis using rigorous data preprocessing. As part of this process, multiple imputation with chained equations was applied to missing values. Despite observed deviations during volatile market periods, Takens'-LSTM model captured stock price trends with high fidelity when compared to real market data. It introduced a practical forecasting tool that may prove useful to a broad range of market participants and contribute to theoretical understanding.

Balamohan & Khanaa [119] introduced a method based on advanced machine learning architectures that attempts to identify underlying dynamics in data. These networks were modeled using LSTM, CNNs, and a hybrid approach using both LSTM and CNN. Evaluation of these models was based on sliding window analysis. This allows the models to be trained continuously, capturing changing market trends. Predictive accuracy was measured by Root Mean Square Error (RMSE) for each forecasting technique. Combining LSTM and CNN effectively captured stock market fluctuations. The hybrid LSTM & CNN models performed better than the LSTM & CNN models separately. The stock market does not follow regular patterns or cycles due to its dynamic nature. The stock market's trends vary by company and sector, so a model that adapts quickly is necessary.

Table 30 shows the comparison between advantages and disadvantages of 2024's articles. Table 31 shows the key attributes and markets of 2024's articles. Table 32 shows the dataset characteristics and feature analysis of 2024's articles. Table 33 shows the comparative analysis and findings overview of 2024's articles.

6 Analysis and Comparison

6.1 Key Reasons for Selecting LSTM Over Alternative Models

In this study, we focus on LSTM networks since they are widely used in stock prediction research for their ability to capture sequential dependencies over a long period of time. Although other models like CNN, RNN, Transformers,

Table 30 Advantages and disadvantages of 2024's articles

Article	Advantages	Disadvantages
[12]	1. Integrative Forecasting Model: Combines Capsule Networks (CapsNets) and LSTM to capture hierarchical relationships in data, enhancing feature extraction.	1. Initial Lack of Applications: CapsNets have limited precedent in financial forecasting, causing uncertainties in their effectiveness.
	2. Multi-Input Multi-Output (MIMO) Approach: Produces multiple forecasts simultaneously for different time horizons, making it efficient for multi-step predictions.	2. Sequential Data Handling: CapsNets may struggle with processing sequential data efficiently, leading to implementation challenges.
	3. Resilience Over Longer Horizons: Demonstrates reduced error rates compared to other models even for extensive forecasting horizons, ensuring consistency.	3. High-Quality Data Requirement: Requires substantial amounts of high-quality historical data, which may not always be accessible.
[107]	1. Multi-Source Data Integration: Merges technical indicators, historical data, and news sentiment, providing a holistic view for better predictions.	1. Dependence on News Quality: The accuracy of predictions can vary greatly based on the relevance and quality of news data.
	2. Adaptive Stock Selection: Uses a heuristic algorithm to dynamically update stock recommendations based on current data trends.	2. Complex Data Processing: The integration of diverse data types requires intricate feature selection and processing.
	3. Empirical Validation: Shows substantial financial gains during market downturns, proving its real-world applicability.	3. High Overfitting Risk: The use of detailed datasets increases the likelihood of overfitting, necessitating careful management.
[11]	1. Data Integrity with Blockchain: Uses blockchain to ensure that data is secure and tamper-proof, enhancing the reliability of predictions.	1. Complex Integration: The combination of blockchain with predictive modeling is complex and may be difficult to implement.
	2. Intraday Trading Focus: Tailored for real-time stock price predictions during the last trading hour, crucial for intraday traders.	2. Scalability Issues: Blockchain technology might face challenges handling the high transaction volumes typical in high-frequency trading.
	3. Comprehensive Performance Assessment: Employs rigorous performance metrics to validate model accuracy.	3. Data Quality Dependency: Relies heavily on the accuracy of historical data; any data inaccuracies can significantly affect predictions.
[108]	1. Bidirectional Context: BiLSTM captures information from both past and future sequences, providing a richer context for predictions.	1. Challenge in Volume Prediction: The model struggles with accurately predicting trading volumes, which are crucial for several strategies.
	2. Real Data Evaluation: Performance validated using real stock data from major companies, demonstrating practical utility.	2. High Resource Requirement: Requires significant computational resources, potentially limiting its use for smaller organizations.
	3. Incorporates Multiple Financial Indicators: Utilizes a variety of stock features, such as price and volume, to improve prediction accuracy.	3. Overfitting Risk: The complexity of the model raises concerns about overfitting, particularly with smaller training datasets.
[109]	1. Hybrid Model Approach: Combines Peephole LSTM with Temporal Attention Layer (TAL) to capture intricate temporal patterns.	1. Technical Indicator Dependence: Reliant on choosing the right technical indicators, which can be subjective and vary over time.
	2. Robust Validation Metrics: Uses significant metrics like MAPE and RMSE to validate the accuracy and robustness of the model.	2. Limited General Applicability: May not generalize well across different markets or stocks without significant adjustments.
	3. Practical Application: Demonstrated effectiveness through empirical studies, reflecting real-world conditions and applicability.	3. Complex Preprocessing: Requires substantial preprocessing and feature engineering, demanding a strong understanding of data manipulation techniques.
[1]	1. Sector-Specific Analysis: Provides detailed insights specific to the transportation sector, making it highly relevant for industry stakeholders.	1. Sector Specificity: The model is tailored to the transportation sector, limiting its applicability to other sectors.
	2. Effective Data Mining Framework: Utilizes the CRISP-DM methodology for structured data management and analysis.	2. Inflation Sensitivity: Faces challenges in adjusting predictions based on inflationary trends and economic changes.
	3. High Predictive Accuracy: Demonstrated strong prediction metrics, indicating reliability in forecasting transportation stock prices.	3. Narrow Range of Hyperparameters: Limited exploration of hyperparameter tuning may prevent achieving optimal model performance.
[3]	1. Integration of Technical Indicators: Utilizes MACD and EMA to significantly enhance predictive modeling.	1. Potential Overfitting: The inclusion of multiple indicators raises the risk of the model becoming too tailored to training data.
	2. Empirical Data Validation: Extensive use of real-world data ensures practical applicability.	2. Data Quality Dependence: Predictions heavily reliant on the quality of the employed historical data.
	3. Multifaceted Evaluation Metrics: Employs diverse measures to thoroughly assess model performance.	3. Limited Generalizability: Results may vary significantly in different market conditions.

Table 30 (continued)

Article	Advantages	Disadvantages
[110]	1. Use of LSTM for Enhanced Prediction Accuracy: LSTM networks excel at capturing complex historical patterns, improving accuracy.	1. Dependency on Historical Data: Model effectiveness heavily relies on the availability of quality historical data.
	2. Integration of Diverse Data Factors: Analyzes multiple factors (highs, lows, volumes) for enriched model performance.	2. Overfitting Risk in Complex Models: The risk of overfitting is significant with complex architectures, impacting performance on unseen data.
	3. Visual Data Representation: A systematic approach aids in data interpretation, making it easier to understand predictions.	3. Limited Scope in Non-linear Market Changes: May struggle to adapt to sudden market changes, affecting prediction reliability.
[111]	1. Enhanced Prediction Accuracy: Exhibits significant performance improvements in capturing trends and exact stock values.	1. Complexity of Implementation: May be less accessible due to optimization schemes utilized in the model.
	2. Application of Deep Learning Techniques: LSTM effectively retains long-term dependencies, improving prediction accuracy.	2. Dependence on Data Quality: Heavily reliant on historical data quality for effectiveness.
	3. Combination of Methods for Robustness: Provides both short and long-term forecasts, enhancing risk management.	3. Gradient Vanishing Problem: LSTM is not immune to vanishing gradients under certain conditions.
[112]	1. Improved Prediction Accuracy: Achieves low error margins, indicating robustness in accuracy.	1. Data Dependency: Heavy reliance on the availability and quality of data; poor data quality can adversely affect outcomes.
	2. Integration of Diverse Data Sources: Combines historical data with technical indicators and news, enhancing prediction quality.	2. Complexity of Implementation: Adds complexity to the model architecture and training process.
	3. Attention Mechanism: Allows the model to focus on relevant past events, improving modeling of complex dependencies.	3. Overfitting Risk: Potential for overfitting on insufficiently diverse datasets, impacting generalization.
[4]	1. Strategic Integration: Combines LSTM predictive capabilities with evolutionary algorithms for optimal decision-making.	1. Demand for Robust Historical Data: Relies on extensive and accurate historical data for meaningful predictions.
	2. Comprehensive Market Analysis: Utilizes multi-layer LSTM for in-depth market understanding.	2. Implementation Complexity in Evolutionary Strategies: Challenges exist in fine-tuning and accessibility for users.
	3. Performance Enhancement: Demonstrates improved exploitation of data-driven strategies in trading.	3. Volatility Sensitivity: Markets' unpredictable behavior potentially affecting financial outcomes adversely.
[113]	1. Attention Mechanism: Enhances accuracy by focusing on important features.	1. Model Complexity: Increases training times.
	2. Bi-Directional Learning: Captures dependencies from both past and future data.	2. Overfitting Risks: Prone with small datasets.
	3. Robust Evaluation: Demonstrates superior forecasting metrics.	3. Data Quality Dependence: Sensitive to noisy datasets.
[7]	1. Innovative Framework: Simulates trading strategies.	1. Complex Implementation: Difficult for deployment.
	2. Dynamic Adaptation: Continuously adjusts with market feedback.	2. Limited Practical Examples: Few real-world applications.
	3. Decision Making Focus: Goes beyond simple predictions.	3. Performance Variability: Setup-dependent results.
[2]	1. Strong LSTM Use: Suited for time series specific to stocks.	1. Training Sensitive: Dependent on training data choice.
	2. Non-Stationary Suitability: Handles market conditions.	2. High Computational Needs: Resource-demanding.
	3. Input Variability: Integrates diverse data sources.	3. Complex Hyperparameter Tuning: Difficult optimization.
[20]	1. Econometrics Integration: Enhances LSTM predictions.	1. Model Complexity: Merges econometrics with machine learning.
	2. Sliding Window Refinement: Tackles traditional issues.	2. Flexibility Limits: Less adaptable beyond stock prices.
	3. Financial Indicator Focus: Balances with financial analysis.	3. Overfitting: Dependent on historical trends.
[5]	1. Attention Mechanism: Prioritizes important features.	1. Resource-Intensive: High computational demand.
	2. Bi-Directional Processing: Captures sequential relationships.	2. Training Duration: Slows market adaptation.
	3. Evaluation Metrics: Detailed performance insight.	3. Parameter Tuning Complexity: Affects performance.
[114]	1. Relevance to Current Events: Focuses on the critical issue of COVID-19's impact on financial markets, providing valuable insights for investors.	1. Limited Generalizability: The reliance on data from a unique period may restrict the model's applicability to future market conditions.
	2. Effective Temporal Analysis: Utilizes LSTM's capabilities to model temporal changes in volatility, enhancing the understanding of market dynamics during crises.	2. Sensitivity to External Shocks: Increased volatility during crises complicates prediction, making the model less reliable during unprecedented events.
	3. Policy Implications: Offers insights that can inform policy decisions regarding market stabilization during similar future events.	3. Data Quality Issues: The quality and availability of data during the pandemic may affect the robustness of the findings.

Table 30 (continued)

Article	Advantages	Disadvantages
[115]	1. High Predictive Accuracy: Achieves a remarkable prediction accuracy of 96.1%, showcasing its effectiveness in real-world scenarios, particularly during volatile periods.	1. Dependence on Data Integrity: The model's performance is heavily reliant on the accuracy and quality of input data; any inconsistencies can lead to significant prediction errors.
	2. Integration of Spatiotemporal Analysis: The model effectively captures nonlinear relationships in financial metrics, addressing limitations seen in traditional prediction models that often rely on linear assumptions.	2. Complexity in Interpretation: The deep learning framework introduces a level of complexity that may hinder stakeholder understanding, making it challenging for non-technical users to interpret the results.
	3. Dynamic Feature Selection: Utilizes XGBoost for selecting significant financial indicators, allowing for a more tailored and efficient modeling process.	3. Overfitting Risk: The intricate architecture, combined with a wide range of features, increases the risk of overfitting, especially if the training dataset lacks diversity.
[116]	1. Effective Combination of Indicators: The integration of EMA and RSI enhances prediction accuracy, making it particularly useful for intraday trading strategies.	1. Sensitivity to Data Quality: The accuracy of predictions is highly dependent on the quality of the input features; poor data can severely affect outcomes.
	2. Robust Framework: Leveraging LSTM's strengths in handling time-series data helps mitigate issues such as the vanishing gradient problem that is common in simpler RNNs.	2. Risk of Overfitting: The complexity introduced by numerous features can lead to overfitting, limiting the model's effectiveness on unseen data.
	3. Practical Focus for Traders: The model is designed with real-world application in mind, aiding active traders in making informed decisions based on reliable forecasts.	3. Adaptability Issues: The model may struggle with sudden market changes or unexpected events that were not present in the training data.
[117]	1. Innovative Use of Sentiment Analysis: The incorporation of social media sentiment data adds a unique layer of context to the price predictions, enhancing their accuracy.	1. Data Processing Complexity: The model requires efficient preprocessing of large tweet datasets, which can be resource-intensive and error-prone.
	2. Bidirectional LSTM Approach: Analyzes sequences in both forward and backward directions, leading to improved understanding of temporal dependencies in the data.	2. Limited Historical Context: Heavy reliance on recent data may overlook important long-term trends that could inform better predictions.
	3. High Accuracy Metrics: Achieves a low RMSE of 0.041, indicating strong predictive capabilities that are crucial for high-stakes trading environments.	3. Dependency on Sentiment Quality: The overall quality of predictions is closely tied to the accuracy of sentiment analysis, which can vary greatly.
[118]	1. Innovative Application of Chaos Theory: Effectively captures the nonlinear dynamics of financial markets, providing a deeper understanding of price movements.	1. Hyperparameter Sensitivity: The model's performance can vary dramatically based on the tuning of hyperparameters, requiring meticulous adjustment.
	2. Advanced Data Handling Techniques: Employs Multiple Imputation by Chained Equations (MICE) for handling missing values, preserving the integrity of the dataset.	2. Complex Phase Space Analysis: The selection of embedding parameters for phase space reconstruction adds to the implementation complexity, necessitating a thorough understanding of the underlying mathematics.
	3. Strong Predictive Performance: Demonstrates low MAPE and other favorable metrics, indicating high reliability in predictions.	3. Generalizability Issues: Focused on specific datasets, which may limit the model's applicability across different market contexts.
[119]	1. Hybrid Model Effectiveness: This approach leverages the strengths of both LSTM and CNN architectures, resulting in improved accuracy for complex patterns often seen in financial data.	1. Dependence on Feature Selection: The model's effectiveness is highly contingent on the relevance and quality of the chosen input features; poor selection can degrade performance.
	2. Handling Nonlinear Dynamics: Specifically designed to manage the intricacies of nonlinear market behavior, allowing for more robust predictions.	2. Training and Optimization Complexity: The hybrid model introduces numerous parameters that complicate the training process, requiring significant computational resources and expertise.
	3. Introduction of Novel Metrics: The incorporation of the PrCA metric provides a more nuanced understanding of classification accuracy in stock predictions.	3. Limited Generalizability: The model's validation primarily on specific datasets may hinder its performance when applied to different markets or conditions.

GNN, GAN, and LLMs also have their strengths, they are currently less established in the context of stock market prediction. While RNNs are essential for understanding temporal relationships, their limitations, such as vanishing gradients, have limited their use in stock prediction models. The LSTM network appears to be a better choice because it addresses these challenges with gate mechanisms. In complex financial time series, CNN models are good at capturing short-term spatial correlations, but they cannot model long-term dependencies. The combination of spatial feature extraction with temporal analysis can improve the accuracy of hybrid models like CNN-LSTM. Modeling sequential data such as news articles or earnings reports is possible using transforms such as BERT and GPT, and large language models like GPT-4. Due to computational demands and domain-specific adaptations,

Table 31 Key attributes and markets of 2024's articles

Reference	Title	Sector	Target	Methods	Market
[12]	1D-CapsNet-LSTM: A deep learning-based model for multi-step stock index forecasting	Market indices	To investigate the potential of integrating a 1D CapsNet with an LSTM network for multi-step stock index forecasting	1D-CapsNet for feature extraction and LSTM for capturing temporal dependencies, using a MIMO strategy for forecasting	United States (S&P 500, DJIA, IXIC, NYSE)
[107]	An Advisor Neural Network framework using LSTM-based informative stock analysis	Stock Market	To develop a framework that provides daily investment advice by predicting stock market trends using LSTM-based models that incorporate diverse financial and contextual data	LSTM for stock trend forecasting combined with contextual information from news articles and technical indicators to provide investment advice	United States (NASDAQ) for stocks and global cryptocurrencies
[11]	An integrated blockchain based real time stock price prediction model by CNN, Bi LSTM and AM	Stock Market	To propose a predictive model integrating CNN, Bi-LSTM, and an attention mechanism (AM) for real-time stock price prediction during the last hour of trading based on historical data	A combination of CNN, Bi-LSTM, and an Attention Mechanism (AM)	New York Stock Exchange (NYSE), specifically analyzing stocks from companies including Apple, Facebook, Nike, and Uber
[108]	DeepInvesting: Stock market predictions with a sequence-oriented BiLSTM stacked model - A dataset case study of AMZN	Stock Market, focusing on specific stocks (Amazon Corporation, AMZN)	To introduce and validate DeepInvesting, a multimodal deep learning model aimed at improving stock price prediction for Amazon's dataset	The model leverages a Sequence-Oriented, Long-Term Dependent (SoLTD) architecture using Bi-LSTM networks to predict stock prices	U.S. Market (Amazon)
[109]	Enhanced prediction of stock markets using a novel deep learning model PLSTM-TAL in urbanized smart cities	Stock Market Prediction, focusing on multiple global indices	To develop and evaluate a hybrid deep learning model (PLSTM-TAL) that accurately predicts stock market direction	Combines Peephole Long Short-Term Memory (PLSTM) with a Temporal Attention Layer (TAL)	U.S., U.K., China, India
[1]	How can we predict transportation stock prices using artificial intelligence? Findings from experiments with Long Short-Term Memory based algorithms	Transportation Sector	Forecast stock prices in Indonesia's transportation sector using artificial intelligence and data mining techniques	LSTM algorithm within the CRISP-DM framework for data mining	Indonesian Stock Market
[3]	Enhancing stock price prediction with MACD and EMA features using LSTM algorithm	Stock Market Prediction	The study aims to enhance the accuracy and robustness of stock market predictions by integrating MACD and EMA indicators with the LSTM algorithm	LSTM algorithm was used in combination with MACD and various EMA indicators (26-day, 100-day, and 200-day) for stock price prediction	Indian Stock Market (NSE)
[110]	Improving Long Short-Term Memory (LSTM)-based stock market price predictions in the machine learning era	Specific Stocks	To enhance the accuracy of stock price predictions using LSTM networks by analyzing historical stock market data	LSTM networks for predicting stock prices using historical data and machine learning methodologies	India
[111]	A prediction of stock time series data using single layer Long Short-Term Memory	Specific Stocks	To forecast stock time series data, specifically predicting the high and close prices for the next day using a Single Layer LSTM model	Utilized a Single Layer LSTM model; comparisons made with other established methods including CNN, RNN, and ANN	Shanghai Composite Index (China)
[112]	Stock market value prediction using attention based Long Short-Term Memory Model	Specific Stocks	To enhance stock market value prediction accuracy using an Attention-LSTM model by incorporating historical data and technical indicators	Used an Attention-based Long Short-Term Memory (Attention-LSTM) model, with comparisons to models like MLP, CNN, RNN, and traditional LSTM	Chinese financial market

Table 31 (continued)

Reference	Title	Sector	Target	Methods	Market
[4]	A novel stock trading utilizing Long Short-Term Memory prediction and evolutionary operating-weights strategy	Automated Stock Trading	To propose an enhanced trading strategy by integrating LSTM predictions with the Evolutionary Operating-Weights (EOW) algorithm to improve investment returns in stock trading	- LSTM for price prediction - Evolutionary Operating-Weights (EOW) algorithm - Reinforcement Learning (Q-learning and Policy Gradient)	China's stock market
[113]	Stock price prediction for new energy vehicle companies based on multi-source data and hybrid attention structure	Specific Stocks	Improving stock price prediction accuracy for new energy vehicle companies by integrating multi-source data and using a novel LSTM-GRU-SA-AM model	LSTM-GRU-SA-AM model; includes LSTM, GRU, Self-Attention (SA), and hybrid attention mechanism	China
[7]	Can Hybrid model improve the forecasting performance of stock price index amid COVID-19? Contextual evidence from the MEEMD-LSTM-MLP approach	Market Indices	To predict the fluctuating stock price index during COVID-19 using a hybrid MEEMD-LSTM-MLP model	1. MEEMD 2. LSTM 3. MLP	China (Shenzhen Component Index) and USA (Dow Jones Industrial Average)
[2]	Deep learning framework for stock price prediction using Long Short-Term Memory	Specific Stocks (focused on major U.S. stock markets: NASDAQ and NYSE)	To develop a model that accurately predicts future stock prices using deep learning techniques, specifically a LSTM network	1. LSTM 2. Data normalization using min-max scaling to improve LSTM performance	United States (NASDAQ and NYSE)
[20]	Improving sliding window effect of LSTM in stock prediction based on Econometrics Theory	Specific Stocks	To enhance the accuracy of stock prediction models by analyzing the impact of different sliding window lengths in LSTM models based on econometric theories	1. LSTM with varying sliding window lengths 2. VAR for comparison	Global markets including SP500, FTSE, FCHI, GDAXI, HSI, N225
[5]	Stock market prediction using Deep Attention Bidirectional Long Short-Term Memory	Specific Stocks	Propose an effective stock market prediction model using a Deep Attention BiLSTM framework optimized with the COOT Birds Algorithm for improved accuracy in closing price predictions	1. Deep Attention Bi-directional Long Short-Term Memory (DABiLSTM) 2. COOT Birds Algorithm for optimization 3. Improved Binary Butterfly Optimization (IBBO) for feature selection	National Stock Exchange (NSE) of India, specifically the Nifty 50 index
[114]	COVID-19 vaccine prediction based on an interpretable CNNLSTM model with threestage feature engineering	Health Technology/ Vaccine Prediction	To accurately predict COVID-19 vaccination rates and interpret prediction results to anticipate vaccine demand in developing countries	A hybrid model combining CNN and LSTM with a three-stage feature engineering process	Focus on developing countries, with case studies on Nigeria, Nepal, and Peru
[115]	Enhancing financial risk prediction using TGLSTM model: An innovative approach with applications to public health emergencies	Financial Risk Management/ Public Health Emergencies	To propose the TG-LSTM model, which combines Time Series Ratio Analysis with LSTM, aimed at providing early financial warnings to enterprises amidst global events	TG-LSTM (Time Series Ratio Analysis combined with LSTM) model; includes Truncated Singular Value Decomposition (TSVD) for enhancing prediction accuracy and XGBoost for feature selection	Primarily the Chinese Stock Market, focusing on datasets like the CSI 300 and SSE 50
[116]	A predictive model of the stock market using the LSTM algorithm with a combination of Exponential Moving Average (EMA) and Relative Strength Index (RSI) indicators	Stock Market Prediction/ Financial Analytics	To develop a predictive model for stock market prices utilizing LSTM combined with technical indicators RSI and various EMAs to improve prediction accuracy	LSTM algorithm combined with Exponential Moving Average (EMA) and Relative Strength Index (RSI) indicators for stock price prediction	India (NSE - National Stock Exchange)

Table 31 (continued)

Reference	Title	Sector	Target	Methods	Market
[117]	A deep learning framework for hourly Bitcoin price prediction using BiLSTM and sentiment analysis of Twitter data	Cryptocurrency Market Prediction	To predict the next hour's Bitcoin price using a model that integrates historical Bitcoin price data along with sentiment analysis from Twitter	Utilizes a Bi-LSTM model along with sentiment analysis to process and predict Bitcoin prices based on preprocessed input data	Bitcoin market, analysis primarily focused on the cryptocurrency landscape
[118]	Chaos Theory enhanced LSTM model of the Philippine stock exchange index	Financial forecasting/ Stock Market Prediction	To improve the accuracy of forecasting the Philippine Stock Exchange Index (PSEi) by integrating chaos theory with LSTM networks to capture nonlinear dynamics effectively	Employs Takens' embedding theorem for phase space reconstruction followed by applying LSTM networks for prediction	Philippine Stock Exchange
[119]	Revolutionizing stock market prediction: Harnessing the power of LSTM and CNN Hybrid approach	Stock Market Prediction/ Financial Analysis	To improve the accuracy of stock market predictions by leveraging stock value data using advanced machine learning architectures, specifically LSTM and CNN	The research employs LSTM networks, CNN, and a hybrid approach combining both for forecasting stock prices	The study focuses on companies listed on the National Stock Exchange (NSE) of India

they have limited real-world application in stock prediction, unlike LSTMs that remain computationally efficient and highly applicable. For modeling graph-structured data, such as capital flows, interconnections between companies, or macroeconomic dependency, GNNs are considered promising. In general, they are paired with LSTM or CNN for comprehensive modeling of graph-structured dependencies, but they aren't directly applicable to stock prediction. As a training and augmentation tool for limited datasets, GANs have been explored primarily for the generation of synthetic data in financial forecasting tasks. Despite their need for adversarial training and instability, they are rarely used as forecasting models, since they need adversarial training and are prone to instability.

6.2 Publication Year

This chart likely represents the distribution of research papers according to their year of publication. This data is important for identifying LSTM's trends in research focus over time. It's been shown that LSTM has become more popular over recent years due to its efficiency and accuracy. Figure 9 summarized the statistics of trends in publication over time.

6.3 Stock Market Data's Countries Distribution

This chart likely categorizes the research by countries whose stock market data has been utilized. The choice of countries for stock market data might be driven by various factors, such as the accessibility of financial data, the sophistication of their stock markets, or their relevance to the research questions being investigated. Figure 10 summarized the statistics of geographical distribution of stock market data in research.

- If certain countries dominate the chart, it suggests that researchers are focusing on well-established stock markets like the USA, China and India.

6.4 Programming Language/Code Used

This chart likely shows the distribution of different programming languages or platforms used by researchers. Figure 11 summarized the statistics of programming languages used in financial research. The choice of programming language in financial research typically depends on the type of analysis required. For example:

- Python: Popular for data analysis and machine learning.
- R: Preferred for statistical analysis and visualization.
- MATLAB: Commonly used in academic research for numerical analysis.
- C++/Java: Used for high-performance computation in financial modeling.

Insights:

- A dominance of Python could indicate a trend towards machine learning and data science in finance.
- The presence of specialized languages like MATLAB may suggest the importance of complex mathematical modeling in some studies.

6.5 Frequency of Data Collection

This chart likely illustrates the frequency with which data is collected in the studies. In financial research, this frequency can vary depending on the research goals. For example:

Table 32 Dataset characteristics and feature analysis of 2024's articles

Reference	Data	Types of datasets	Features	Period	Frequency	Feature selection/ extraction techniques
[12]	Historical daily closing prices from Yahoo Finance	Univariate financial time series data	High-level features through capsules capturing complex patterns in stock index values	January 1, 2010, to December 31, 2019	Daily	1D CapsNet for automatic feature extraction through dynamic routing among capsules
[107]	Historical stock prices and news sentiment data obtained from Yahoo Finance and EOD Historical Data platform	Time series data for stock prices and sentiment analysis from news articles	Historical OHLC (Open, High, Low, Close) prices, Daily Return (DR), the number of news articles, sentiment scores, and seasonal data (e.g., Day of the Week, Month)	November 1, 2019, to October 31, 2022 for stock and cryptocurrency data	Daily	Comprehensive feature selection process integrating technical indicators, sentiment analysis, and seasonal data features
[11]	Stock price data obtained from Yahoo Finance for the selected companies	Time series data consisting of minute-level stock prices, specifically for an 8-hour trading period	Input features include minute-level stock prices for Apple, Facebook, Nike, and Uber, along with extracted relevant features by CNN and parameters processed through Bi-LSTM and AM	Data collected for the day of April 14, 2014, with a training period of 7 h and a testing period of 1 h	Minute-level data (1-minute intervals)	Utilization of CNN for feature extraction followed by processing with Bi-LSTM and AM to enhance predictions
[108]	Sourced from Yahoo Finance; contains features like Opening, Closing, High, Low, Volume, and Adjusted Close prices for AMZN	Time-series dataset with daily stock prices	Input features include Open, High, Low, Close, Adjusted Close prices, and Trading Volume of Amazon stocks	Data includes a comprehensive span of Amazon's stock market activity, though the exact years are unspecified in the summary	Daily	Data was preprocessed using Min-Max scaling and organized in sequences for processing by the BiLSTM model
[109]	Historical daily stock data from Yahoo Finance	Time series dataset consisting of daily stock prices	Open, High, Low, Close (OHLC) prices, Volume, and technical indicators	2005 to 2022	Daily	Utilizes Ensemble Empirical Mode Decomposition, Contractive Autoencoder, and Bayesian optimization
[1]	Daily stock data for several transportation companies (e.g., Air Asia Indonesia) sourced from Yahoo Finance	Time series dataset capturing daily stock prices	Open, High, Low, Close, Adjusted Close, Volume	April 1, 2011, to April 1, 2023	Daily	Mean imputation for handling missing values
[3]	Historical stock price data from Yahoo Finance for companies like RELIANCE, KOTAK BANK, INFOSYS, ASIAN PAINT, HINDUSTAN UNILEVER, HDFC BANK, VEDANTA, and ICICI BANK	Time series dataset containing stock price features (OHLCV: Open, High, Low, Close, Volume)	Open, High, Low, Close (OHLC) prices, MACD (Moving Average Convergence Divergence), 26-day EMA, 100-day EMA, 200-day EMA	Data used from January 1, 2020, to December 30, 2023	Daily	Feature engineering with MACD and EMA derived from historical stock price data
[110]	Historical stock data from Google Finance covering Google's stock prices from 2010 to 2020	Time series dataset (daily stock prices and trading volume)	Date, daily high, daily low, open price, close price, daily traded volume, and adjusted close price	2010 to 2020	Daily	Feature selection for price prediction and data preprocessing steps such as filling missing values and scaling

Table 32 (continued)

Reference	Data	Types of datasets	Features	Period	Frequency	Feature selection/ extraction techniques
[111]	The dataset used is sourced from the Shanghai Composite Index	Time-series dataset containing stock price data	Input features include close price, high price, open price, volume, and adjusted close price	Not explicitly mentioned for specific years but involves a historical dataset used for computing stock prediction	Daily data frequency based on stock trading days	Implemented Exponential Smoothing to eliminate noise and split data into training (80%) and testing (20%) sets
[112]	Data sourced from Chinese internet financial media, including 2,351 update articles and 33,500 forum postings, from July 1, 2017, to April 30, 2020	Time-series dataset comprising stock market historical data and technical indicators	Includes closing value, trading volume, open, highest, and lowest stock values, along with technical indicators like RSI, William's %R, and Stochastic Oscillator	Data from July 1, 2017, to April 30, 2020	Daily data frequency based on trading days	Computed technical indicators to construct the prediction feature set with preprocessing to remove unnecessary data
[4]	Stock data from the UQER platform, specifically focusing on the 500ETF from January 4, 2005, to September 29, 2022	Time series data consisting of historical stock prices	- Previous closing price - Opening price - Highest price - Lowest price - Daily turnover rate - Price to book ratio - Change percentage	January 4, 2005, to September 29, 2022	Daily	Sliding window technique for feature extraction and normalization of features to facilitate analysis
[113]	Stock market trading data from Wind Database and financial news from Eastmoney.com	Multi-source datasets consisting of 27 features including stock prices and technical indicators	Opening price, closing price, highest and lowest prices, volume, percent change, swing, turnover, MACD, RSI, CCI, sentiment probabilities from news	January 2, 2020, to December 30, 2022	Daily data (728 trading days)	Emphasis on temporal features and long-distance dependencies captured by hybrid attention mechanisms
[7]	Daily closing prices from: - Shenzhen Component Index (SZI), - Dow Jones Industrial Average (DJIA). Sources: Investing.com	Time Series Data for stock price indices	Daily closing prices, high-frequency and low-frequency components extracted through MEEMD	From January 5, 2015 to November 28, 2022 for main analysis; and January 21, 2020 to November 28, 2022 for COVID-19 impact analysis	Daily	The MEEMD method decomposes the stock price indices into intrinsic mode functions (IMFs) and a residual component to improve forecasting accuracy
[2]	Historical stock price data from eight companies, sourced from Yahoo Finance, for the period of January 2009 to December 2018	Time Series Data (daily stock prices, including volume and high, low, opening, and closing prices)	Ten technical indicators derived from daily stock prices: SMA, EMA, RSI, MACD, stochastic %K and %D, ROC, Larry Williams %R	2009 to 2018	Daily	Creation of feature vectors based on ten technical indicators; normalization of inputs to a 0 to 1 range
[20]	Daily stock index prices from January 2010 to December 2019, sourced from the CSMAR database	Time Series Data (daily stock index prices with open, high, low, close values)	Stock price indices in O, H, L, C format	January 2010 to December 2019, with training data from 2010–2016 and testing data from 2017–2019	Daily	Evaluation of the impact of sliding window sizes on input data for LSTM model accuracy

Table 32 (continued)

Reference	Data	Types of datasets	Features	Period	Frequency	Feature selection/ extraction techniques
[5]	NSE stock dataset (Nifty 50)	Time Series Data (includes technical indicators for stock prices)	Technical indicators such as RSI, EMA, MACD, SMI, and others	Data collected over several years leading up to 2024 (exact range not specified)	Daily	Feature selection via IBBO algorithm to enhance model accuracy and reduce computational time
[114]	Historical vaccination data and information on disease spread and government policies affecting vaccination rates from the mentioned countries	Multivariate time series datasets including vaccination rates, COVID-19 case numbers, and government policy metrics	Disease spread statistics, government policy strictness, historical vaccination data, and temporal factors affecting vaccination rates	While the exact period is not specified, the data pertains to the COVID-19 pandemic era	Weekly, bi-weekly, and monthly data predictions	A three-stage feature engineering approach comprising feature extraction and construction, preprocessing, and selection using domain knowledge
[115]	Real-world financial data sources include the CSI 300 and SSE 50 datasets	Time series datasets related to financial performance indicators (ratios reflecting aspects like profitability, growth prospects, etc.)	30 financial indicators, including growth rates, return on assets, cash flow metrics, and profit margins influencing risk predictions	Data relevant to the contemporary economic context, particularly COVID-19 pandemic effects	Likely daily or monthly basis typical in financial analytics	Features selected using the Boruta algorithm for finding significant indicators
[116]	The dataset was sourced from Yahoo Finance, specifically using data from the NSE India	Time series dataset capturing stock prices and relevant indicators (OHLCV + RSI + EMA values)	Input features include Open, High, Low, Close, Volume (OHLCV), along with technical indicators like RSI and different EMAs (50-day, 100-day, 150-day)	January 1 st, 2012, through December 31 st, 2023	Likely daily	Utilizes technical indicators (RSI and EMA) in addition to necessary feature engineering of OHLCV data for improved prediction capability
[117]	Approximately 6 million data points derived from historical Bitcoin prices and sentiment data sourced from social media (Twitter)	Time series dataset of Bitcoin price data and sentiment scores calculated from social media	Input features include historical Bitcoin prices (OHLC), trading volume, and sentiment scores derived from Twitter data	Recent data from the last 90 days was used for Bitcoin price prediction	Hourly frequency for Bitcoin price prediction	Data preprocessing techniques including data cleansing, sentiment analysis, and chunking the data for manageable processing which includes removing duplicates and unnecessary details
[118]	Dataset sourced from Yahoo Finance, encompassing historical transaction records for the PSEi from January 1, 2014, to December 29, 2023	The dataset includes daily financial indicators including "Date," "Open," "High," "Low," "Close," "Adj Close," and "Volume"	Input features consist of daily closing prices of the PSEi, focusing on the "Close" price due to its relevance in financial forecasting	Historical data spans from January 1, 2014, to December 29, 2023, totaling 2,564 data points	Daily frequency, incorporating daily closing prices of the PSEi	Uses Multiple Imputation by Chained Equations (MICE) for handling missing values, Predictive Mean Matching (PMM) for imputation, and Takens' embedding theorem for phase space reconstruction
[119]	Utilizes stock value data from NSE-listed companies; specific dataset sources are not mentioned	Time series stock price data characterized by high volatility typically encountered in financial markets	Input features likely include historical stock prices derived from the value data, though specifics are not detailed	2012 to 2021	The analysis might be conducted on daily data, given the stock market prediction context	A sliding window approach is utilized for model training, capturing evolving market trends over time while accounting for variations in the dataset

Table 33 Comparative analysis and findings overview of 2024's articles

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[12]	Compared with LSTM, RNN, and CNN-LSTM models	RMSE, MAE, MAPE, TIC; the model outperformed baseline models with lower error rates	Multi-step stock index forecasting	Superior performance in forecasting, slower error increase over time, effectively captures dependencies in financial data	Model complexity, longer training times; future work on configuration optimization, applying model to other domains, addressing training adaptation to market changes
[107]	Compared with various state-of-the-art methods like traditional econometric models and machine learning techniques (e.g., LSTM, GRU, CNN)	MAE, MAPE, RMSE. Results included a 41.21% cumulative gain in stocks and 39.38% in cryptocurrencies over the test period	Daily investment advice based on stock price predictions and trend forecasting	The LSTM model incorporating contextual information outperformed other methods and achieved significant financial gains; effective in predicting daily returns and generating reliable investment advice	Future work aims to include transaction costs, evaluate investment risks, and expand datasets by integrating various markets and years, along with enhancing trading strategies for better financial gains
[11]	Compared alongside six other machine learning models previously proposed for stock prediction on similar datasets, including ARIMA, CNN-LSTM, FBProphet, and LSTM models	The model's performance was evaluated using RMSE and computational time. Average RMSE results were: Apple: 0.021, Facebook: 0.014, Nike: 0.042, Uber: 0.023	The primary focus was on stock price prediction, specifically predicting the next minute prices for intraday trading	The proposed CNN-Bi-LSTM-AM model achieved better accuracy and lower RMSE compared to traditional models, showcasing its effectiveness for real-time stock price prediction	The paper mentions that while the model shows great potential, future work will focus on refining performance parameters for improved accuracy and applying it to other domains beyond stock predictions, such as time series forecasting for different areas like weather and gold prices
[108]	Compared to models such as KNN, LSTM, RNN, CNN, and ANN	Performance measured using metrics like MAPE, Root Mean Squared Percentage Error (RMSPE), and R-squared (R^2); results include MAPE (Close: 0.0176), RMSPE (Close: 0.0238), R^2 (Close: 0.9923) for Amazon's closing prices	Stock price prediction, specifically for Amazon	The DeepInvesting model significantly outperformed traditional models in prediction accuracy with lower errors and higher forecast reliability	Identified challenge in accurately predicting trading volumes; future work aims to improve this aspect and enhance the model's overall performance
[109]	Compared with CNN, LSTM, SVM, and RF models	Accuracy, precision, recall, F1-score, AUC-ROC, PR-AUC, and MCC; specific accuracies include 96% for U.K., 88% for China, 85% for U.S. and India	Prediction of stock market trends and directions	Demonstrates high prediction accuracy, with attention layer enhancing model performance by identifying long-term dependencies	Challenges with market complexities and external factors; future work includes refinement of the model and exploration in other areas of financial forecasting
[1]	Various activation functions (ReLU, sigmoid, etc.) and optimizers compared within the LSTM framework	MAE: 0.0092918, MAPE: 0.06422, R-Squared: 96%	Prediction of stock prices in the transportation sector	ReLU activation function and ADAM optimizer proved most effective; significant growth observed in TMAS.JK stock	Market unpredictability poses a challenge; future work suggests integrating more datasets and advanced analytics techniques for improved prediction accuracy

Table 33 (continued)

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[3]	Comparison of various LSTM model configurations and performance with those of other studies	RMSE, R^2 score, Accuracy, Precision, Recall, F1 Score. Results showed R^2 scores for training data ranging from 0.75 to 0.95, with lower RMSE values indicating better performance	Stock price and trend prediction (uptrend/downtrend classification)	Integration of MACD and EMA features with LSTM significantly improves prediction accuracy. The model achieved high precision, recall, and F1 scores, effectively capturing positive and negative events	Limitations include possible overfitting and the nonstationary nature of financial data. Future work aims to integrate advanced techniques or additional features to enhance prediction performance in volatile markets
[110]	Comparison of LSTM performance with traditional prediction methods	Evaluated using MSE, and RMSE, noting considerable error reduction	Price prediction	LSTM improves prediction accuracy in volatile markets	Overfitting risks and dependence on hyperparameter tuning; future work involves using larger datasets and advanced technologies to enhance predictive accuracy
[111]	Compared the Single Layer LSTM model performance with CNN, RNN, DNN, and ANN models	Metrics include MAE, RMSE, and R^2. - MAE: 26.961 - RMSE: 12.112 - R^2: 0.987	Price prediction of stocks, specifically focusing on forecasting next-day high and close prices	The Single Layer LSTM model demonstrates lower MAE and RMSE, making it more effective than other models tested	The model may struggle with complex relationships in multivariate data. Future work plans to explore enhancing the model with multi-layer architectures
[112]	Compared the Attention-LSTM model with MLP, CNN, RNN, and LSTM models	Used metrics like MAE, MSE, RMSE, and MAPE. Results indicated lower errors with Attention-LSTM.	Focused on stock price prediction, particularly forecasting the closing price of stocks	Attention-LSTM showed superior performance over traditional models, effectively reducing errors and supporting better decision-making for investors	Recognized limitations in model robustness and prediction cycle analysis. Future work aims at improving model adaptability and exploring different domains
[4]	Compared with strategies like buy-and-hold, Turtle Trading Strategy, Moving Average Strategy, Q-learning Strategy, and Policy Gradient Strategy	Evaluated using Cumulative Return, Annualized Rate of Return, Sortino Ratio, Maximum Drawdown, Sharpe Ratio, showing enhanced performance over comparison strategies	Stock trading performance through price prediction and optimal trading strategy development using machine learning and evolutionary algorithms	The combination of LSTM predictions with the EOW strategy results in improved trading performance compared to existing benchmarks	Model faces challenges with market randomness affecting prediction accuracy. Future work could focus on enhancing the evolutionary algorithm and exploring other machine learning frameworks for robustness
[113]	Compared with LSTM, GRU, Transformer-Encoder, LSTM-GRU, LSTM-GRU-AM, and LSTM-GRU-SA baseline models	RMSE: 7.727 ($\pm$0.050), MAE: 5.835 ($\pm$0.030), MAPE: 2.115 ($\pm$0.011), R^2: 0.955 ($\pm$0.001) for the proposed model on BYD stock. Outperformed baseline models	Price prediction for new energy vehicle companies	Multi-source data improves prediction accuracy significantly; the LSTM-GRU-SA-AM model showed superior performance over baseline models	Limitations include possible neglect of temporal information. Future work includes exploring diverse features from financial news and novel temporal attention structures beyond the matrix representation
[7]	Compared against: 1. LSTM 2. MLP 3. SVR 4. BP 5. MEEMD-LSTM 6. MEEMD-MLP 7. CNN-LSTM 8. GRU-CEEMDAN-wavelet	1. RMSE 2. MAE 3. MAPE 4. Determination Coefficient (R^2) 5. Diebold-Mariano (MDM) test	Price prediction of stock price indices, particularly during COVID-19 disruptions	The MEEMD-LSTM-MLP model significantly improves forecasting performance in turbulent markets compared to traditional models, particularly during the COVID-19 pandemic	Limitations in model influence factor selection; future work aims to incorporate more factors that impact stock prices and improve model applicability to various economic conditions

Table 33 (continued)

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[2]	Compared with models such as Standard Recurrent Neural Network (SRNN), Autoregressive Moving Average-RNN (ARMR-RNN), and LSTM using technical indicators	MSE, MAE, RMSE, Symmetric Mean Absolute Percent Error (SMAPE), and Average Relative Value (ARV)	Stock price prediction (aimed at forecasting the next day's closing stock price)	The LSTM model demonstrated lower prediction error metrics, outperforming several traditional models, indicating higher accuracy in stock price forecast	The study suggests incorporating external data, such as news and social media (e.g., Twitter), to enhance the model's predictive capability and adaptability in diverse market conditions
[20]	Compared performance of LSTM and Vector Autoregression (VAR) models with different sliding window lengths	MAE, RMSE And MAPE	Prediction of stock prices using LSTM and comparison with VAR models	Stationary data improves LSTM performance; optimal sliding window length varies by dataset; LSTM does not always outperform simpler models like VAR	Future work suggests integrating econometric principles into deep learning frameworks and extending analysis on data transformations and model adaptability
[5]	Compared DABiLSTM against other techniques such as traditional prediction methods and existing deep learning models	Precision, Recall, F1 Score, MSE, RMSE, MAPE	Closing price prediction of stocks	DABiLSTM-COOT model provides more accurate closing price predictions compared to existing models by effectively utilizing technical indicators and optimized feature selection	Limitations include reliance on minimal databases reducing accuracy. Future work involves exploring additional predictive techniques and datasets
[114]	Compared the CNN-LSTM model against traditional statistical models such as ARIMA and standalone LSTM models based on historical rates	Mean Absolute Percentage Error (MAPE) showed an average 30% reduction with feature engineering; a 16% improvement noted with CNN and LSTM integration	Prediction of vaccination rates for COVID-19 in developing countries to aid vaccine supply planning	The interpretability of the CNN-LSTM model helps understand significant features influencing vaccination rates, informing government policy decisions	Challenges with the black-box nature of machine learning models; future work aims to enhance data inclusion and refine modeling processes for better predictions
[115]	Compared with traditional prediction models, evaluating effectiveness based on RMSE, MAE, and MAPE metrics	Achieved an accuracy rate of 96.1%, with lowest RMSE, MAE, and MAPE values	Financial risk prediction for early warning systems during economic hardships such as public health emergencies	TG-LSTM model outperforms traditional methodologies, processing complex financial data while improving accuracy in risk prediction	Need for model enhancement to maintain accuracy amid dynamic economic changes; future research may involve integrating different data sources for improved predictive power
[116]	The document does not specify models against which the LSTM + EMA + RSI approach was compared, but it notes that previous studies using traditional models had less accuracy	Metrics include RMSE, MAPE, and R^2 score. The proposed model showed improvements in R^2 and reductions in RMSE and MAPE	Stock price prediction, specifically the closing prices for subsequent days and trends for intraday trading	The integration of technical indicators with the LSTM model significantly enhances stock closing price prediction accuracy, benefiting intraday trading	Limitations of the model are not explicitly mentioned; future work may explore optimization of prediction accuracy further and the incorporation of additional datasets or variables
[117]	The Bi-LSTM model has been compared implicitly to other models by highlighting its enhanced prediction capabilities over traditional models; specific comparisons not detailed in the provided text	Metrics include evaluation based on prediction accuracy and the ability of the model to capture volatility in Bitcoin prices, although specific numerical metrics are not stated	Price prediction of Bitcoin for the upcoming hour using deep learning techniques and sentiment analysis	The model demonstrates that incorporating sentiment analysis significantly improves the prediction accuracy of Bitcoin prices, leading to closer predictions to actual prices	Limitations include the complexity of handling vast amounts of data and potentially unobservable market factors. Future work may involve integrating additional data sources or enhancing sentiment analysis mechanisms

Table 33 (continued)

Reference	Comparisons	Performance Metrics	Research object	Key Findings	Limitations/Future Work
[118]	Performance evaluated against traditional forecasting methods; specific models for comparison are not explicitly mentioned	Performance evaluated based on the model's capability to capture stock price trends with high fidelity, particularly in volatile market periods	Price prediction of the Philippine Stock Exchange Index (PSEi) using advanced forecasting techniques	The integration of chaos theory with LSTM improves predictive accuracy by capturing the complex, nonlinear dynamics of the PSEi	Limitations include challenges with accounting for high market volatility. Future work may explore further integrations of additional data sources or modeling assumptions
[119]	The study explicitly compares the performance of LSTM, CNN, and the hybrid model based on their forecasting abilities	The main performance metric is the RMSE, assessing predictive accuracy	Focuses on price prediction of stocks in the Indian market through advanced machine learning techniques	The introduction of machine learning algorithms, particularly the hybrid LSTM-CNN approach, aims to uncover underlying dynamics within stock price data, enhancing forecasting accuracy	Limitations are unspecified, but typical challenges include data quality and market anomalies; future work may enhance data integration strategies

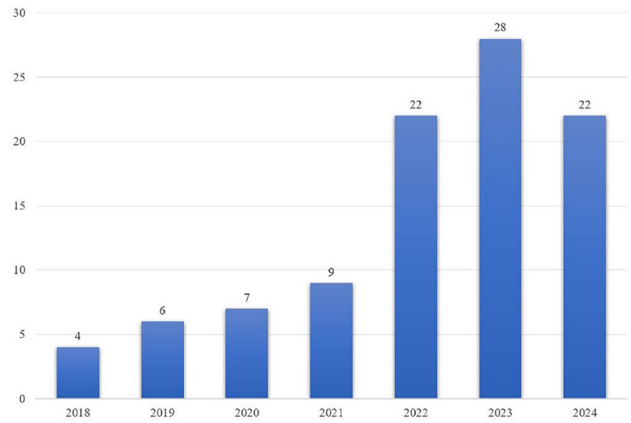


Fig. 9 Trends in publication over time

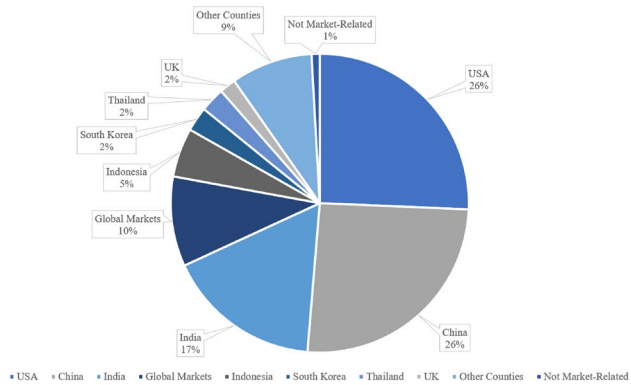


Fig. 10 Geographical distribution of stock market data in research

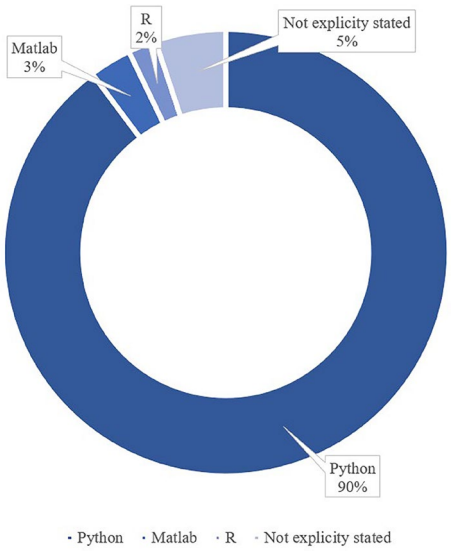


Fig. 11 Programming languages used in financial research

- High-frequency data: Such as minute-by-minute or second-by-second data, is often used in algorithmic trading research or high-frequency market analysis.
- Low-frequency data: Such as daily, weekly, or monthly data, might be used for long-term investment studies or macroeconomic modeling.

The most common frequency in our references was daily frequency since the statistics of stock market are mostly divided by day. Figure 12 summarized the statistics of frequency of data collection in financial research.

6.6 Period of Collected Data

This chart likely shows the time span of data used in the studies, which can vary widely. Researchers might use:

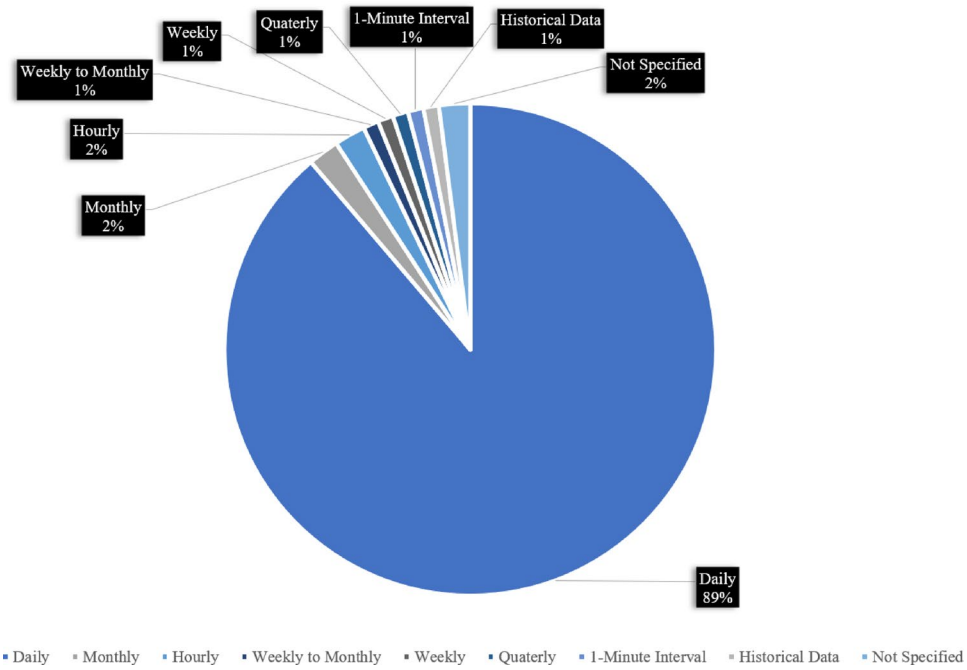


Fig. 12 Frequency of data collection in financial research

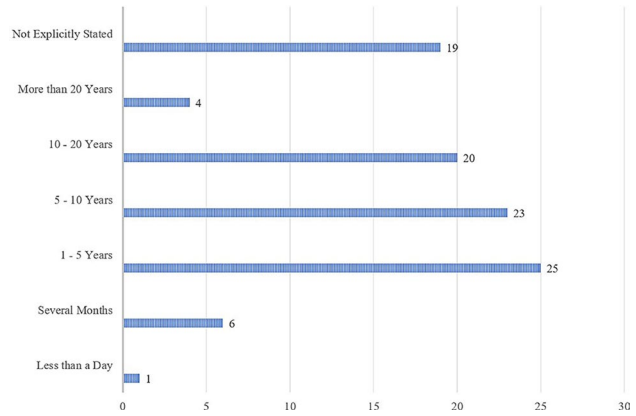


Fig. 13 Time period of collected data in financial studies

- Short-term data: Focused on a few months or years, often used in short-term market behavior analysis.
- Long-term data: Spanning decades or even centuries, useful for understanding long-term trends, asset price movements, and economic cycles.

Insights:

- A predominance of short-term data may indicate a focus on market volatility, speculative behavior, or algorithmic trading.
- Long-term data could suggest research on asset valuation, economic forecasting, or fundamental analyses that require historical perspective.

Figure 13 summarized the statistics of time period of collected data in financial studies.

6.7 Performance Metrics

Financial models and strategies can be evaluated using different types of performance metrics. Figure 14 summarized the statistics of performance metrics in financial research.

- Root Mean Square Error (RSME): In most regression models, Root Mean Square Error (RMSE) is used to evaluate their accuracy. Additionally, it emphasizes larger errors, making it particularly useful for identifying areas where models may fall short of results. The RSME measures how well a regression line fits the data points. In residuals, RMSE can also be referred to as Standard Deviation. It refers to the magnitude of the difference between predicted values (such as the results of a model) and observed values (the actual results). Essentially, it measures how well a model predicts numeric outcomes.
- Mean Absolute Error (MAE): This method calculates the average difference between calculated and actual values. Machine learning produces scale-dependent accuracy by calculating the error in observations.
- Mean Absolute Percentage Error (MAPE): Mean absolute percentage error measures the average magnitude of error produced by a model, or how far off predictions are on average.
- Mean Squared Error (MSE): Mean Squared Error (MSE) plays an important role in assessing the accuracy

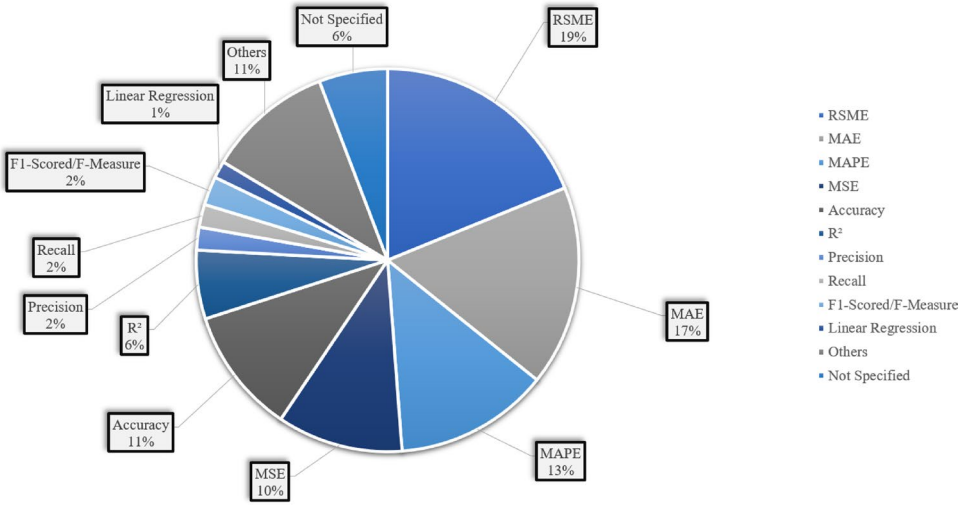


Fig. 14 Performance metrics in financial research

of predictive models in statistics and machine learning. This parameter is used to calculate the model's accuracy. In this variable, the average square difference between the predicted and actual values is calculated.

- Accuracy: In terms of correct predictions, classification accuracy is a fundamental metric used to evaluate the performance of a classification model.
- R-squared: The R-squared coefficient measures how much variation in the dependent variable (y) is explained by the regression line as opposed to the variance explained by the mean. It measures how accurately the regression line predicts each point's value compared to simply using y's average value.
- Precision: Positive predictions with a high percentage. The ratio can be calculated by subtracting the number of true positive predictions from the number of false positive predictions.
- Recall: A model's recall measures its ability to identify positive cases. Divide the true positives by the true positives (including false positives and false negatives).
- F1-Score/F-Measure: This means offers a balance between precision and recall. The F-Measure formula facilitates better machine learning decision-making by balancing precision and recall.
- Linear regression: Statistical modeling of relationships between independent variables and dependent variables uses linear regression. As a result, it provides useful insights for predicting and analyzing data.

6.8 Hybrid Models

Hybrid models generally offer improved predictive accuracy and robustness by combining the strengths of different model types. They can also provide a balance between performance and interpretability and potentially reduce biases and variances. Hybrid models are often more complex to design, implement, and maintain. They can also be prone to overfitting and require careful parameter tuning. Managing the complexity and potential interactions between different model components can be challenging. Complex hybrid models often require larger datasets for effective training and to avoid overfitting. Some hybrid models, especially those involving deep learning components and optimization algorithms, can be computationally demanding. There is often a trade-off between model interpretability and predictive accuracy. Simpler models are generally easier to interpret, while complex hybrid models might achieve higher accuracy at the cost of interpretability.

In addition to linear and non-linear aspects of financial time series, models like AWT-LSTM-ARIMA-GARCH and MEEMD-LSTM-MLP are also available. By using signal decomposition techniques before feeding data to the LSTM, CEEMDAN-LSTM and VMD-LSTM can reduce noise to a large extent. By combining CNNs to extract spatial features and LSTMs to learn temporal relationships, CNN-LSTM and CNN-Bi-LSTM-Attention excel. In complex financial markets, where many factors influence prices, MV-LSTM is specifically designed to handle multiple input features. In situations where interpretability and computational efficiency are more important than maximum accuracy, or when datasets are smaller, LSTM-DNN and LSTM-LightGBM may be preferable. Volatility forecasting models like AWT-LSTM-ARIMA-GARCH explicitly handle volatility. The use of robust models such as LSTM-BOXGBoost could also be beneficial. LSTM-GRU-SA-AM is a sentiment analysis model integrated with LSTM that may capture the changes in market sentiment driving volatility. LSTM-LightGBM, LSTM-DNN, or even a basic LSTM-SGD model might suffice in less volatile markets. In scenarios with low volatility, overly complex models may overfit to noise.

Sentiment data is explicitly integrated into BERT-LSTM and LSTM-GRU-SA-AM. Text features can be extracted

from texts using BERT for sentiment analysis. Input features such as technical indicators can be added to any LSTM-based model. Multiple technical indicators would be particularly well suited to MV-LSTM. The BERT-LSTM model may be more effective in sectors that are highly sensitive to news and public sentiment (e.g., technology, consumer discretionary). The LSTM-ARIMA variant might be preferred for sectors driven by macroeconomic factors and historical patterns (e.g., energy, materials). High-frequency trading requires models like LSTM-LightGBM or LSTM-XGBoost that offer a balance of speed and accuracy. Long-term investment strategies require models that are robust to changes in the market and take into account long-term dependencies, such as Bi-LSTM or Attention-LSTM. To predict and capitalize on volatility, models like AWT-LSTM-ARIMA-GARCH or models incorporating GARCH components are specifically designed. Table 34 describes the advantages and limitations of hybrid models.

6.9 Models and Market Conditions

The LSTM has been gaining a lot of attention in stock market prediction because it captures long-term dependencies in nonlinear and dynamic financial data, but some models outperform it. Under certain market conditions, certain models may perform better than LSTM because its performance is highly context-dependent. LSTMs achieve error rates below 2% in low-volatility contexts, whereas DBNs are more robust to high-volatility predictions (PTT and SCC). In such environments, DBNs excelled since they were able to extract complex hierarchical features from financial datasets that captured abrupt variations better than LSTMs. For LSTM models, market conditions with limited training data or sparse datasets are challenging. Due to their robust feature extraction mechanisms, models like CNN and evolutionary hybrids can outperform standalone LSTMs.

When training data is insufficient, CNNs excel at preprocessing trend-based data with high robustness. Through the optimization of spatial and temporal representations, CNN's integration with LSTM demonstrated improved performance across short-term and trend-based predictions. The prediction accuracy of CNN-LSTM models was found to be 19.3% better than LSTM and SVR models alone, according to a study. A distinct advantage of PSO-based and GA-based optimization techniques is the reduction of error rates compared to traditional neural network methods, particularly for highly non-stationary datasets. Additionally, traditional techniques like ARIMA and XGBoost may be superior for short-term predictions or simplified datasets requiring rapid computation. An attention-based mechanism or autoencoder-enhanced methodology often perform better than standard LSTMs under extreme market conditions. Particularly during times of market distress or political turmoil, attention-based LSTM models can enhance prediction accuracy by focusing on relevant data segments. The Autoencoder-LSTM approach also outperforms generative adversarial networks when analyzing cluster stocks in volatile scenarios. While LSTM is an effective tool, hybrid architectures often outperform or complement it when facing disordered market behavior or poor data quality. According to this comparative analysis, forecasting models should be tailored to the particular characteristics of datasets and market conditions.

6.10 Stock Data and time Points

Based on the 98 articles reviewed, we discovered a multidimensional perspective about the relevance of different models, the stock data used in the studies, and the choice of time points for stock market prediction. Several studies have addressed the inherent complexity of stock price movement, which emphasizes the role of advanced deep learning techniques in modeling these dynamics due to their nonlinear, volatile, and unpredictable patterns. The studies utilized various datasets, including the S&P 500, NASDAQ, NYSE, and NSE stock indices, the CSI 300 for Chinese markets, and various cryptocurrency datasets such as Bitcoin. In order to ensure that insights are applicable globally, researchers choose datasets based on geographical and economic diversity. Deep learning models like LSTM can inherently capture sequential dependencies in high-frequency datasets (intraday, hourly, daily), while statistical models like ARIMA can study overarching trends in long-term datasets (weekly, monthly). Investor sentiment is increasingly revealed by combining data from platforms such as Twitter, Weibo, and News outlets such as The New York Times with historical stock price data, which plays a crucial role in predicting sudden market shifts.

The choice of time points revealed a clear dichotomy. Trading decisions are made faster when short-term predictions are based on daily or intraday data. In these studies, CNNs, LSTMs, and CNN-LSTMs are heavily favored because they are able to capture finer-grained temporal patterns. In contrast, macroeconomic forecasting and portfolio allocation are well suited to long-term predictions based on monthly or annual data. In order to deduce statistical relationships in stock price trends, models like ARIMA and SVR are often applied to longer datasets. As a result of the COVID pandemic's repercussions, researchers have become increasingly dependent on short-term datasets to study market dynamics.

Data granularity and market dynamics played a major role in determining the effectiveness of the models used. Recurrent neural networks, such as LSTMs, are famous for

Table 34 Hybrid models advantages & limitations

Hybrid Model	Advantages	Limitations
CNN-LSTM	Captures spatial (CNN) and temporal (LSTM) features, robust to noise, effective in feature extraction.	Complex to train, computationally intensive, potential overfitting, requires large datasets.
LSTM-DNN	Combines sequence learning (LSTM) with deep feature learning (DNN), versatile.	Can be complex to optimize, DNN can overfit, requires careful architecture design.
LSTM-BO-LightGBM	Optimized by Bayesian Optimization (BO), efficient gradient boosting (LightGBM), fast and accurate.	BO can be computationally expensive, LightGBM sensitive to hyperparameters, might not capture complex non-linearities.
LSTM-BOXGBoost	Robust gradient boosting (XGBoost), handles outliers well, accurate.	XGBoost can be prone to overfitting, complex parameter tuning, computationally intensive for very large datasets.
LSTM-XGBoost	Combines sequence learning (LSTM) with powerful gradient boosting (XGBoost), high accuracy.	XGBoost can be prone to overfitting, complex parameter tuning, computationally intensive for very large datasets.
LSTM-LightGBM	Efficient gradient boosting (LightGBM), fast training, good generalization.	LightGBM sensitive to hyperparameters, might not capture very complex patterns.
GRU-LSTM	GRU for faster training, LSTM for long-term dependencies, complementary strengths.	Increased complexity, potential for redundancy in learning, requires careful architecture design.
Bi-LSTM	Captures bidirectional dependencies, improved context understanding.	More complex than unidirectional LSTM, computationally more intensive, can overfit.
SBU-LSTM	Enhances LSTM's memory or processing, potentially improved accuracy.	May add complexity.
EA-LSTM	Optimized architecture or parameters.	EA can be computationally expensive.
AE-LSTM	AE for feature extraction/dimensionality reduction, improved LSTM efficiency.	AE adds complexity, information loss during dimensionality reduction, requires careful AE design.
AWT-LSTM-ARIMA-GARCH	Captures linear (ARIMA), non-linear (LSTM), and volatility (GARCH) aspects, comprehensive modeling.	Very complex, requires expertise in multiple models, computationally intensive, potential for error propagation.
CEEMDAN-LSTM	CEEMDAN for noise reduction and signal decomposition, improved LSTM performance on noisy data.	CEEMDAN adds preprocessing complexity, can introduce artifacts, LSTM still needs careful tuning.
BERT-LSTM	BERT for sentiment analysis and feature extraction from text, LSTM for time series integration.	Primarily for sentiment-driven forecasting, BERT is computationally expensive, requires large text datasets.
MLCA-LSTM	Potentially improved pattern recognition.	MLCA can be complex to design and interpret.
TLBO-LSTM	Optimized by TLBO, efficient optimization, potentially robust.	TLBO might converge to local optima, LSTM still requires tuning.
MV-LSTM	Multi-Variate LSTM, handles multiple input features effectively, captures complex relationships.	Requires more data for training, increased complexity, potential for feature redundancy.
SVM-LSTM-GRU	SVM for classification/regression, LSTM and GRU for sequence learning, ensemble of different model types.	Very complex, potential for model conflict, requires careful integration and tuning of three models.
K-means-LSTM	K-means for data segmentation/clustering, LSTM for pattern learning within clusters, improved localized predictions.	K-means clustering might not be optimal, LSTM performance depends on cluster quality, added preprocessing step.
PSO-LSTM	Optimized by PSO, efficient parameter search, potentially robust.	PSO might get stuck in local optima, LSTM still needs tuning.
GA-LSTM-CNN	Optimized by GA, CNN for feature extraction, LSTM for temporal patterns, robust optimization.	GA can be computationally expensive, complex to set up, CNN and LSTM need individual tuning.
VMD-LSTM	VMD for signal decomposition, LSTM for learning decomposed components.	VMD adds preprocessing complexity, parameter selection for VMD is crucial, LSTM performance depends on VMD output.
GCN-LSTM	GCN for network feature extraction, LSTM for temporal dynamics in networks.	Requires data representable as graphs, GCN adds complexity, LSTM needs tuning, specialized for network data.
Confrontation-LSTM (CL-Net)	Potentially improved robustness and adversarial resistance.	may increase training complexity.
CNN-Bi-LSTM-Attention	CNN for feature extraction, Bi-LSTM for bidirectional context, Attention for focusing on important parts.	Very complex, computationally intensive, attention mechanism adds parameters, potential overfitting.

Table 34 (continued)

Hybrid Model	Advantages	Limitations
W-LSTMG	Potentially improved sensitivity to important features.	Might add complexity in weighting mechanism.
1D-CapsNet-LSTM	1D-CapsNet for feature extraction, robust to viewpoint variations, LSTM for temporal modeling.	CapsNet is complex and computationally intensive, LSTM adds to complexity, requires careful architecture design.
Attention-LSTM	Attention mechanism for focusing on relevant time steps, improved long-term dependency learning.	Attention adds complexity, computationally more intensive, can be harder to interpret.
MEEMD-LSTM-MLP	MEEMD for non-linear decomposition, MLP for final prediction.	MEEMD adds significant preprocessing, MLP might not capture temporal dependencies as well as LSTM alone.
TG-LSTM	Potentially improved temporal gating mechanism.	May add complexity in gating mechanism.
LSTM-SGD	LSTM trained with SGD, basic and widely applicable.	SGD can be slow to converge, sensitive to learning rate, can get stuck in local minima.
LSTM-kNN	kNN for non-parametric regression/imputation, LSTM for time series modeling.	kNN can be computationally expensive for large datasets, LSTM and kNN might not integrate seamlessly.
LSTM-ARO	Potentially improved generalization.	Might add complexity in regularization/optimization.
LSTM-GA	Optimized by GA, robust parameter search, potentially global optimum.	GA can be computationally very expensive, complex to set up, LSTM still needs tuning.
LSTM-UP	Potentially improved uncertainty estimation.	Might add complexity in uncertainty quantification.
LSTM-RF-MDBN	RF for feature selection/ensemble, LSTM for temporal modeling, MDBN for complex pattern recognition.	Very complex, requires expertise in multiple models, computationally intensive, potential for overfitting.
LSTM-IARO	Optimized by IARO, enhanced generalization.	Might be computationally more expensive than standard optimization.
LSTM-GRU-SA-AM	GRU and LSTM for sequence learning, Sentiment Analysis (SA) integration, AM for focus, comprehensive model.	Extremely complex, computationally very intensive, requires large datasets, potential for overfitting, difficult to interpret.

their ability to maintain long memories and perform well across multiple datasets and time points. LSTM outperforms GARCH for short-term predictions. By leveraging spatial and temporal data features, hybrid methods like CNN-LSTM and Attention-LSTM improved accuracy across multiple datasets. In one study, LSTM was used to combine technical features such as the golden cross and death cross with stock prices, resulting in enhanced predictive accuracy with reduced error rates. In addition, LSTM models using sentiment analysis from news headlines or social media have shown to improve forecast accuracy when combined with stock data.

The computational intensity of deep learning approaches is a recurring theme across these articles, despite their powerful effectiveness. There remains a challenge in tuning hyperparameters such as hidden layer sizes, batch sizes, and learning rates, as noted in studies, which highlight the variance in results due to resource limitations like Tensor processing units. Hybrid models, which combine deep learning with statistical or machine learning techniques, offer potential solutions to make more robust predictions across volatile and stable datasets.

Additionally, these studies highlighted the nuanced relationship between predictive factors and model performance. Despite what many people think, incorporating more predictive factors does not always lead to better results, as evidenced by experiments that demonstrate that fewer, but more meaningful inputs produce better results; for example, studies that compare single-factor and multi-factor LSTM applications with historical prices and social media sentiment. Furthermore, technical analysis indicators like RSI, MACD, and stochastic crossovers improve prediction accuracy when their value transformations are fed into models, highlighting the importance of feature engineering. Several critical points are highlighted by these studies:

- It is important to choose stock data and time points that align with the model and prediction objective, regardless of whether it is short-term volatility or long-term trends.
- Hybrid models, which incorporate domain-specific preprocessing techniques, provide compelling alternatives to deep learning models in noisy or volatile markets.
- Sentiment analysis combined with historical data will provide greater insight into human behavior and financial data, which contributes to stock market forecasting.

Table 35 The summery of stock datasets

Stock Index/Market	Frequency of Use Across Articles	Most Common Dataset Source	Additional Observations
S&P 500	25 articles	Yahoo Finance	Often combined with technical indicators; analyzed for long-term trends.
NASDAQ	15 articles	Historical stock data	Used for volatile and nonlinear stock behavior studies.
NYSE	19 articles	NASDAQ and Yahoo Finance	Focused on short-term (daily/hourly) price movements.
DJIA (Dow Jones)	17 articles	Financial news-based datasets	Frequently used with sentiment analysis approaches.
NSE (India)	20 articles	Nifty 50 dataset via Yahoo Finance	Aimed at local prediction studies; integrates event-driven data.
CSI 300 (China)	8 articles	Local financial institutions	Focused on hybrid model implementations with LSTM.
Cryptocurrencies (Bitcoin)	7 articles	Social media-driven sentiment	Evaluated for hourly volatile trends via Bi-LSTM and CNN.

Technical analysis, sentiment modeling, and deep learning innovations are increasingly interconnected, paving the way for more sophisticated and accurate forecasting. The studies examined a variety of stock markets and indices. Table 35 summarizes the aggregated stock datasets.

Table 36 summarizes the range of time periods for analyzed stock data based on the model's objective and data availability.

Table 37 contains some information and its key points related to the choice of the time points and the stock data involved.

6.11 Models and Parameter Designs

It is interesting to note that although LSTM models are similar in architecture across 98 studies, their parameter designs differ considerably based on the dataset, the prediction task, and the research objectives. This article discusses a number of differences, similarities, and common parameter choices, along with their reasoning. The different parameters in LSTM are explained.

Table 36 Time period of the stock data involved in the study

Study Objective	Time Period Analyzed	Granularity (Daily/Hourly/Intra)
Short-term price movements	2015–2023 (most common)	Hourly/Daily
Long-term event-driven trends	2000–2020	Weekly/Monthly
Post-COVID-related financial forecasting	2020–2022	Monthly intervals
Cryptocurrency-specific market predictions	Mostly 2–5 years of data	Intraday

Number of Layers and Neurons *Studies have found different layers and configurations of neurons.*

- Single-layer models: This method is used in simpler setups with limited data or short-term predictions. According to Gülmez [16], a one-layer LSTM with 20 neurons was used to capture basic trends in price fluctuations over intraday data.
- Multi-layer models: More complex tasks often require two- and three-layer configurations. For datasets with long time periods and higher noise, several studies used 500 neurons per layer across three layers (e.g., NASDAQ and CSI 300).

Epochs and Batch Size *Different studies have demonstrated a wide variation in epoch counts and batch sizes.*

- Epoch Count: In a dataset with a large number of epochs and limited computational resources, there are wide variations in epoch counts. For small datasets, such as predicting stock price trends, 5–20 epochs are sufficient. It is often necessary to use 100–500 epochs in large datasets with long historical windows.
- Batch Size: is recommended that high-frequency intraday datasets be analyzed in batches as small as 16, while long-term monthly datasets should be analyzed in batches as large as 64.

Activation Functions *Different models have different activation functions based on their objectives.*

- There are two common activation functions, sigmoid and tanh, which can constrain values between specific ranges. In LSTM architectures, the sigmoid is particularly effective for gates such as the forget gate.
- In some cases, gradients vanished when ReLU was used to speed up computations.

Table 37 The effectiveness of some of the common models

Model	Best MAE	Best RMSE	Key points
LSTM	0.047	0.0050	Captures long-term dependencies, robust for sequential data. Computational expense for processing nonlinearities. Outperforms ARIMA and SVR in volatile and nonlinear markets.
CNN	Variable	Variable	Handles spatial patterns effectively in data. Limited in handling time-sequential dependencies. Performs well in forecasting multiple stock indices with spatial features.
ARIMA	0.1904	0.0380	Good for linear trends and time-series forecasting. Struggles with volatile, nonlinear datasets. Good for monthly trend analysis.
SVR	0.096	Variable	Accurate for regression-based analytical tasks. Lacks robustness for high variations in stock prices. Effective for regression tasks but performs poorly with noisy data compared to deep learning models.
CNN-LSTM	<0.05	~ 0.04	Combines temporal and spatial feature extraction; excels in volatile market scenarios. Delivers excellent results for high-frequency, intraday data modeling.
Attention-LSTM	0.0479	0.0050	Integrates feature relevance dynamically; superior for sparse datasets with noise. Achieves precision by focusing on critical data features dynamically.

Optimization Algorithms The Adam Optimizer is the most commonly used optimization algorithm, since it can handle sparse gradients and has an adaptive learning rate. In a few studies, however, moment-based optimizers were used to speed up convergence, which is usually appropriate for noisy datasets. Several studies have used RMSprop to accelerate convergence, but most use moment-based optimizers.

Sliding Window Size *Studies typically used sliding windows of 30 days, but approaches that used dynamic windowing (e.g., adjusting for events such as COVID-19-induced volatility) yielded better results.*

Hybrid Variants of LSTM *Attention-based LSTMs and CNN-LSTMs were extensively used in specialized tasks.*

- The attention mechanisms introduce a layer which focuses on relevant features dynamically, improving predictions using datasets with heterogeneous features, such as sentiment analysis and technical indicators.
- CNN-LSTM hybrids incorporate spatial feature extraction for volatile markets, such as cryptocurrency, using datasets such as candlestick charts.

The following examples illustrate how parameter choices are similar after examining the differences:

Universal Features in Cell Design *The fundamental cell structure of LSTM remains consistent across studies, regardless of hyperparameter variability.*

- Forget Gate: Regulates irrelevant information during training.
- Input Gate: Selects relevant information for updating the cell state.
- Output Gate: Filters processed information for prediction.

In contrast to classical RNNs, LSTM models effectively capture sequential dependencies and mitigate vanishing gradients.

Feature Scaling *Most of the studies apply Min-Max normalization to scale input features between $[0, 1]$ or $[-1, 1]$, to ensure faster model convergence and better compatibility with activation functions.*

Dataset Division *In order to ensure robust generalization, it is usually a good idea to split datasets into 70% for training and 30% for testing.*

Performance Metrics *Common evaluation metrics includes RMSE, MAE, MAPE, and Accuracy; with RMSE being the most frequently used due to its sensitivity to large errors.*

Table 38 shows the most common parameters across our reviewed papers.

The design of LSTM parameters varies significantly across studies based on the complexity of the dataset, the prediction horizon, and the domain-specific challenges; however, there are some consistent similarities in the use of optimization techniques, activation functions, and general architectural choices. It is common to choose Adam optimizer, Sigmoid/tanh activation, and window sizes of 30 days in studies, which reflect that tested practices in this field have been widely adopted. In order to improve prediction accuracy, advanced techniques such as Attention-LSTM and CNN-LSTM hybrids dynamically focus on input relevance or incorporate spatial information. Both theoretical guidelines and domain-specific experiments drove continual evolution in parameter designs as researchers sought to balance computational efficiency with prediction performance.

Table 38 Common parameters across studies

Parameter	Most Common Choices	Reasoning Behind Parameter Choices
Layers	1–3 layers. Single-layer for small tasks; multi-layer (2–3) for complex data.	Fewer layers and neurons reduce computational complexity, suitable for simpler datasets.
Neurons	Range between 20–500 per layer. Larger models used for noisy datasets.	More layers provide deeper feature extraction, while larger neuron counts handle noise and complexity in larger datasets.
Activation	Sigmoid, tanh (most common). Occasionally ReLU for faster computation.	Sigmoid and tanh are favored for maintaining stability in gate outputs, ensuring long-term dependencies are effectively captured. The bounded output range prevents exploding gradients.
Optimizer	Adam was widely preferred, followed by RMSProp.	Adam is universally preferred for its computational efficiency and adaptability across convex and non-convex problems. It also minimizes learning rate fluctuations, rectifying vanishing gradients.
Batch Size	16–64 based on dataset complexity.	Smaller batch sizes such as 16 are used for complex datasets with high variability; and larger batch sizes such as 64 are suitable for simpler datasets
Learning Rate	$0.001-0.0001$ for adaptive learning.	Higher learning rate (e.g., 0.001) speeds up training and is good for early stages or simpler problems; while lower learning rate (e.g., 0.0001) allows finer adjustments and is better for complex problems.
Dropout	0.2–0.4 to prevent overfitting.	Dropout reduces overfitting by randomly ignoring subsets of neurons during training, ensuring model generalization.
Window Size	Fixed (e.g., 30 days) or dynamic windows based on financial events.	Sliding windows segment time-series data, enabling models to track relevant price movements while balancing computational load.

6.12 Analyzing the Effects of Different Features for Models

According to a careful analysis of the 98 articles reviewed, the input features of stock price prediction models, especially those based on LSTM, profoundly influence their accuracy and effectiveness. The choice of features, whether based on raw price data, technical indicators, or sentiment analysis, affects the model's ability to detect patterns, adapt to market dynamics, and produce reliable predictions. The following section examines how features differ across studies, the reasoning behind their selection, and their impact on prediction quality.

Historical Stock Data (Basic Features) *This approach uses primary stock metrics such as.*

- Opening price.
- Closing price.
- High and low prices.
- Volume (OHLCV).

Many studies use historical stock price data to predict future prices, such as NASDAQ and CSI 300 indices. Especially in stable markets, raw price information can provide accurate predictions within short periods of time. As a result, results suffer in volatile conditions without additional contextual or explanation information. When only OHLCV features are used, metrics such as RMSE and MAE are often higher, since these features do not capture external factors such as investor sentiment or macroeconomic events.

Technical Indicators *Common indicators include:*

- Moving averages (e.g., EMA, SMA).
- Relative Strength Index (RSI).
- Bollinger Bands.
- MACD.

In one study, 50-day, 100-day, and 150-day EMAs are combined with RSI into LSTM models in NSE datasets, which results in significant improvements in prediction reliability. Indicators help to capture stock trends, momentum, and volatility better than raw historical data alone. The smoothing of noisy price movements makes models more predictable. In studies that used technical indicators in conjunction with OHLCV data, RMSE and R2 scores were consistently lower.

Predictions are improved both in stable markets and in markets with high volatility as a result of this combination.

Sentiment Analysis (External Features) *Data is extracted from the following sources to determine emotional responses and market sentiment.*

- Social media (e.g., Twitter, Weibo).
- News headlines and text features (e.g., New York Times sentiment scores).

The BERT sentiment analysis model combines quantified sentiment scores and OHLCV data to predict Chinese stocks with higher accuracy. Price movements are driven by behavioral factors such as speculative behavior or macroeconomic events. Models that integrate sentiment features are particularly effective at predicting dynamic intraday fluctuations. Sentiment-based models outperform models based only on historical data or technical indicators in volatile markets (for example, the S&P 500 during the COVID-19 disruption).

Macro-Financial Indicators *These features include global economic measures like.*

- Inflation rates.
- Exchange rates.
- Policy uncertainty indices such as GARCH-MIDAS models related to stock volatility.

Studies that used macroeconomic indicators and LSTM to predict BTC prices versus USD exchange rates showed better modeling of long-term financial trends. It is possible for macro-level features to introduce noise in short-term predictions, but they are better for long-term horizons. FOREX rates and commodities are great examples of assets that can benefit from these features. Macro indicators will reduce errors such as MAE when paired with raw stock data, especially for multi-year datasets.

Hybrid Features (Combined Models) *Multiple feature categories are combined.*

For example, OHLCV + Technical indicators + Sentiment data + Macroeconomic indicators.

In predicting stock indices across multiple regions, CNN-LSTM models integrating wavelet-transformed technical indicators, sentiment scores, and commodity prices perform better than standalone models. By using hybrid features, models are more robust to market changes across a variety of market dynamics. In addition to reducing noise, these features can also address market complexity, leading to advancements in RMSE, MAE, and prediction latency in articles that employ these features.

Table 39 explained some key points related to different feature types. Here we examine some of those points.

In stock market models, different features do significantly affect prediction results. Diversity of features tends to outperform relying on single features. According to articles, including technical or sentiment indicators reduces prediction errors and increases generalization. It is important to align features with the dynamics of the market and the prediction task at hand. The integration of sentiment scores with OHLCV is almost always beneficial for short-term predictions, whereas long-term trends require technical and macroeconomic indicators. Hybrid models consistently outperform traditional models; for example, articles utilizing CNN-LSTM and Attention mechanisms thrive by taking advantage of both external and internal factors. In addition to basic features, advanced technical indicators, sentiment analysis, and macroeconomic data can greatly enhance the performance and robustness of prediction models. The accuracy of hybrid feature models is highest when all relevant inputs are integrated, as described in several articles, thereby balancing short- and long-term volatility. Model success is therefore determined by feature selection tailored to prediction tasks.

6.13 Stock Movement Categorizing

LSTM networks were primarily utilized to predict stock prices. In the studies that have been reviewed, historical price data, technical indicators, and other input features such as market sentiment from news and social media were used to predict stock movements. Regression analysis frameworks are heavily used in the studies that were reviewed. In these frameworks, numerical values of stock prices were predicted (e.g., open, close, etc.) and trends were predicted. These studies seek to optimize prediction accuracy and computational models rather than categorize stock movements. We summarized 22 studies from our references that categorize up and down movements of stock prices:

1. Thakkar & Chaudhari [17]: CREST predicts price trends in Indian stock exchanges (NSE and BSE) based on LSTM models.
2. Latif et al. [109]: For improved classification accuracy, the PLSTM-TAL model uses denoising methods (EEMD) and feature extraction (CAE) to classify trends as upward or downward.
3. He & Xu [90]: CL-Net is a model that aims to predict the direction of stock prices in long time-series stock data (rise or fall). According to the study's fluctuation

Table 39 Different feature types and their key points

Feature Type	Advantages	Disadvantages	Performance Impact	Reasons Behind Feature Selection
Basic Stock Data	Simple and computationally efficient; works well on stable datasets.	Limited in capturing external drivers (e.g., sentiment).	Moderate prediction capacity; high error in noisy markets.	Crucial for identifying recurring price patterns across stable time windows. Commonly used when computational simplicity is required.
Technical Indicators	Captures momentum and volatility trends effectively; smooths noise.	Lagging in detecting abrupt changes (e.g., news-driven events).	Consistent improvements in RMSE and R^2.	Helps overcome unpredictability in noisy datasets by smoothening patterns. Preferred for intraday trading or medium-term predictions.
Sentiment Analysis	Reflects behavioral drivers; accounts for emerging trends (e.g., crisis).	Challenges in extracting reliable features with noisy sentiment data.	Significant improvements in dynamic market prediction.	Essential during periods of socio-political instability or major economic events. Real-time prediction benefits from investor sentiment dynamics.
Macro Indicators	Excellent for long-term global predictions.	Adds complexity, may dilute prediction accuracy for short-term.	Best for multi-year horizons; limited scalability short-term.	Supports better calibration for global market fluctuations.
Hybrid Features	Holistic modeling for intricate dynamics; maximizes feature synergy.	Computationally expensive; risk of overfitting with excess features.	Performs best; combines advantages of all inputs.	When multiple datasets (historical prices, sentiment, macroeconomic indices) interact, combining features produces the most reliable outputs.

range, samples are categorised into positive and negative movements.

4. Dhokane & Agarwal [3]: It explicitly discusses the use of a Confusion Matrix to classify stock trends into uptrends (class 1) and downtrends (class 0). The model evaluates the accuracy of the predictions for stocks like Reliance, Kotak Bank, Infosys, etc., based on precision, recall, and F1-scores.
5. Liu et al. [49]: It explicitly categorizes training samples into ups and downs based on daily stock movements, labeled as "true" (up) and "false" (down).
6. Wang [97]: Discusses directional movement prediction, which predicts up-and-down directions for intraday trading using LSTMs and Random Forests.
7. Cheng et al. [50]: This paper explicitly involves a multi-class categorization study of stock price movements. The model classifies stock movements into 9 categories. The output from the attention-based LSTM model predicts these classes, which are then used to generate trading strategies (buy for increasing classes and sell for decreasing classes).
8. Faraz et al. [55]: Stock price movements are explicitly categorized in this paper using multiple classes. Stock movements are categorized into nine categories based on the model. Attention-based LSTM models predict these classes, which are then used to generate trading strategies (buy for increasing classes and sell for decreasing classes).
9. Nashir et al. [9]: This study incorporates sentiment analysis from news articles (using GDELT dataset) and correlates them with price movements of the Indonesian Stock Index (JKSE).
10. Wijaya et al. [78]: The purpose of this paper is to categorize stock price movement indicators (Golden Cross/Death Cross) to reflect upward and downward trends. As a result, stock movements can be predicted more accurately than directly using raw technical indicator values.
11. Chang et al. [62]: In this paper, up-and-down amounts are used as input features to predict price trends.
12. Thakkar and Chaudhari [79]: This paper includes trend prediction, which inherently involves categorizing stock prices' directional movements.
13. Wang et al. [14]: In this article, technical indicators such as Relative Strength Index (RSI) are used to categorize up-and-down movements of stock prices. Stock price movements are predicted using RSI to identify overbought and oversold conditions. It is effective to use this categorization in order to speculate on price trends and momentum.
14. Chen et al. [81]: K-means clustering is used to categorize historical stock price trends in terms of their movement patterns over time (including ups and downs). In order to apply the prediction model, this clustering helps capture inherent similarities or differences in stock price dynamics.

15. Huang et al. [4]: In this article, stock trends are explicitly categorized as oscillating, rising, and falling trends. Prediction accuracy is evaluated using these categories, and strategies adapted to the types of trends are developed.
16. Banik et al. [67]: This study directly relates to swing trading, which involves the upward and downward movements of stock prices. This study uses LSTM to predict swing trading patterns by distinguishing support and resistance levels, and trend reversals, all of which fall under the category of categorizing price movement directions.
17. Gao & Zhang [26]: The purpose of this study is to investigate the correlation and causality between investor sentiment and stock price movements, analyzing short-, medium-, and long-term fluctuations using VMD and LSTM. Stock indices are explicitly related to upward or downward movements in investor sentiment.
18. Swathi et al. [83]: Tweets are classified into positive and negative sentiments using sentiment analysis on Twitter data. The sentiment polarity is then used to predict upward or downward trends based on stock price movements.
19. Shi et al. [13]: This article uses GCN and LSTM to forecast stock price movements (up and down). Relationship-based graphs are constructed and price directions (upward or downward) are predicted.
20. Thu et al. [86]: The purpose of this paper is to highlight price movements with significant changes using a dynamic sliding window method. It uses LSTM primarily to predict stock prices, but also categorizes changes in stock prices over time.
21. Noubakhsh & Habibi [84]: It discusses trend prediction, or predicting stock price movements upwards or downwards. Although the methodology emphasizes fundamental indicators, such as profitability, it also involves understanding trends and categorizing movement.
22. Balamohan & Khanaa [119]: LSTM and CNN are integrated to analyze stock price trends. Trends can be used to distinguish upward and downward movements in stock prices.

In the reviewed studies, LSTMs consistently outperform traditional statistical methods for modeling the temporal dynamics of financial time-series data. Studies like Thakkar & Chaudhari [17] used LSTMs to predict stock trends (upward or downward) in the Indian market, leveraging their ability to deal with temporal inputs effectively. Additionally, Latif et al. [109] incorporated feature extraction (CAE) and denoising (EEMD) into their PLSTM-TAL model for binary trend classification, which substantially improves accuracy. He & Xu [90] also used Confrontation-LSTM Networks (CL-Nets) to categorize stock price fluctuations (positive versus negative movements), making them useful for long-term and highly volatile markets. Dhokane & Agarwal [3] demonstrated how to incorporate technical indicators to achieve highly precise predictions of uptrends and downtrends by integrating MACD and EMA features along with LSTM, which underscores the importance of technical integration for stock movements prediction.

Liu et al. [49] and Faraz et al. [55] emphasized binary outcomes like “up” or “down”, which ensures practical utility for real-time decisions. Cheng et al. [50] enhanced it by integrating an attention mechanism within LSTM to strengthen its multi-class categorization capabilities. Balamohan & Khanaa [119] and Shi et al. [13] proposed hybrid approaches that include both sequential dependencies and complex relationships in price data. In Nashir et al. [9], Gao & Zhang [26], and Swathi et al. [83], sentiment analysis emerges as another powerful enhancement. As a result, social media and news data (e.g., Twitter polarities or GDELT data) were combined with LSTM models to reflect external market sentiment, which resulted in a more comprehensive forecast.

K-means clustering was mentioned by Chen et al. [81] and advanced architectural approaches, such as Thakkar and Chaudhari [79], demonstrated innovative preprocessing and feature grouping to deal with the noisy and non-stationary nature of stock market data before LSTM modeling. According to these studies, grouping price movements into clusters or trends (e.g., oscillating, rising, falling) refines feature extraction, resulting in higher accuracy. Banik et al. [67] explored advanced adaptive techniques such as swing trading models to identify support and resistance thresholds, which focus on medium-frequency trading. Furthermore, Wijaya et al. [78] and Wang et al. [14] used technical analysis indicators such as RSI (Relative Strength Index) and Golden Cross/Death Cross patterns to improve predictions about trend reversals and directional movements, which demonstrates their effectiveness in speculative trading scenarios.

Despite these challenges, Wang [97] identified a trade-off between computational intensity and prediction accuracy, particularly when additional machine learning models like Random Forests are incorporated. Thu et al. [86] demonstrated how adaptive sliding window techniques enhance temporal segmentation for better response to abrupt market changes. These advancements, however, are limited by the availability of data, the diversity of features, and the requirement to tune hyperparameters. According to Huang et al. [4], hybrid evolutionary strategies offer improved outcomes by combining trained LSTM models with algorithmic trading, however, they will face certain constraints in responding to market randomness. It is also noted that

historical price data cannot always accommodate sudden or external economic or political changes.

LSTM networks are capable of incorporating hybrid methodologies and enhanced features, which has been found to be a strong point across these studies. Noubakhsh & Habibi [84] developed LSTM-CNN hybrid models for analyzing stock prices, focusing on profitability indicators to understand long- and short-term trends. Chang et al. [62] took a straightforward approach by relying solely on trend-labeled data (such as up-and-down magnitudes) as input. Balamohan & Khanaa [119] and other hybrid studies emphasized the importance of leveraging both long-term and localized dependencies when analyzing transactional data. In stock price forecasting, these innovations demonstrate the growth of interdisciplinary efforts.

7 Concluding Remarks

This review summarized 98 studies on Long Short-Term Memory (LSTM) networks and stock market prediction. LSTM networks are significant when combined with other techniques such as sentiment analysis and hybrid model architectures, as we show in our analysis to provide a general overview of this continuously developing field. As a result of these integrations, predictive accuracy or robustness are improved. This review indicates that LSTM networks provide highly accurate predictions, making them an effective tool for optimizing portfolios, reducing risks, and developing strategies within the financial industry, while also empowering investors to make more informed decisions. LSTM offers lower-risk options that yield consistent returns by predicting low-volatility stocks, which is ideal for conservative investment strategies. For stock predictions, LSTM has the potential to process large-scale, nonlinear data, and by combining diverse data streams, such as historical stock prices and external sentiment, it facilitates robust forecasting frameworks when integrated into hybrid models with attention mechanisms or CNNs. Hybrid models like these have been used in practical scenarios, such as using attention-based LSTM networks to forecast price movements and assist traders. In addition to providing portfolio managers with data-driven insights to mitigate risks and optimize asset allocation during uncertain times, this capability democratizes financial decision-making by reducing entry barriers for retail investors through intuitive dashboards and tools. Additionally, the dual purpose of these models, which are designed to meet long-term investment objectives as well as short-term trading goals, makes them more inclusive for a variety of investor profiles than traditional methods like ARIMA and SVR. As a result, financial institutions can incorporate LSTM models into real-time monitoring systems to identify irregular trading patterns and anticipate market disruptions proactively. Adding attention mechanisms further enhances these benefits by enabling analysts to refine their analyses of sentiment data from social media and news outlets, giving them a more nuanced tool for anticipating market reactions. As computational efficiency and model interpretability continue to improve, LSTM-based frameworks are expected to have greater impacts on risk management, asset placement, and investor education. As a result, predictive practices across the financial sector will be revolutionized and systemic risks will be mitigated. This review emphasizes the critical role of data preprocessing, feature engineering, and architectural considerations in optimizing LSTM performance. Challenges persist despite recent advances. Financial markets present ongoing challenges due to noise and non-stationarity. In future research, LSTM should be applied to emerging financial domains and more sophisticated hybrid models, addressing computational complexity associated with deep learning models. The purpose of this review is to provide a robust foundation for future work, guiding researchers in the development and application of LSTM-based approaches. Figure 15 illustrates the relative strengths of LSTM against other models (like ARIMA and SVR).

8 Future Works

Several promising research directions are highlighted in the article on LSTM-based execution price forecasting. The purpose of this article is to summarize insights from 98 studies that highlight both the advantages and challenges associated with the use of LSTM. It offers a solid foundation for identifying research directions that can advance the field further. It is also possible to greatly increase the impact of research by promoting interdisciplinary collaboration. Economics, psychology, and data science can be used to predict stock prices. Future research can facilitate partnerships between these disciplines. It combines behavioral and context analysis to enhance predictive models. It suggests several research avenues for understanding the complex dimensions of LSTM network stock price prediction. The development of preprocessing techniques and ethical considerations for financial AI can advance financial prediction. In the modern stock market, investors will be able to make more informed decisions by improving predictive analysis methods and investing skills.

Future research should focus on the following areas:

- Advanced LSTM Architectures: Predictive accuracy in complex financial environments can be enhanced by

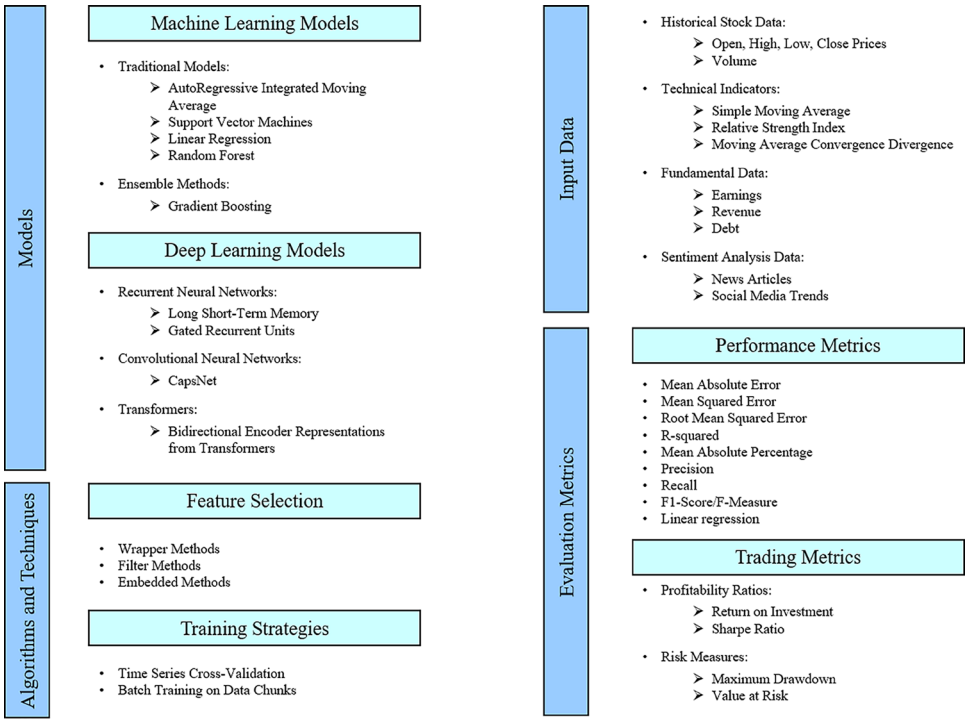


Fig. 15 Overview of machine learning and deep learning techniques for stock price prediction

evolving LSTM architectures. It is often difficult to deal with long-range dependencies and noisy data when using traditional LSTMs. In order to capture a more comprehensive view of temporal relationships, Bi-LSTMs process data in both forward and backward directions. Data analysis of multivariate time series is made easier by convolutional LSTMs, which integrate convolutional layers for extracting both spatial and temporal features. The dynamic weighting of inputs in attention-based models like EA-LSTM and DABiLSTM further refines feature extraction, reducing noise and improving model focus. In order to build more robust and reliable financial forecasting models, these advanced architectures are essential.

- Hybrid Modeling Techniques: The use of LSTM in combination with other machine learning techniques can offer a synergistic approach to financial forecasting. With CNN-LSTM hybrid models, for example, spatial features are extracted from financial data and then fed to LSTMs to model temporal dependencies. By integrating sentiment analysis with LSTM, models can incorporate external factors, such as news and social media trends, to better understand market sentiment. The prediction accuracy and robustness of hybrid models are increased by incorporating CNNs, GRUs, and DBNs. In order to create more comprehensive and adaptable forecasting systems, researchers need to combine diverse modeling approaches.
- Refined Data Preprocessing: To ensure model accuracy, financial data must be preprocessed with sophisticated techniques due to its inherent noise and non-stationarity. It is crucial to identify the most relevant variables for prediction, and automated feature extraction can reduce manual intervention and maximize model efficiency. It is essential to prepare data for LSTM models by normalizing, detrending, and removing outliers to ensure clean, consistent, and representative input data. The performance and reliability of LSTM-based forecasting models can be significantly improved by refining data preprocessing methods.
- External Data Integration: Many external factors influence financial markets, from news events to social media sentiment to macroeconomic indicators. Integrating these diverse data sources into LSTM models enhances predictive accuracy and model robustness by providing a holistic view of market dynamics. LSTM models can capture sentiment impacts on stock prices while macroeconomic indicators provide insight into broader economic trends. It is possible to create more comprehensive and context-aware forecasting models by incorporating external data sources.
- Ethical and Regulatory Frameworks: Increasingly, AI-driven trading systems have ethical implications that must be addressed. For these systems to be used responsibly, issues such as fairness, transparency, and accountability must be carefully considered. It is imperative that future research on AI in finance focus on developing ethical guidelines and regulatory frameworks so that these technologies can benefit society as a whole. To create a more equitable and trustworthy financial ecosystem,

we must address issues such as algorithmic bias, data privacy, and market manipulation.

- Model Validation Strategies: Validation methods must be robust in order to prevent overfitting and ensure generalizability. In real-world applications, backtesting and cross-validation techniques ensure LSTM models are reliable and effective by evaluating their performance on unseen data. In cross-validation, data are split into multiple subsets for training and testing, while backtesting simulates the performance of a trading strategy on historical data. Financial forecasts are more reliable and trustworthy when researchers use rigorous validation strategies to ensure their models are accurate and robust.
- Computational Efficiency: Real-time applications pose a significant challenge to LSTM networks due to their computational complexity. High-frequency trading and other time-sensitive scenarios require LSTM networks to be optimized for faster training and deployment. In order to make LSTM models more practical for real-time deployment, pruning, quantization, and distributed training can be used. As a result of pruning, the model's parameters are reduced in precision, whereas quantization involves removing unnecessary connections. Multiple machines can be used to train the model, significantly reducing training time.
- Large-Scale Datasets: Deep learning models perform best when high-quality, large-scale datasets are available. In order to train robust and accurate LSTM models, it is essential to acquire and curate such datasets. It is also possible to augment existing datasets and improve the performance of models by adding noise or transforming existing data. The complexity of financial markets can be captured by LSTM models trained with large-scale datasets.
- Explainability and Interpretability: LSTM networks have a black-box nature that makes it difficult to understand how they come up with predictions. In order to build trust and confidence in LSTM-based models, their interpretability must be enhanced. LSTM models can become more transparent and trustworthy by using techniques such as SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations). In SHAP values, input features are quantified as contributions to the model's predictions, while LIME provides local approximations. In order to make more informed decisions, researchers can use these techniques to gain a deeper understanding of the inner workings of LSTM models.
- Cross-Domain Applications: LSTM networks are well-suited to modeling temporal dependencies beyond finance. LSTM networks can be applied to other domains, such as healthcare, transportation, and e-commerce, to capture temporal dependencies across various applications. Models trained on financial data can be adapted to other domains using transfer learning techniques, resulting in fewer data sets needed. LSTM networks can be exploited in various fields by exploring cross-domain applications.
- Real-Time Prediction Systems: In order to develop real-time trading systems using LSTM networks, it is necessary to develop pipelines that process data efficiently and with low latency. In real-time applications, optimizing LSTM models can provide traders with immediate market insights, enabling quick response to changing market conditions. It requires robust risk management strategies, efficient data processing pipelines, and low-latency model inference. The development of more sophisticated and responsive trading systems requires optimizing LSTM models for real-time deployment.
- Risk Management: The incorporation of risk assessment metrics into LSTM models is essential for safer investment strategies. Different investment decisions can be evaluated based on volatility, correlation, and tail risk measures. It is possible to make more informed decisions and manage risk more effectively by integrating these risk metrics into LSTM models. Market events may surprise the model's assumptions, the quality of the input data, and the model's assumptions.
- User-Friendly Tools: Non-experts may have difficulty utilizing LSTM networks because of their complexity. LSTM-based models should be deployed in financial decision-making using accessible platforms to bridge the theoretical research and practical implementation gap. Model output should be visualized clearly with these tools, allowing users to understand and interpret model predictions. The financial industry can make more informed decisions through the use of user-friendly interfaces for LSTM-based models.
- Comparative Studies: In order to identify the strengths and weaknesses of different approaches, LSTM needs to be compared to other models comprehensively. The relative merits of different modeling techniques can be better understood when models are evaluated against a variety of datasets and performance metrics. Additionally, comparative studies can provide practitioners with valuable insight into the conditions under which LSTM models outperform other approaches. Various financial forecasting tasks can be identified using these studies.
- Novel Loss Functions: It is possible that the standard loss functions used in machine learning are not optimal for financial time series data. In order to improve LSTM model performance, it is possible to design custom loss functions tailored to financial time series data. Loss functions can reward accurate trend predictions or penalize

large prediction errors. The development of novel loss functions can help researchers optimize LSTM models for specific financial forecasting tasks, improving their accuracy and reliability.

- Transfer Learning: In data-scarce environments, the cost of training deep learning models from scratch can be prohibitive. Pre-trained LSTMs can be adapted to new datasets using transfer learning techniques, reducing the amount of data needed to train new models, increasing their versatility and efficiency. In transfer learning, knowledge is transferred from one dataset to another dataset in order to improve the performance of a model. LSTM training is now faster and more efficient, allowing a greater range of users to benefit from it.
- Ensemble Methods: Machine learning techniques such as combining multiple LSTM models to improve prediction accuracy and robustness are well established. It combines the strengths of individual models to improve accuracy and robustness. Ensembles of LSTM models can be created using bags and boosts, improving their performance and reducing overfitting risks. Ensemble methods can improve the accuracy and reliability of LSTM-based financial forecasting.
- Anomaly Detection: Financial data can be used to identify outliers and unusual patterns. In order to detect anomalies and outliers, LSTM networks can be used. The LSTM model can detect deviations in financial data that could indicate anomalous behavior by training to recognize normal patterns. Financial systems are protected from fraud by anomaly detection techniques.
- Global Market Analysis: In order to analyze data from multiple markets simultaneously, models must be able to take into account the interconnectedness of global financial markets. In addition to dealing with data from different markets, LSTM applications are required to handle exchange rates and other factors when extending to global stock markets. Cross-border investments can be made more informed with the help of LSTM models. Financial markets are complex, so processing and analyzing data from diverse sources is required.
- Long-Term Forecasting: Investors and policymakers must be able to predict long-term market trends. It is necessary to create models that can capture long-range dependencies in financial data in order to improve LSTM's ability to predict long-term market trends. It may involve developing new LSTM architectures or incorporating external data sources that provide insight into long-term market dynamics. In order to make strategic planning and investment decisions, long-term forecasting models can provide valuable insights.
- Integration of Alternative Data: Market dynamics may not always be captured fully by traditional financial data sources. It may be beneficial to integrate alternative data sources, such as satellite imagery, credit card transactions, and geolocation data, in order to strengthen predictive accuracy through additional context and insights into market behavior. Financial markets may be influenced by factors such as consumer behavior, economic activity, and a number of other factors. Alternative data sources can provide valuable insights into these factors. Researchers can improve forecasting accuracy by incorporating alternative data sources into LSTM models.
- Quantum Computing: With quantum computing, complex problems can be solved more efficiently than they would with classical computers. The possibility of using quantum computing to accelerate LSTM training and improve model performance in the future is a promising area for research in the field of financial forecasting in the future. There is a possibility that quantum algorithms may be able to train LSTM models faster and more accurately than classical algorithms, enabling the development of more sophisticated and powerful forecasting systems. In spite of the fact that quantum computing is still in its infancy, it does have the potential to revolutionize the process of financial modeling and forecasting in the near future.
- Causal Inference: Machine learning models traditionally focus more on identifying correlations between variables than causal relationships. In order to improve prediction accuracy, causal inference techniques can be used to identify causal relationships between different market variables. The use of causal inference techniques can help researchers disentangle cause-and-effect relationships, enabling them to create forecasting models that are more accurate and reliable. Future research in financial modeling could benefit from this approach.

It can be concluded that LSTM networks provide an effective approach to predicting stock markets. They will achieve their full potential only if they are able to address existing challenges and pursue the future research directions outlined. In addition to improving investment strategies, these developments will help develop LSTM-based models that are more reliable, scalable, and ethically sound. These research avenues promise to transform financial forecasting.

9 Conclusion

The purpose of this article has been to provide a thorough examination of LSTM networks and their pivotal role in transforming stock market prediction. In this review, insights from a broad range of research studies are synthesized to illustrate how LSTM models are effective at capturing

complex temporal dynamics in financial time series data. In addition to highlighting LSTM's strengths in modeling complex relationships and long-term dependencies, this exploration also underscored the value of methodological rigor. We can enhance current models by acknowledging their limitations, including the difficulties associated with hyperparameter tuning and computational resource requirements. In financial forecasting, LSTM appears to have a very promising future. In addition to developing novel architectures, refining data preprocessing techniques, and exploring more advanced hybrid models, real-time data integration presents major opportunities. To maintain trust in automated systems and ensure the responsible advancement of these powerful technologies, transparency, accountability, and fairness must also be considered in the deployment of AI in financial markets. Both researchers and practitioners can use this article as a guide to consolidate existing knowledge. In order to navigate the complexities of the stock market with more accuracy and reliability, we must bridge the gap between theoretical advancements and real-world applications. As a result, investors will have access to insights and tools that will enable them to make more informed financial decisions. Financial forecasting will be transformed by the continued advancement of LSTM methodologies, which will enable more accurate and reliable predictions.

Acknowledgements Not applicable.

Author Contributions Melika Shafiei Hafshejani: Design the study, Gathering data for the article, Interpretation of the resultsNajme Mansouri: Writing- reviewing and editing, Ideas.

Funding Not applicable.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical Approval Not applicable.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

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